10 CHAPTER

Enzyme Science and Technology

Introduction

Enzymes are biological catalysts. They increase the rate of chemical reactions taking place within living cells without themselves suffering any overall change. The reactants of enzyme-catalysed reactions are termed substrates and each enzyme is quite specific in character, acting on a particular substrate or **substrates** to produce a particular product or products.

All enzymes are proteins. However, without the presence of a non-protein component called a cofactor, many enzyme proteins lack catalytic activity, when this is the case, the inactive protein component of an enzyme is termed the **apoenzyme**, and the active enzyme, including **cofactor**, the **holoenzyme**. The cofactor may be an organic molecule, it is known as a **coenzyme**, or it may be a metal ion. Some enzymes bind cofactors more tightly than others. When a cofactor is bound so tightly that it is difficult to remove without damaging the enzyme it is sometimes called a **prosthetic group**.

In 1878 German physiologist Wilhelm Kuhne first used the term enzyme. In 1897 Eduard Buchner began to study the ability of yeast extracts that lacked any living yeast cells to ferment sugar. He named the enzyme that brought about the fermentation of sucrose "zymase". In 1907 he received the Nobel Prize in Chemistry "for his biochemical research and his discovery of cell-free fermentation."

In 1926, James B. Sumner showed that the enzyme urease was a pure protein and crystallized it; Sumner did likewise for the enzyme catalase in 1937. The conclusion that pure proteins can be enzymes was definitively proved by Northrop and Stanley, who worked on the digestive enzymes pepsin (1930), trypsin and chymotrypsin. These three scientists were awarded the 1946 Nobel Prize in Chemistry.

This discovery that enzymes could be crystallized eventually allowed their structures to be solved by x-ray crystallography. This was first done for lysozyme, an enzyme found in tears, saliva and egg whites that digests the coating of some bacteria; the structure was solved by a group led by David Chilton Phillips and published in 1965.

Characteristics of enzymes : Enzymes being proteins exhibit all properties of proteins. They have their specific isoelectric points at which they are least soluble. Like proteins, they can be denatured by changes in pH and temperature.

Enzymes undergo physical changes during the reaction but revert to their original form at the end of the reaction. Enzymes exhibit enormous catalytic power. The rates of enzymatically catalysed reactions are 10^6–10^{12} times greater than those of the corresponding uncatalysed reactions and several times greater than those of the corresponding chemically catalysed reactions.

Specificity of the enzymes : One of the characteristic features which distinguish enzymes from catalysts is their specificity. Enzymes are specific in the reaction catalysed and in their choice of substrates. It usually catalyses a single chemical reaction or a set of closely related reactions. Three kinds of specificities are observed.

(*i*) **Absolute specificity :** When enzymes catalyse only one particular reaction they are said to exhibit absolute specificity e.g. Urease acts only on urea.

(*ii*) **Group specificity :** Enzymes acting on a group of substances that possess a particular type of linkage common to that group of substances are said to exhibit group specificity. Amylase hydrolyses the group of substances like starch, dextrin and glycogen, which have the same type of glycosidic linkages (α 1,4).

(*iii*) **Optical specificity :** Almost all enzymes show a high degree of optical specificity. There are certain enzymes which catalyse the hydrolysis of same group of substances possessing same optical activity i.e. D-amino acid oxidase acts on D-amino acid and L-amino acid oxidase acts on L-amino acid. Maltase catalyses the hydrolysis of α-but not β-glycosides.

Classification of Enzyme

Enzymes are classified and named by the commission on Biochemical Nomenclature of the International Union of Biochemistry. An enzyme may be denoted in one of the following three accepted ways: (*i*) a four number code following the letters EC (for Enzyme Commission) *e.g.*, EC 1.1.1.3 (the first number refers to enzyme class, the second to subclass, the third to sub-subclass and the fourth to the serial number of the enzyme within a sub-subclass), (*ii*) its systematic name based on the above classification, and (*iii*) its recommended name. The enzymes have been classified in the following 6 classes.

Class 1. Oxidoreductases are involved in redox ractions, *i.e.*, transfer of hydrogen or oxygen atoms between molecules. This class includes dehydrogenases (hydride transfer), oxidases (e transfer to O_2), oxigenases (oxygen atom transfer from O_2). and peroxidases (e transfer to peroxides). Example, glucose oxidase (EC 1.1.3.4).

Class 2. Transferases catalyse the transfer of an atom or group of atoms (like acyl-, alkyl-and glycosyl-groups) between two molecules. The transferred groups are different from those transferred by the other classes of enzymes like oxidoreductases, etc. Example, aspartate aminotransferase (EC 2.6.1.1).

Class 3. Hydrolases are those enzymes, which catalyze hydrolytic reactions (and their reversals); this class includes esterases, glycosidases, proteases and lipases. Example, chymosin or rennin (EC 3.4.23.4).

Class 4. Lyases are involved in elimination reactions resulting in the removal of a group of atoms from the substrate molecule. This class includes aldolases, decarboxylases, dehydratases and some pectinases. Example, histidine ammonia lyase (EC 4.3.1.3).

Class 5. Isomerases catalyse the formation of isomers of molecules; they include epimerases, racemases and intramolecular transferases. Example, xylose isomerase (EC 5.3.1.5).

Class 6. Ligases, Synthases or Synthetases catalyze the formation of covalent bonds between two molecules utilizing the energy obtained from hydrolysis of a nucleoside triphosphate like ATP or GTP. Example, glutathione synthase (EC 6.3.2.3).

Mechanism of Enzyme Action

Energy of activation : Enzymes increase the speed of a chemical reaction. They lower the energy of activation of a reaction, thus enabling it to occur at ordinary physiological temperatures. When reactions proceed from one directon to another they have to overcome an energy barrier called the *activation energy*. Normally only a small part of the total number of molecules in a compound contain enough energy to react. Application of heat enables a larger proportion of molecules to overcome the activation energy barrier. However, living systems are relatively isothermal, *i.e.*, they do not have large temperature differences. They therefore employ enzymes to activate the molecules of the reacting compounds. When the enzyme combines with the substrate to form the enzyme-substrate complex the energy level of the substrate is raised, and it reacts faster. Enzymes increase the speed of a chemical reaction thousands of times by bringing about mutual contact of reacting compounds. The coming together of these compounds is no longer a matter of chance, but becomes a certainty.

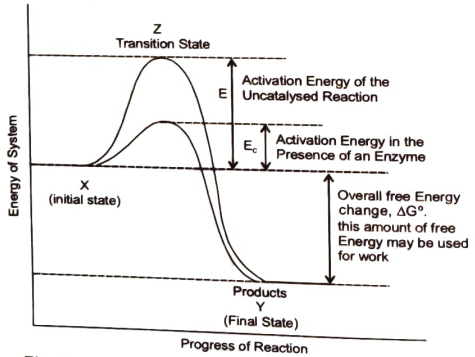


Fig. Energy profile of uncatalysed and enzymic reactions

In Fig., the chemical compound X is in a metastable state and it requires E as energy of activation to convert the reactant in a new activated state Z. The reaction then becomes spontaneous, converting Z into Y and energy E is liberated. A further amount of energy is released according to the difference in enthalpy (ΔH) of X and Y. Thus enzymes increase the rate of conversion of X into Y with a lower energy of activation. For example, the energy of activation for the decomposition of H_2O_2 is 18 K cal $mole^{-1}$, while in the presence of enzyme catalyses it is only 6.4 K cal $mole^{-1}$. Thus, the energy of activation in this case becomes about one-third in the presence of enzyme.

Active Site : The functional groups that are essential for the formation of ES complex occur at a specific location on the surface of the enzyme molecule. This section of enzyme where substrate binding and transformation of substrate to product occurs is called as active site. Many attempts have been made to implicate specific amino acid residues (side chain or R groups) as being part of the active site of various enzymes. Some of the amino acids occurring at the active site of enzymes are hydroxyl group of serine, sulfhydryl group of cysteine, imidazole group of histidine and carboxyl group of aspartic acid. Two theories were proposed to explain the mechanism of enzyme action.

Fischer's lock and key theory (Rigid template model) : According to this theory proposed by Emil Fischer during 1890s, the active site possesses a unique conformation which is complementary to the structure of the substrate thus enabling the two molecules to fit together in much the same way as a key fits into a lock. An unfortunate feature of this model is the implied rigidity of the catalytic site. However, while this model explains enzyme specificity, it fails to explain the stabilization of the transition state that enzymes achieve. The "lock and key" model has proven inaccurate and the induced fit model is the most currently accepted enzyme-substrate-coenzyme figure.

Koshland's induced fit model : In 1958, Daniel Koshland suggested a modification to the lock and key model: since enzymes are rather flexible structures, the active site is continually re-shaped by interactions with the substrate as the substrate interacts with the enzyme. As a result, the substrate does not simply bind to a rigid active site, the amino acid side chains which make up the active site are moulded into the precise positions that enable the enzyme to perform its catalytic function. In some cases, such as glycosidases, the substrate molecule also changes shape slightly as it enters the active site. The active site continues to change until the substrate is completely bound.

Enzyme Kinetics

The Michaelis-Menten Equation : The study of enzyme kinetics began in 1902 when Adrian Brown reported an investigation of the rate of hydrolysis of sucrose as catalyzed by the yeast enzyme **invertase** (now known as β-fructofuranosidase) :

$$\text{Sucrose} + H_2O \rightarrow \text{glucose} + \text{fructose}$$

Brown demonstrated that when the sucrose concentration is much higher than that of the enzyme, the reaction rate becomes independent of the sucrose concentration; that is, the rate is **zero order** with respect to sucrose. He therefore proposed that the overall reaction is composed of two elementary reactions in which the substrate forms a complex with the enzyme that subsequently decomposes to products and enzyme.

$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_2} P + E$$

Here E, S, ES, and P symbolize the enzyme, substrate, **enzyme-substrate complex**, and product, respectively According to this model, *when the substrate concentration becomes high enough to entirely convert the enzyme to the ES form, the second step of the reaction becomes rate limiting and the overall reaction rate becomes insensitive to further increases in substrate concentration.*

The general expression for the velocity (rate) of this reaction is

$$v = \frac{d[P]}{dt} = k_2[ES]$$

The overall rate of production of [ES] is the difference between the rates of elementry reactions leading to its appearance and those resulting in its disappearance:

$$\frac{d[ES]}{dt} = k_1[E][S] - k_{-1}[ES] - k_2[ES] \qquad \text{....(1)}$$

This equation cannot be explicity integrated, however, without simplifying assumptions. Two possibilltes, are:

1. Assumption of Equilibrium

In 1913, Lenor Michaelis and Maude Menten, building upon earlier work by Victor Henri, assumed that $k_{-1} >> k_2$, so that the first step of the reaction achieves eqilibrium.

$$K_s = \frac{k_{-1}}{k_1} = \frac{[E][S]}{[ES]}$$

Here K_s is the dissociation constant of the first step in the enzymatic reaction. Although this assumption is not often correct, in recognition of the importance of this pioneering work, the noncovalently bound enzyme-substrate complex ES is known as the Michaelis complex.

2. Assumption of steady state

The rate of synthesis of ES must be equal to its rate of consumption over most of the course of the reaction; i.e. [ES] maintains a steady state. One can therefore assume with a reasonable degree of accuracy that [ES] is constant; i.e.

$$\frac{d[ES]}{dt} = 0$$

This so-called steady-state assumption was first proposed in 1925 by G.E. Briggs and James B.S. Haldane.

In order to be of use, kinetic expressions for overall reactions must be formulated in terms of experimentally measurable, quantities. The quantities [ES] and [E] are not, in general, directly measurable but the total enzyme concentration is usually readily determined.

$$[E]_T = [E] + [ES]$$

The rate equation for our enzymatic reaction is then derived as follows. Combining equation with the steady state assumption equation, and the conservation condition, equation (1) gives

$$k_1([E]_T - [ES])[S] = (k_{-1} + k_2)[ES]$$

which upon rearrangement becomes

$$[ES](k_{-1} + k_2 + k_1[S]) = k_1[E]_T[S]$$

Dividing both sides by k_1 and solving for [ES], we get

$$[ES] = \frac{[E]_T[S]}{K_M + [S]}$$

where K_M, which is known as the Michaelis constant, is defined as

$$K_M = \frac{k_{-1} + k_2}{k_1}$$

The initial velocity of the reaction from equation can then be expressed in terms of the experimentally measurable quantities $[E]_T$ and [S].

$$v_0 = \left(\frac{d[P]}{dt}\right)_{t=0} = k_2[ES] = \frac{k_2[E]_T[S]}{K_M + [S]} \quad ...(i)$$

The use of the initial velocity (operationally taken as the velocity measured before more than ~ 10% of the substrate has been converted to product) rather than just the velocity minimizes such complicating factors as the effects of reversible reactions, inhibition of the enzyme by product, and progressive inactivation of the enzyme.

Maximal velocity of a reaction, V_{max}, occurs at high substrate concentrations when the enzyme is saturated, that is, when it is entirely in the ES form

$$V_{max} = k_2[E]_T \quad ...(ii)$$

Combining equations (*i*) and (*ii*), we get

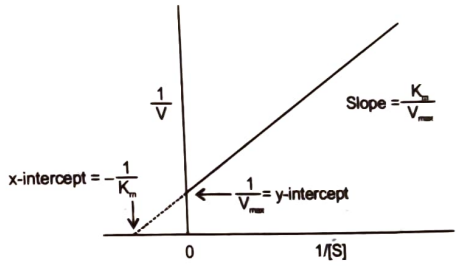
$$v_0 = \frac{V_{max}[S]}{K_M + [S]}$$

This expression, the **Michaelis-Menten equation**, is the basic equation of enzyme kinetics. The Michaelis constant, K_M, has a simple operational definition. At the substrate concentration where $[S] = K_M$, Equation yields $v_0 = \frac{V_{max}}{2}$ so that K_M is the substrate concentration at which the reaction velocity is half-maximal. Therefore, if an enzyme has a small value of K_M, it achieves maximal catalytic efficiency at low substrate concentrations.

Analysis of Kinetic Data : A better method for determining the values of V_{max} and K_M, which was formulated by Hans Lineweaver and Dean Burk, uses the reciprocal of Equation

$$\frac{1}{v_0} = \left(\frac{K_M}{V_{max}}\right)\frac{1}{[S]} + \frac{1}{V_{max}}$$

This is a linear equation in $1/v$ and $1/[S]$. If these quantities are plotted, the so-called **Lineweaver-Burk** or **double reciprocal plot**, the slope of the line is K_M/V_{max}, intercept is $1/V_{max}$, and the extrapolated intercept is $-1/K_M$.



Hanes Plot : The Lineweaver-Burk equation may be rearranged to yield the linear equation for the Hanes plot.

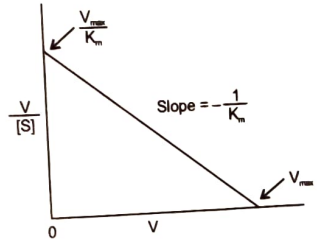
$$\frac{[S]}{V} = \frac{1}{V_{max}}[S] + \frac{K_m}{V_{max}}$$

Thus, a plot of [S]/V versus [S] is linear with a slope of $1/V_{max}$. The intercept on the [S]/V axis gives K_m/V_{max}.

Eadie-Hofstee Plot : Another widely used linear form of the Michaelis-Menten equation is the Eadie-Hofstee plot, which is described by

$$\frac{V}{[S]} = -\frac{V}{K_m} + \frac{V_{max}}{K_m}$$

The plot of V/[S] versus V is linear with slope of $-1/K_m$. The intercept on the V axis gives V_{max}.



An Eadie-Hofstee plot

Factor Affecting Enzymatic reaction

The factors that mainly influance any enzyme catalised reaction are

(1) Substrate concentration (2) Enzyme concentration (3) Temperature (4) pH (5) Inhibitors

Other factors such as state of enzyme (oxidation), time and activators also affect enzyme-catalysed reaction to certain extent.

Substrate concentration : Keeping the factors such as pH, temperature and enzyme concentration at optimum levels, if the substrate concentration is increased, the velocity of the reaction recorded a rectangular hyperbola. At very low substrate concentration the initial reaction velocity (v) is nearly proportional to the substrate concentratrion (first order kinetics). However, if the substrate concentration is increased the rate of increase slows down (mixed order kinetics). With a further increase in the substrate concentration the reaction rate approaches a constant (zero order-reaction where velocity is independent of substrate concentration).

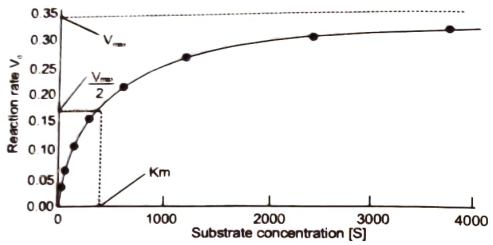


Fig. Saturation curve for an enzyme reaction showing the relation between the substrate concentration

Significance of Km : Km is (roughly) an inverse measure of the affinity or strength of binding between the enzyme and its substrate. Lower the Km, greater the affinity (so the lower the concentration of substrate needed to achieve a given rate)

Enzyme concentration : When compared to substrate concentration, the concentration of enzyme is always very very low on molar basis. Hence, increasing the enzyme concentration will always increase the reaction rate.

Temperature : Over a limited range of temperature, the velocity of enzyme-catalysed reactions roughly doubles with a 10°C rise in temperature. Enzymes, being proteins, are denatured by heat and become inactive as the temperatuer increases beyond a certain point. Most of the enzymes are inactivated at temperatures above 60°C. The temperature, at which the reaction rate is maximum, is known as optimum temperature.

pH : Most enzymes have a characterstic pH at which their activity is maximum; above or below this pH, the activity declines.

Activity of Enzymes

The activity of enzymes can be expressed in a number of ways:

1. The recommended unit of enzyme activity is the **katal**, which is the amount of activity that converts 1 mol of substrate per second. Frequently, the activities are expressed in microkatals (μkat), nanokatals (nkat) etc. corresponding to reaction rates of μmol, nmol s^{-1} respectively.
2. An older term, the International Enzyme Unit (IU), is defined as the amount of enzyme which will catalyse the transformation of 1 μmol of substrate min^{-1} under standard conditions.

Other related terms are:

(*i*) The specific activity of an enzyme preparation is kat kg^{-1} of protein or IU mg^{-1} of protein.

(*ii*) The molar activity is kat mol^{-1} of enzyme.

(*iii*) The turnover number of an enzyme is the number of molecules of substrate transformed per active site of the enzyme min^{-1}.

Catalytic Efficiency (k_{cat}/K_M)

$$k_{cat} = \frac{V_{max}}{[E]_T}$$

This quantity is also known as the turnover number of an enzyme because it is the number of reaction processes (turnovers) that each active site catalyzes per unit time.

Equation indicates that for the Michaelis-Menten model, $k_{cat} = k_2$.

When $[S] << K_M$, very little ES is formed. Consequently, $[E] = [E]_T$, so that equation reduces to a second-order rate equation.

$$v_0 = \left(\frac{k_2}{K_M}\right)[E]_T\,[S] = \left(\frac{k_{cat}}{K_M}\right)[E]\,[S]$$

k_{cat}/K_M is the apparent second-order rate constant of the enzymatic reaction; the rate of the reaction varies directly with how often enzyme and substrate encounter one another in solution. *The quantity k_{cat}/K_M is therefore a measure of an enzyme's catalytic efficiency.*

Enzyme Inhibition

Inhibitors are substances which tend to decrease the rate of an enzymne-catalysed reaction. Although some act on a substrate or cofactor.

Reversible inhibitors bind to an enzyme in a reversible fashion and can be removed by dialysis (or simply dilution) to restore full enzymic activity.

Irreversible inhibitors cannot be removed from an enzyme by dialysis. Sometimes it may be possible to remove an irreversible inhibitor from an enzyme by introducing another component to the reaction mixture, but this would not affect the classification of the original interaction.

Reversible Inhibition

(*a*) Competitive

(*b*) Uncompetitive

(*c*) Non-competitive

(*d*) Mixed

(*a*) Competitive Inhibition : In competitive inhibition, the inhibitor and substract compete for the enzyme (i.e., they can not bind at the same time). Often competitive inhibitors strongly resemble the real substrate of the enzyme. For example, methotrexate is a competitive inhibitor of the enzyme dihydrofolate reductase, which catalyzes the reduction of dihydrofolate to tetrahydrofolate. Note that binding of the inhibitor need not be to the substrate binding site (as frequently stated), if binding of the inhibitor changes the conformation of the enzyme to prevent substrate binding and vice versa. In competitive inhibition the maximal velocity of the reaction is not changed, but higher substrate concentrations are required to reach a given velocity, increasing the apparent K_m.

(*b*) Uncompetitive Inhibition : In uncompetitive inhibition the inhibitor can not bind to the free enzyme, but only to the ES-complex. The EIS-complex thus formed is enzymatically inactive. This type of inhibition is rare, but may occur in multimeric enzymes.

(*c*) Non-competitive Inhibition : Non-competitive inhibitors can bind to the enzyme at the same time as the substrate, i.e. they never bind to the active site. Both the EI and EIS complexes are enzymatically inactive. Because the inhibitor can not be driven from the enzyme by higher substrate concentration, (in contrast to competitive inhibiton) the apparent V_{max} changes. But because the substrate can still bind to the enzyme, the K_m stays the same.

(*d*) Mixed inhibition : This type of inhibition resembles the non-competitive, except that the EIS-complex has residual enzymatic activity.

In many organisms inhibitors may act as part of a feedback mechanism. If an enzyme produces too much of one substance in the organism, that substance may act as an inhibitor for the enzyme at the beginning of the pathway that produces it, causing production of the substance to slow down or stop when there is sufficient amount. This is a form of negative feedback. Enzymes which are subject to this form of regulation are often multimeric and have allosteric binding sites for regulatory substances. Their substrate/velocity plots are not hyperbolar, but sigmoidal (S-shaped).

The coenzyme folic acid and the anti-cancer drug methotrexate are very similar in structure. As a result, methotrexate is a competitive inhibitor of many enzymes that use folates.

Irreversible inhibitors react with the enzyme and form a covalent adduct with the protein. The inactivation is irreversible. These compounds include eflornithine a drug used to treat the parasitic disease sleeping sickness. Penicillin and Aspirin also act in this manner. With these drugs, the compound is bound in the active site and the enzyme then converts the inhibitor into an activated form that reacts irreversibly with one or more amino acid residues.

Uses of inactivators : Inhibitors are often used as drugs, but they can also act as poisons. However, the difference between a drug and a poison is usually only a matter of amount, an example of an inactivator being used as a drug is aspirin, which inhibits the COX-1 and COX-2 enzymes that produce the inflammation messenger prostaglandin, thus suppressing pain and inflammation. The poison cyanide is an irreversible enzyme inactivator flat combines with the copper and iron in the active site of the enzyme cytochrome c oxidase and blocks cellular respiration.

Mechanism of Reversible Inhibition

Mechanism	Equation	Diagram	V_0 versus [S] plot	Line weaver-Burk plot
Competitive	E+S⇌ES⇌P+E + I ⇵k_1 EI (Enzyme-inhibitor complex)	+ Substrate ⇌ Enzyme Inhibitor	No. I, V, +1 K_m= K [S]	$\frac{1}{V_0}$, +1, No. I, $\frac{1}{V}$ = $\frac{1}{V}$ $-\frac{1}{K_m}=-\frac{1}{K_m}$ $\frac{1}{S}$
Non-competitive	E+S⇌ES⇌P+E + + I I ⇵k_1 ⇵K_1 EI (Enzyme-inhibitor subtrate complex)	⇌ ⇌	V_0, No. I, V=V, +1 K_m= K_n[S]	$\frac{1}{V_0}$, +1, No. I, $\frac{1}{V}$, $\frac{1}{V}$ $=\frac{1}{K_m}=\frac{1}{K_m}$ $\frac{1}{S}$
Uncompetitive	E+S⇌ES⇌P+E + I ⇵k_1 EIS	⇌	V_0, No. I, V > V, +1 K_m= K_n[S]	$\frac{1}{V_0}$, +1, No. I, $\frac{1}{V}$, $\frac{1}{V}$ $=\frac{1}{K_m}=\frac{1}{K_m}$ $\frac{1}{S}$

Intercepts on Axes of A Lineweaver-Burk Plot In the Absence and Presence of Inhibitors

	Intercept on 1/n axis	*Intercept on 1/[S] axis*	*Slope*
No inhibitor	$\frac{1}{V}$	$\frac{-1}{K_m}$	$\frac{K_m}{V}$
Competitive inhibitor	$\frac{1}{V}$	$\frac{-1}{K_m\left(1+\frac{[1]}{K_i}\right)}$	$\frac{K_m}{V}\left(1+\frac{[1]}{K_i}\right)$
Non-competitive* inhibitor	$\frac{1}{V}\left(1+\frac{[1]}{K_i}\right)$	$\frac{-1}{K_m}$	$\frac{K_m}{V}\left(1+\frac{[1]}{K_i}\right)$
Uncompetitive inhibitor	$\frac{1}{V}\left(1+\frac{[1]}{K_i}\right)$	$\frac{-1}{K_m\left(1+\frac{[1]}{K_i}\right)}$	$\frac{K_m}{V}$

* K may be K'_i or K_1 depending on mechanism

Irreversible Inhibiton

An irreversible inhibitor binds to the active site of enzyme by an irreversible reaction :

$$E + I \rightarrow EI$$

and hence cannot subsequently dissociate from it. A covalent bond is usually formed between inhibitor and enzyme. The inhibitor may act by preventing substrate-binding or it may destroy some component of the catalytic site.

Many irreversible inhibitors attack-SH groups (in cysteine side chains) which are often found at the active sites of enzymes. Important examples are alkylating agents, such as iodoacetate and iodoacetamide, which form covalent linkages with essential-SH groups.

$$E - SH + ICH_2.CO^-_2 \rightarrow E - S - CH_2.CO^-_2 + HI$$

enzyme iodoacetate

Irreversible inhibitors are useful in the investigation of an enzyme, since the inhibitor, unlike the substrate, will remain firmly bound to one of the amino acids of the enzyme and thus act as a marker to enable it to be identified.

Multisubstrate Reactions

Kineticists recognize three general mechanisms for multi-substrate enzyme systems. Two of these mechanisms, termed ordered and random, require that all substrates must be added to the enzyme before any products can be released. In the third mechanism, called ping pong, one or more products is released from the enzyme before all the substates have reacted with the enzyme.

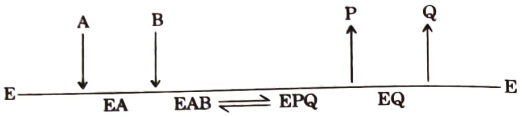
All substrates, if not specifically stated, are designated as A, B, C and D. All products are designated P, Q, R. Different forms of the enzyme are represented by E, F, G, E is the form of the enzyme that is free or at least most nearly free of any form of the substrates or products. The short-hand notation is applied to each of the following three general mechanisms.

(1) **Ordered Mechanism :** In this general mechanism, there is precise order in which substrates associate with the active sites of the enzyme. All of the products are released in a specific order and only after all of the substrates have reacted.

The reaction

$$E + A \rightleftharpoons EA \overset{B}{\rightarrow} EAB \rightleftharpoons EPQ \overset{P}{\rightarrow} EQ \rightleftharpoons E + Q$$

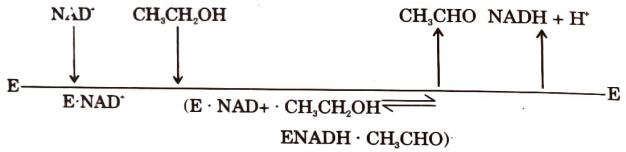
shows A as the first substrate to react with the enzyme E. Only then can B react to form the complex EAB. As catalysis occurs, first P and then Q are released, and in that order. The short-hand notation of this reaction is



This specific type of reaction is referred to as having an **ordered Bi, Bi mechanism**. The term **Bi** indicates two substrates or two products, so an ordered Bi, Bi mechanism requires two substrates and two products. **Uni** indicates a single substrate or a single product, and **Ter** three substrates or three products. A good example of an ordered Bi, Bi mechanism is that catalyzed by alcohol dehydrogenase. The two substrates are ethanol and NAD^+, and the two products are acetaldehyde and NADH. NAD^+ and NADH are nucleotide coenzymes. The proton that is produced in the reaction can be ignored if the reaction is carried out in a buffered solution, that is, constant H^+ concentration.

$$CH_3CH_2OH + NAD^+ \xrightleftharpoons{\text{Alcohol dehydrogenase}} CH_3CHO + NADH + H^+$$

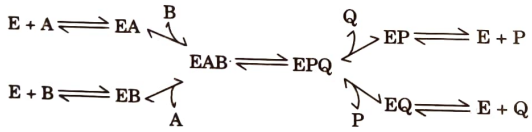
Experimental kinetic analyses of alcohol dehydrogenase revealed that the mechanism is ordered Bi, Bi, with the specific shorthand notation of



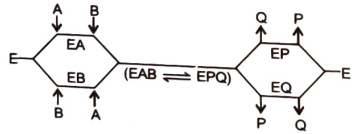
(2) Random Mechanism

When two substrates A and B add to an enzyme in random order and two products P and Q are released in a random order, such a sequence is designated as a random Bi, Bi mechanism.

Thus, a general reaction would be



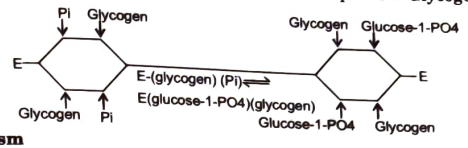
The short-hand notation is



The random Bi, Bi mechanism describes the raction that catalyzed by glycogen phosphorylase. In the equation here, inorganic phosphate is abbreviated Pi, and the product glycogen is, of course, one glucose unit smaller than the substrate glycogen.

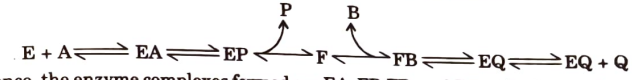
$$\text{Glycogen} + Pi \xrightleftharpoons{\text{glycogen phosphorylase}} \text{glucose-1 – phosphate + glycogen}$$

$$\text{Glycogen} + Pi \xrightleftharpoons{\text{Phosphorylase}} \text{Glucose-1 – Phosphate + Glycogen}$$

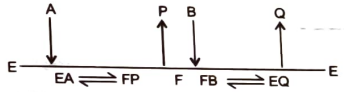


(3) Ping Pong Mechanism

For two substrates and two products, the general mechanism is depicted as



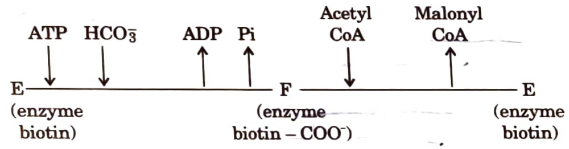
In this sequence, the enzyme complexes formed are EA, FP, FB, and EQ with A being first converted to P and then B to Q. F designates a modified enzyme (i.e., X-enzyme where X might be a phosphorylated, carboxylated, or other functional group attached to the enzyme transiently). F combines with B, and subsequently, X is transferred to B to form the second product Q and regenerate the original form of the enzyme, E. The short-hand notation for a Bi, Bi ping pong mechanism is



An example of a reaction of the ping pong type is that catalyzed by rat liver acetyl CoA carboxylase.

$$\text{Acetyl CoA} + ATP + HCO^-_3 \rightarrow \text{Malonyl CoA} + ADP + Pi$$

This is designated as a Bi, Bi, uni, uni ping pong mechanism because first two substrates add to the enzyme, then two products are released, then another substrate adds the final product is released.



Acetyl CoA carboxylase catalyzes a coupled reaction, it mediates the energetically unfavorable formation of a carbon-carbon bond by coupling the reaction to the structurally unrelated but energetically favorable hydrolysis reaction of ATP to ADP and inorganic phosphate.

Kinetics of Bi Bi Reactions

The rate equations for the above-described bisubstrate mechanisms in the absence of products are given below in double reciprocal form.

For ordered Bi Bi reactions:

$$\frac{1}{v_0} = \frac{1}{V_m} + \frac{K_M^A}{V_m[A]} + \frac{K_M^B}{V_m[B]} + \frac{K_S^A K_M^B}{V_m[A][B]}$$

For random Bi Bi reactions:

The rate equation for the general random Bi Bi reaction is quite complicated. However, in the special case that both substrates are in rapid and independent equilibrium with the enzyme; that is the EAB-EPQ interconversion is rate determining, the initial rate equation reduces to the following relatively simple form. This mechanism is known as the *rapid equilibrium random Bi Bi mechanism* :

$$\frac{1}{v_0} = \frac{1}{V_m} + \frac{K_S^A K_M^B}{V_m K_S^B[A]} + \frac{K_M^B}{V_m[B]} + \frac{K_S^A K_M^B}{V_m[A][B]}$$

For ping pong Bi Bi reactions :

$$\frac{1}{v_0} = \frac{K_M^A}{V_m[A]} + \frac{K_M^B}{V_m[B]} + \frac{1}{V_m}$$

Physical Significance of the Bisubstrate Kinetic Parameters

The parameters in the equations describing bisubstrate reactions have meanings similar to those for single-substrate reactions. V_m is the maximal velocity of the enzyme obtained when both A and B are present at saturating concentrations; K_M^A and K_M^B are the respective concentrations of A and B, necessary to achieve $\frac{1}{2}V_m$ in the presence of a saturating concentration of the other; and K_S^A and K_S^B are the respective dissociation constants of A and B from the enzyme, E.

Allosteric Enzymes

A modulator is a metabolite which, when bound to the allosteric site of an enzyme, alters its kinetic characteristics. The modulators for allosteric enzymes may be either stimulatory or inhibitory. A stimulator is often the substrate itself. The regulatory enzymes for which substrate and modulator are identical are called **homotropic**. When the modulator has a structure different that the substrate, the enzyme is called **heterotropic**. Some enzymes have more than one modulators.

The allosteric enzymes are structurally different from the simple nonregulatory enzymes. Besides the presence of an active or catalytic site, the allosteric enzymes also have one or more regulatory or allosteric sites for binding the modulator. Just as an enzyme's active site is specific for its substrate, the allosteric site is specific for its modulator. Enzymes with several modulators generally have different specific binding sites for each. By contrast, in homotropic enzymes, the active site and allosteric site are the same. Most of them have two or more polypeptide chains or sub-units, for example, *asparate transcarbamoylase* has 12 subunits.

Other differences between nonergulatory enzymes and allosteric enzymes involve kinetic properties. The allosteric enzymes often display sigmoidal plots of the reaction velocity, V versus substrate concentration, [S], rather than the hyperbolic plots predicted by Michaelis-Menten equation. In allosteric enzymes, the binding of substrate to one active site can affect the properties of other active sites in the same enzyme molecule. The activity of an allosteric enzyme may be altered by regulatory molecules that are bound to the allosteric sites, just as oxygen binding to haemoglobin is affected by H^+, CO_2 and BPG.

Isoenzymes

Enzymes which exist in multiple forms within a single species of organism or even in a single cell are called isoenzymes or isozymes. Such multiple forms can be detected and separated by gel electrophoresis of cell extracts. Since they are coded by different genes, they differ in amino acid composition and thus in their isoelectric pH values. Lactate dehydrogenase is an example for the isoenzymes which occur as five different forms in the tissues of the human and other vertebrates. All the five isozymes catalyze the same reaction.

$$\text{Lactate} + NAD^+ \longrightarrow \text{Pyruvate} + NADH + H^+$$

Abzymes

Abzymes are antibodies that catalyze specific chemical reactions, i.e. function as enzymes. Antibodies, by definition, have evolved to recognise and bind to the ground states of the molecules they are specific to. In contrast, enzymes have binding sites that preferentially bind to the transition states of their substrate molecules. For this reason, abzymes are not produced naturally. A catalytic antibody is produced in response to molecules that have a structure similar to the proposed/expected transition state of the substrate for the reaction to catalyse which the antibody (abzyme) is sought. Such molecules are, in fact, stable analogues of the transition states of the concerned substrates. The fact that such catalytic antibodies are in fact produced is the strongest evidence in support of the transition theory of enzyme action. Abzymes have been produced that catalyze such reactions as hydrolysis, e.g., of esters and carbonates, cleavage of amide bond, and biomolecular chemical reactions. Some abzymes incorporate metal ion cofactors like conventional protein enzymes.

Ribozymes

RNA molecules that have the capability of catalysing chemical reactions are called *ribozymes*. Ribozymes are true catalysts in that they enhance the rate of chemical reactions without any net change to themselves. They are capable of turnover, i.e., recycling, and show kinetics typical of enzymes. Ribozymes are so far known to catalyze reactions like cleavage of RNA, cleavage of DNA, etc. Ribozymes are able to cut and splice themselves into a form that can catalyze the cleavage of other RNA/DNA molecules. The catalytic ability of ribozymes arises due to their three dimensional structures, which is able to generate in them the substrate-specific binding site. This is similar to enzymes in which the substrate specific binding site is produced by their three-dimensional structure. It may be emphasized that all biological catalysts must possess the ability to bind to their specific substrates.

Ribozymes offer exciting possibilities of varied applications, e.g., in human, animal and plant disease control. It has been suggested that RNA was the first macromolecule on earth, which catalyzed its own replication, and it subsequently developed other catalytic capabilities. Much later, RNA acquired the ability to catalyze protein synthesis. In due course of time, proteins became the primary biological catalysts since they were capable of catalyzing a much larger variety of reactions more efficiently than ribozymes. The first DNA molecules are thought to have originated by reverse transcription of RNA molecules. Ultimately, DNA became the primary genetic material in view of its greater stability than RNA. Thus, according to this hypothesis, RNA has been dethroned from its princely status by the same two macromolecules, i.e., proteins and DNA, the synthesis of which it either catalyzed (proteins) or provided the primary information for (DNA by reverse transcription); RNA is now generally relegated to a mere secondary (although essential) role as the link between DNA and proteins.

Lysozyme

1. Lysozyme is an enzyme present in tears, nasal mucus, sputum, tissues, gastric secretions, milk and eggwhite.
2. It catalyzes the hydrolysis of β-1, 4 link-ages of N-acetylneuraminic acid in mucopolysaccharides or mucopeptides:
3. It destroys the cell walls of many airborne grampositive bacteria in tears and nasal mucus.
4. Its molecular weight is about 15,000.
5. It consists of a single polypeptide chain of 129 amino acid residues having no coenzymes or metal ion cofactors.
6. As the lysozyme moelcule may readily be unfolded and refolded, catalysis and specificity as well as 3-dimensional structure are determined solely by these residues. Recent research work states that the central feature of its mechanism is the formation of an ion pair intermediate, rather than a covalent species.
7. There are little α-helix and large regions without secondary structure.
8. It has a central catalytic site with 6 subsites which bind various substrates or inhibitors.
9. The residues responsible for bond cleavage lie between sites D and E, close to the carboxyl groups of ASP. 52 and Glu. 35.

Ribonuclease (RNase)

Ribonuclease a small globular protein, is an enzyme secreted by the pancreas into the small intestine, where it catalyzes the hydrolysis of certain bonds in ribonucleic acids present in ingested food.

Immobilization of Enzymes

Since most enzymes are globular protein, they are soluble in water. Therefore, it is very difficult or impractical to separate the enzyme for reuse in a batch process. Enzymes can be immobilized on the surface of or inside of an insoluble matrix either by chemical or physical methods. They can be also immobilized in their soluble forms by retaining them with a semipermeable membrane.

The materials used for immobilization of enzymes, called carrier matrices, are usually inert polymers of inorganic materials.

The ideal carrier matrix has the following properties :

(*i*) low cost (*ii*) inertness (*iii*) physical strength (*iv*) stability (*v*) regenerability after the useful lifetime of the immobilized enzyme (*vi*) enhancement of enzyme specificity (*vii*) reduction in product inhibition (*viii*) a shift in the pH optimum for enzyme action to the desired value for the process (*ix*) reduction in microbial contamination and nonspecific adsorption

Clearly, most matrices possess only some of the above features. Therefore, carrier matrix for the immobilization of an enzyme must be chosen with care keeping in view the properties and limitations of various matrices.

Immobilization Techniques

Immobilization techniques can be classified by basically two methods :

(*i*) Chemical method

(*ii*) Physical method

The former is covalent bond formation dependent and the latter is noncovalent bond formation dependent.

(*i*) Chemical Method

Covalent Attachment : The covalent attachment of enzyme molecules via nonessential amino acid residues (that is, amino acids minus water) to water insoluble, functionalized supports are the most widely used method for immobilizing enzymes, which can be illustrated as

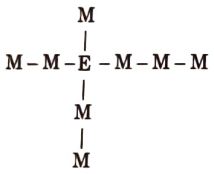
$$|\text{–X} + \text{E} \rightarrow |\text{–E} + \text{X}$$

where E is enzyme.

Another variation of immobilization by covalent attachment is the *copolymerization* of the enzyme with a reactive monomer (M) such as

$$n\text{M} + \text{E} \rightarrow \text{M}_n\text{E}$$

where, M_n E may have the following structure.



Commonly employed water-insoluble supports for the covalent attachment of enzymes include :

(*i*) *Synthetic supports* such as acrylamide-based polymers, maleic anhydride-based polymers, methacrylic acid-based polymers, styrene-based polymers, and polypeptides.

(*ii*) *Natural supports* such as agarose (Sepharose), cellulose, dextran (Sephadex), glass and starch.

Already active polymers such as maleic anhydride copolymers will be simply mixed with enzymes to produce immobilized enzymes. Normally, natural or synthetic polymers need to be activated by treating them with reagents before adding the enzyme. The activation involves the chemical conversion of a functional group of the polymer. The enzyme's active site should not be involved in the attachment, in which case the enzyme would lose its activity upon immobilization.

(*ii*) Physical Method

Adsorption : This method is the simplest way to immobilize enzymes. Enzymes can be adsorbed physically on a surface-active adsorbent by contacting an aqueous solution of enzyme with an adsorbent.

Commonly employed adsorbents are alumina, aninon-exchange resins, calcium carbonate, carbon, action-exchange resins, celluloses, clays, collagen colloid-ion, conditioned metal, glass plates, diatomaceous earth and hydroxyapatite.

Entrapment : Enzymes can be entrapped within cross-linked polymers by forming a highly cross-linked network of polymer in the presence of an enzyme. This method has a mojor advantage in the fact that there is no chemical modification of the enzyme, therefore, the intrinsic properties of an enzyme are not altered. Howeveer, the enzyme may be deactivated during the gel formation. Enzyme leakage is also a problem. The most commonly employed cross-linked polymer is the polyacrylamide gel system. This has been used to immobilize alcohol dehydrogenase, glucose oxidase, amino acid oxidase, hexokinase, glucose isomerase, urease, and many other enzymes.

Microencapsulation : Enzymes can be immobilized within semipermeable membrane microcapsules. This can be done by the interfacial polymerization technique. Organic solvent containing one component of copolymer with surfactant is agitated in a vessel and aqueous enzyme solution is introduced. The polymer membrane is formed at the liquid-liquid interface.

Applications of Enzymes

1. **Wine Manufacturing :** Pasteur was perhaps best known to the French nation as the "saviour of the wine industry" because his pasteurization process salvaged an ailing industry be set with problems of microbial contamination. *Papain* is used in brewing industry as a stabilizer for chill-proof beer, because it removes small amounts of protein that cause turbidity in chilled beer.
2. **Cheese Making :** Since long the animal *rennin* (or rennet) is employed in making cheese. The enzyme rennet is obtained on a comercial scale from the fourth or true stomach of the unweaned calves which are specifically slaughtered for this purpose. One calf produces only 5 to 10 gm of rennet. The enzyme helps in coagulating the casein of milk. Certain preservatives (boric acid, benzoic acid or sodium chloride) are, sometimes, added to prevent decomposition of the enzyme preparations by bacteria. An enzyme lipase is added to cheese for imparting flavour to it. Many vegetarians are unware that the cheese made in india contains animal rennet. However, an international charitable trust concerned with the welfare of animals, the Beauty Without Cruelty (BWC) has, with the help of Aurey Dairy, Mumbai, undertaken successful experimental trials in cheese making using non-animal rennet.
3. **Candy Making :** An enzyme, *invertase* helps preventing granulation of sugars in soft-centred candies. Another enzyme, lactase prevents formation of lactose crystals in ice cream which would otherwise not allow the product seem sandy in texture.
4. **Bread Whitening :** *Lipoxygenase* is used for whitening the bread.
5. **Clarifying Fruit-juices :** The enzymes are being used in processing of fruit juices such as apple and grapejuice juice. The juices are clarified by adding a mixture of *pecticenzymes* which hydrolyze the pectic substances causing turbidity.
6. **Tenderizing Meat :** Because hydroxyprolyl residues create bends in collagen helices, which contribute to the tough and rubbery texture often associated with cooked meat, treating the meat with a protease (*bromelain or papain*) prior to its cooking hydrolyzes peptide bonds, and thus tenderizes it.
7. **Desizing Fabrics :** The woven fabrics are sized by applying starch to the warp (lengthwise) threads to strengthen the yarn before weaving. But when these fabrics are printed or dyed, the sizing should be removed. Desizing may be done by acids, alkalies or enzymes. Enzymatic desizing is, however, preferred as it does not weaken the fabrics. Enzymes for this purpose are obtained from a variety of sources including bacteria, fungi and malt.
8. **Destaining Fabrics :** In drycleaning, the stains due to glue, gelatin or starch are removed by employing certain enzymes, such as *alcalase*.
9. **Dehairing Hide :** In the manufacture of leather, the hide is made free from hair. This is done by employing *pancreatic enzymes* which hydrolyze the proteins of the hair follicles, thus freeing the hair so that it may be easily scraped off from the hide.
10. **Recovering Silver :** Pepsin in used to digest gelatin in the process of recovering silver from photographic films.
11. **Correcting Digestion :** When the enzymes are present insufficiently in the body, certain digestive disorders come up. These may be cured by supplying the lacking enzymes. Pepsin, papain and amylases aid digestion in the stomach while pancreatic enzymes act in the duodenum.
12. **Wound Healing :** *Proteolytic enzymes* from pig pancreas are used to alleviate skin diseases, bed sores and sloughing wounds. These enzymes act by destroying porteolytic enzymes of man, that prevent the healing of such wounds. The enzymes commonly used for wound debridement are the proteases such as *streptodornase, ficin* and *trypsin*.
13. **Analyzing Biochemicals :** Certain enzymes are used in clinical analysis. For example, *uricase* and *urease* are employed in the deteremination of uric acid and urea respectively in blood. Besides, sucrose and raffinose contents in sugar mixtures are determined by polarimetry before and after treating the solutions with the enzymes, *sucrase* and *melibiase*.
14. **Dissolving Blood Clot :** The enzyme urokinase, which is manufactured from urine, is being used effectively in Japan in the treatment of blood clot in brain, artery and other circulatory diseases. A team of Soviet scientists led by Yevgeni Chazov, Director of the National Cardiological Research Centre, Moscow, have, in

1982, developed an effective enzyme *streptodekase*, which can dissolve blood clots in vessels. The new enzyme is particularly useful in preventing heart attackes as clots are responsible for 9 out of 10 fatal cases or cardiac arrests.

15. **Diagnosing Hypertension :** A new method called radio immunoassay procedure for diagnosing cases of hypertension had been developed by Bhabha Atomic Research Centre (BARC). In it, the activity of renin, a proteolytic enzyme secreted by the kidneys, is calculated indirectly by measuring angiotensin-I which is formed by the action of renin. Renin acts as part of a complex feedback meachanism for regulating blood volume and pressure.

16. **Augmenting Surgery :** A technique using the enzyme trypsin as an adjunct to cataract* surgery. The hollow needle is used to inject a microscopic amount of trypsin a digestive enzyme secreted by the pancreas. Trypsin digests and liquifies the semisolid interior of the lens without harming other parts of the eye. Once the enzyme liquifies the lens-which takes from a few hours to overnight-the lens is removed by suction through the same hollow needle.

17. **Syrup Manufacturing :** These days, immobilized glucose isomerase is being successfully used in the production of high-fructose corn syrup, in the United States.

A list of some important applications of immobilized enzymes

Enzyme	Substrate	Enzyme Reactor used*	Product
Aminoacylas (immobilized on anion exchange resins)	N-acyl-DL-amino acids	PBR	L-amino acids
Aspartate ammonialyase	Fumaric acid + NH_4^+	PBR	L-aspartic acid
Cyanidase	Cyanide present in industrial waste, food or feed	–	Formic acid
Glucoamylase	Dextrins produced by α-amylase		D-glucose
Glucose isomerase (immobilized with glutaraldehyde by cross-linking)	D-glucose in glucose syrup	PBR	High fructose corn syrup
Invertase	Sucrose	PBR	Invert sugar
Lactase (immobilized in cellulose triacetate fibers)	Milk and whey	STR	Lactose-free milk and whey
Lipase	Vegetable oils, e.g., palm oil		Coca butter substitute
Nitrile hydratase (immobilized cells)	Acrylonitrile		Acrylamide
Penicillin amylases	Penicillin G and penicillin V	STR, PBR	Penicillins
Raffinase (immobilized cells)	Raffinose in beet juice, soyabean milk	STR	Raffinose-free solutions
Thermolysin	L-aspartic acid and L-phenylalanine methyl ester	–	Aspartame

*PBR, packed bed reactor; STR, stirred tank reactor.

A list of some enzymes used for commercial activities

Enzyme	Source	Application	Use in the form
α-Amylase	1. Bacillius *licheniformis* 2. *B. amylolique faciens*	hydrolysis of starch to *dextrans*	Extracellular, soluble
Glucoamylase	*Aspergillus niger*	Dextran hydrolysis to glucose	Extracellular, soluble
Xylose (Glucose) isomerase	*Bacillus coagulans**	Pure glucose to equilibrium brium mixture of glucose + fructose	Immobilized whole cells
Alkaline protease	*B. Lichen:formis, B. Subtilis*	Detergents (protein Digestion)	Extracellular, soluble
Neutral protease	*B amyloliquefaciens, B. subtilis*	Protein digestion in brewing substances	Extracellur, soluble
Acid protease, e.g., rennet	*A. niger***	Milk coagulation and cheese flavour enhancement	Extracellular, soluble
Pectinase	*A. niger, B. subtilis*	Pectin hydrolysis in fruit juices	Extracellular, soluble
Glucanase	*A. niger, B. subtilis*	In fruit juices	Extracellular, soluble
Cellulase	*A. niger, Tricho derma reesei*	Cellulose hydrolysis, in Fruit juices	Extracellular, soluble
Lipase	*Rhizopus species**	Detergents, lipid hydrolysis	Extracellular, soluble
Lactase	*A. niger*	Milk loctose hydrolysis to glucose + galactose	Immobilized on glass beads

EXERCISE – I

MCQ TYPE QUESTIONS

1. Who coined the term zymase for enzymes in yeast
   (a) Kuhne (b) Summer
   (c) Louis Pasteur (d) Edward Buchner
2. Enzymes are basically or all enzymes contain
   (a) Sugars (b) Proteins
   (c) Fats (d) Vitamins
3. Who got the Nobel prize working on enzymes in the year 1978
   (a) W. Arber and D. Nathans
   (b) Nass and Nass
   (c) R. Mishra
   (d) H.G. Khorana
4. Enzymes were discovered for the first time in
   (a) Yeast (b) Maize
   (c) Cteria (d) Algae
5. In plants, enzymes are present
   (a) Only in flowers
   (b) only in leaves
   (c) In all the living cells of the plant body
   (d) only in parenchyma
6. Enzymes are the polymers of
   (a) Hexose carbon (b) fatty acids
   (c) Amino acids (d) organic phosphate
7. Enzymes are absent in
   (a) Algae (b) Fungi
   (c) Bacteria (d) Virus
8. Which of the following is not an attribute of enzymes
   (a) They are proteinaceous in nature
   (b) They speed up the rate of biochemical reaction
   (c) They are specific in nature
   (d) They are used up in reactions
9. Which of the following is not a part of enzyme but it activates the enzyme
   (a) K (b) Zn
   (c) Mg (d) Mn
10. Which of the following enzymes is not used in making detergents?
    (a) Amylase (b) Cellulose
    (c) Protease (d) Peptidase
11. An enzyme can be synthesized by chemically bonding together molecules of
    (a) Carbohydrates (b) Amino acids
    (c) Lipases (d) CO
12. Which one is cofactor of carbonic anhydrase?
    (a) Fe (b) Zn
    (c) Cu (d) Mg

13. Non- protein part of an enzyme is known as
(a) Holoenzyme (b) Apo enzyme
(c) Coenzyme (d) All the above

14. The enzyme which combines with non-protein part to form a functional enzyme is
(a) Co-enzyme (b) Holoenzyme
(c) Apo enzyme (d) Prosthetic group

15. The permeases are
(a) Structural membrane proteins
(b) Enzymatic membrane proteins
(c) Carrier membrane proteins
(d) None of these

16. FAD or FMN is a coenzyme. Which vitamin is incorporated into structure?
(a) Vitamin C (b) Vitamin B
(c) Vitamin B6 (d) vitamin B2

17. Which one of the following metallic ion does not occur as prosthetic group in any enzyme?
(a) Mg (b) Cu
(c) Zn (d) Ag

18. Decline in the activity of the enzyme hexokinase by glucose 6-phosphate is caused by
(a) Non-competitive
(b) Competitive
(c) Allosteric modular
(d) Denaturation of enzymes.

19. Which one of the following enzyme contains Mo metallic ion as the prosthetic group
(a) Phosphatase (b) Dehydrogenase
(c) Peptidase (d) Nitrate reductase

20. An enzyme acts by
(a) Reducing the energy of activation
(b) Increasing the energy of activation
(c) Decreasing the pH
(d) Increasing the pH

21. Km is related to
(a) Morphology (b) ABO blood group
(c) ES complex (d) Chromatography

22. At the time of cotton seeds germination, the stored food is digested by
(a) Diastase (b) Maltase
(c) Lipase (d) Amylase

23. The enzyme which converts glucose into ethyl alcohol (C_2H_5OH) is
(a) Diastase (b) Maltase
(c) Zymase (d) Invertase

24. As temperature changes from 3°C to 45°C, the rate of enzyme activity will
(a) Decrease and then increase
(b) Increase and then decrease
(c) Increase only
(d) Decrease only

25. Enzymes are sensitive to Rainfall
(a) light (b) pH
(c) wind velocity (d) None of these

26. Zymogens are
(a) Enzyme acting upon starch
(b) Group of zymase enzymes
(c) Inactive enzyme precursors
(d) None of the above

27. Enzymes which are slightly different in molecular structure but can perform identical activity are called
(a) Isoenzymes (b) Holoenzymes
(c) Apo enzymes (d) Coenzymes

28. Lactic dehydrogenase (LDH) which catalyzes pyruvate to lactate is an example of
(a) Apo enzyme (b) Antienzyme
(c) Isoenzyme (d) Coenzyme

29. Which of the following enzyme was first isolated and purified in the form of crystals
(a) Amylase (b) Urease
(c) Ribonuclease (d) Pepsin

30. Enzymes generally have
(a) Same pH and temperature optima
(b) Same pH but different temperature optima
(c) Different pH but same temperature optima
(d) Different pH and different temperature optima

31. Most of the biochemical reactions differ from those occurring in the non-living world in
(a) Requiring energy (b) Releasing energy
(c) Being enzymatic (d) Being spontaneous

32. How many genes determine synthesis of one enzyme?
(a) One (b) Two
(c) Four (d) Sixteen

33. The most important property of an enzyme is its
(a) Composition (b) Thermal denaturation
(c) Specificity (d) Solubility

34. Repressible enzymes are formed
(a) In the absence of co repressor
(b) In the presence of co repressor
(c) In the presence of apressor
(d) All of the above

ASSERTION AND REASON

Read the assertion and reason carefully to mark the correct option out of the options given below:
(a) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(b) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(c) If the assertion is true but the reason is false.
(d) If both the assertion and reason are false.
(e) If the assertion is false but reason is true.

35. **Assertion:** Enzymes have active sites and substrates reactive sites, on their surfaces respectively.
Reason: Active and reactive sites push the enzyme and substrate molecules away from each other:

36. **Assertion:** Enzyme substrate complex remains throughout the reaction.
Reason: The greater the affinity of the enzyme for a substrate, the higher is the catalytic activity.

37. **Assertion:** Desmolysing enzymes are those which catalyse the reactions by hydrolysis.
Reason: Digestive enzymes are hydrolysing in nature.

38. **Assertion:** Coenzymes are also called prosthetic groups.
Reason: Coenzymes and prosthetic groups are cofactors.

39. **Assertion:** Enzymes are defined as biological proteins.
Reason: Chemically all enzymes are globular proteins.

40. **Assertion:** DNA molecules and RNA molecules are found in the nucleus of cell.
Reason: On heating, enzymes do not lose their specific activity.

41. **Assertion:** The higher the turn-over number the more efficient an enzyme is.
Reason: It is not dependent upon the number of active sites present over an enzyme.

42. **Assertion:** Allosteric enzymes show feedback inhibition.
Reason: The inhibitor is competitive

43. **Assertion:** Enzyme becomes inactive below minimum temperature.
Reason: The inactivity of the enzymes is due to denaturation.

44. **Assertion:** Enzymes lower the activation energy.
Reason: A substrate molecule can be acted upon by a particular enzyme.

45. According to Holum, the cofactor may be
(a) Co enzyme (b) Prosthetic group
(c) Metal ion activator (d) All of these

46. The type of specificity where enzyme will catalyze only one reaction
(a) Absolute specificity
(b) Group specificity
(c) Stereochemical specificity
(d) Linkage specificity

47. A RING domain is an example of
(a) DNA ligase
(b) Ubiquitin ligase
(c) Reverse transcriptase
(d) Alkaline phosphatase

48. Transaminase is involved in
(a) Transfer of radical (b) Transfer of electron
(c) Hydrolysis (d) Splits C-C bond

49. Cholesterol side-chain cleavage enzyme, also called
(a) CYP17A1
(b) 17, 20 desmolase
(c) 17 alpha - hydroxylase
(d) 20, 22-Desmolase

50. Zero order reaction rate is independent of
(a) Substrate (b) Product
(c) Coenzyme (d) Cofactors

51. The order of reaction in which the rate is proportional to the first power of each of two reactants
(a) Zero order (b) First order
(c) Second order (d) Third order

52. Proofreading mechanisms are found in
(a) RNA polymerase
(b) Aminoacyl tRNA synthetases
(c) Ribosomes
(d) All of these

53. Heme is an example of
(a) Organic co factor (b) Inorganic co factor
(c) Holoenzyme (d) Co enzyme

54. According to Buchner yeast can ferment sugar to
(a) H_2O and CO_2
(b) Alcohol and CO_2
(c) Carboxylic acid and H_2O
(d) Acetic acid

55. The qualitative test for tyrosine is
(a) Milons test (b) Biuret test
(c) Nitroprusside test (d) Hopkins - cole test

56. Reverse transcriptase is also called as
(a) RNA directed DNA polymerase
(b) RNA dependent RNA polymerase
(c) DNA dependent DNA polymerase
(d) RNA dependent DNA methylation

57. One gene and one enzyme hypothesis was conducted on
(a) Agrobacterium
(b) Halobacterium
(c) Neurospora
(d) Mycoplasma

58. Alpha amylase is also called as
(a) Ptyallin
(b) Amylopectin
(c) 1, 4-a-beta-glucan maltohydrolase
(d) Saccharogen amylase

59. Steady state implies
(a) The rate of formation of the Es complex is exactly double by its rate of decay to free enzyme and products
(b) The rate of formation of the ES complex is exactly matched by its rate of decay to free enzyme and products
(c) The rate of decay of free enzyme and products is exactly double by its rate of formation of the ES complex
(d) The rate of formation of the ES complex is independent by its rate of decay to free enzyme and products

60. The catalytic efficiency of an enzyme is
(a) $k_{cat} * k_m$ (b) k_{cat}/k_m
(c) $k_{cat}[S]/k_m$ (d) $k_{cat}/k_m[S]$

61. For an enzyme in solution, the rate-determining step in catalysis will be
(a) k1
(b) Rate of ES formation
(c) Multiple steps contributing to kcat
(d) All of these

62. Number of molecules of substrate converted to product by one enzyme site per second is
(a) Turn over number (b) Reynold's number
(c) Enzyme efficiency (d) Catalytic efficiency

63. Y = (Vmax * X)/(Km + X)
In the above reaction X represents?
(a) Product (b) Substrate
(c) Enzyme (d) Enzyme velocity

64. Vmax is measured as
(a) Moles of enzyme produced per minute
(b) Moles of substrate produced per minute
(c) Moles of product produced per minute
(d) Moles of enzyme substrate complex produced per minute

65. The value of slope formed in lineweaver - Burk plot taking 1/V on y axis and 1/S on x-axis?
(a) k_m/V_{max} (b) $1/V_{max}$
(c) $1/k_m$ (d) $-1/k_m$

66. In Lineweaver-Burk plot the most accurate values of KM and Vmax are obtained by
(a) Linear regression (b) Nonlinear regression
(c) Transformation (d) All of these

67. The term which is used for binding of substrate to one binding site increases the activity of neighboring sites is
(a) No cooperativity
(b) Positive cooperativity
(c) Multiple cooperativity
(d) Negative cooperativity

68. The unit used to measure k_{cat}
(a) sec-1 (b) sec
(c) moles/sec2 (d) moles/sec

69. The overall free energy change for a reaction is related to the
(a) Rate constant (b) Equilibrium constant
(c) Transition state (d) Dynamic equilibrium

70. Small k_m value indicates
(a) Weak binding
(b) Tight binding
(c) Depletion of substrate
(d) Depletion of enzyme

71. Reversible inhibitors interact with an enzyme via
(a) Non covalent association
(b) Covalent association
(c) VanderWaal's association
(d) Ionic association

72. Turn over number is formulated as
(a) kcat = vmax/[E]t (b) kcat = vmax/[S]
(c) kcat = vmax/[ES] (d) kcat = vmax/[P]

73. Chymotrypsin suitable pH is at
(a) Stomach (b) Small intestine
(c) Liver (d) Pancrease

74. RNA catalysts are composed of
(a) Nucleotides (b) Aminoacids
(c) Both (a) & (b) (d) Kinases

75. An inhibitor binding at an allosteric site can be
(a) Noncompetitive (b) Uncompetitive
(c) Competitive (d) Feedback

76. Inhibitor that chemically react at a substrate binding site irreversibly inactivate the enzyme and are also very effective inhibitors are
(a) Active-site-directed
(b) Suicide inactivators
(c) Both (a) & (b)
(d) Competitive inhibitors

77. In biochemistry, eicosanoids are signaling molecules made by
(a) Oxidation of 20 carbon essential fatty acids
(b) Reduction of 20 carbon essential fatty acids
(c) Oxidation of 18 carbon essential fatty acids
(d) Oxidation of 20 carbon non essential fatty acids

78. Eicosanoids derive from
(a) Omega-3 essential fatty acids
(b) Omega-6 essential fatty acids
(c) Both (a) & (b)
(d) Omega-3 nonessential fatty acids

79. Covalently binding inhibitors are grouped into
(a) Mechanism-based inhibitors
(b) Affinity labels
(c) Pseudoirreversible
(d) All of these

80. Inhibition of the enzyme catechol-O-methyltransferase (COMT) is an important approach in the treatment of
(a) Albinism (b) Parkinson's disease
(c) Leukemia (d) Down's syndrome

81. Predict the siope for a graph drawn Ki on y-axis with various concentration of inhibitor on x-axis
(a) Km/Ki (b) Ki/Km
(c) –Ki/Km (d) –Km/Ki

82. An inhibitor that binds covalently to irreversibly inactivate the enzyme is called
(a) Irreversible inhibitor
(b) Inactivator
(c) Both (a) & (b)
(d) IC50

83. If an inhibitor binds reversibly to the same site as the substrate, the inhibition will be
(a) Competitive
(b) Non competitive
(c) Uncompetitive
(d) Mixed

84. Reversible inhibitors bind to enzymes with
(a) Hydrogen bonds
(b) Hydrophobic interactions
(c) Ionic bonds
(d) All of these

85. In uncompetitive inhibition
(a) Vmax and Km decreases
(b) Vmax decreases and Km increases
(c) Vmax and Km increases
(d) Vmax increase and Km decreases

86. The type of inhibition where the initial enzyme-inhibitor complex EI undergoes isomerisation to a second more tightly held complex is
(a) Substrate inhibition
(b) Product inhibition
(c) Slow tight inhibition
(d) Competitive inhibition

87. Some protein kinase inhibitors have chemical structures that are similar to
(a) ATP (b) NAD^+
(c) Polymerase (d) Ligase

NUMERICAL TYPE QUESTIONS

1. An average living cell contains about
2. The ratio of the enzyme to substrate molecule can be as high as

EXERCISE – II

(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2016

1. Which one of the following statements is **NOT** true?
(a) In competitive inhibition, substrate and inhibitor compete for the same active site of an enzyme
(b) Addition of a large amount of substrate to an enzyme cannot overcome uncompetitive inhibition
(c) A transition state analogue in enzyme catalyzed reaction increases the rate of product formation
(d) In non-competitive inhibition, K_m of an enzyme for its substrate remains constant as the concentration of the inhibitor increases.

2. Choose the **CORRECT** combination of True (T) and False (F) statements about microcarriers used in animal cell culture.
P. Higher cell densities can be achieved using microcarriers
Q. Microcarriers increase the surface area for cell growth
R. Microcarriers are used for both anchorage- and nonanchorage-dependent cells
S. Absence of surface charge on microcarriers enhances attachment of cells
(a) P-T, Q-F, R-T and S-F
(b) P-T, Q-T, R-F and S-F
(c) P-F, Q-F, R-T and S-T
(d) P-F, Q-T, R-F and S-T

3. Analysis of a hexapeptide using enzymatic cleavage reveals the following result:
   - Amino acid composition of the peptide is: 2R, A,V, S, Y
   - Trypsin digestion yields two fragments and the compositions are: (R, A, V) and (R, S, Y)
   - Chymotrypsin digestion yields two fragments and the compositions are: (A, R, V, Y) and (R, S)
   - Digestion with carboxypeptidase A yields no cleavage product.

   Given:

   Trypsin cleaves at carboxyl side of R.

   Chymotrypsin cleaves at carboxyl side of Y.

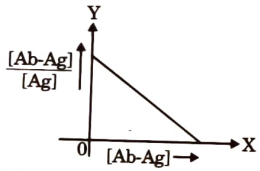
   Carboxypeptidase A cleaves at amino side of the C-terminal amino acid (except R and K) of the peptide.

   The correct amino acid sequence of the peptide is:

   (a) RSYRVA
   (b) AVRYSR
   (c) SRYVAR
   (d) SVRRYA

2015

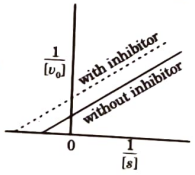
4. The interaction between an antigen (Ag) and a single-chain antibody (Ab) was studied using Scatchard analysis. The result is shown below.

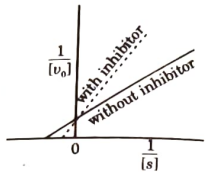


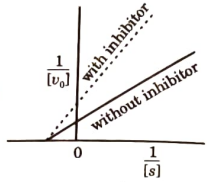
The affinity of interaction and the total concentration of antibody, respectively, can be determined from

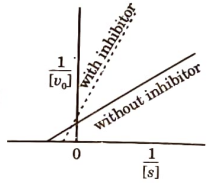
(a) slope and Y-intercept
(b) Y-intercept and slope
(c) X-intercept and slope
(d) slope and X-intercept

5. Which one of the following graphs represents uncompetitive inhibition ?

(a)


(b)


(c)


(d)


2014

6. Which of the following is employed for the repeated use of enzymes in bioprocesses?

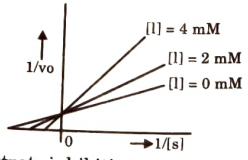
   (a) polymerization (b) immobilization
   (c) ligation (d) isomerization

2013

7. The catalytic efficiency for an enzyme is definedas

   (a) k_{cat} (b) $\frac{v_{max}}{k_{cat}}$
   (c) $\frac{k_{cat}}{k_m}$ (d) $\frac{k_{cat}}{v_{max}}$

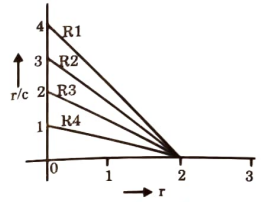
8. The activity of an enzyme was measured by varying the concentration of the substrate (S) in the presence of three different concentrations of inhibitor (I) 0, 2 and 4 mM.The double reciprocal plot given belowsuggests that the inhibitor (I) exhibits



(a) substrate inhibition
(b) uncompetitive inhibition
(c) mixed inhibition
(d) competitive inhibition

Common Data for Questions 9 and 10:

The binding efficiency of three different receptorsR1, R2 and R3 were tested against a ligand using equilibrium dialysis, with a constant concentration of receptor and varying concentrations of ligand. The Scatchard plot of receptor titration with different concentration of ligand is given below (r is moles of bound ligand per moles of receptor and c is concentration of free ligand)



9. The number of ligand binding sites present on receptors R1 and R3, respectively are

   (a) 1 and 4 (b) 1 and 1
   (c) 4 and 1 (d) 2 and 2

10. Which one of the receptors has thehighestaffinity for the ligand?

    (a) R1 (b) R2
    (c) R3 (d) R4

2012

11. The activity of an enzyme is expressed in International Units (IU). However, the S.I. unit for enzyme activity is Katal. One Katal is

    (a) 1.66×10^4 IU (b) 60 IU
    (c) 6×10^7 IU (d) 10^6 IU

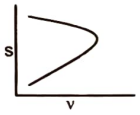
12. Identify the statement that is NOT applicable to an enzyme catalyzed reaction

    (a) Enzyme catalysis involves propinquity effects
    (b) The binding of substrate to the active site causes a strain in the substrate
    (c) Enzymes do not accelerate the rate of reverse reaction
    (d) Enzyme catalysis involves acid-base chemistry

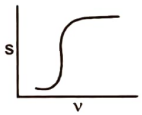
13. An enzymatic reaction is described by the following rate expression.

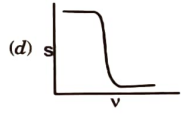
$$v = \frac{v_m s}{k_m + s + \frac{s^2}{k_s}}$$

Which one of the following curves represents this expression?

(a)


(b)

(c)


(d)


Directions (Q. 14 – 15) : *The purification data for an enzyme is given below*

	Step	Volume (ml)	Total protein (mg)	Total activity (Units)	Specific activity (Units/mg)
P	Cell-free extract	17	177	102	0.58
Q	Q-Sepharose	14	18.8	72	3.83
R	Phenyl Sepharose	26	9.2	45	4.89
S	Sephacryl S-200	7	4.1	30	7.32

14. The fold purification for each step is

    (a) P – 0.1, Q – 0.66, R – 0.84, S – 1.26 (b) P – 1.0, Q – 0.52, R – 0.67, S – 0.8
    (c) P – 1, Q – 6.6, R – 8.4, S – 12.6 (d) P – 100, Q – 66, R – 84, S – 12

15. The yield (%) for each step is

    (a) P – 10, Q – 7.2, R – 4.5, S – 2.0 (b) P – 34, Q – 24, R – 15, S – 1
    (c) P – 3.4, Q – 2.4, R – 1.5, S – 0.1 (d) P – 100, Q – 71, R – 44, S – 29

Directions (Q. 16 – 17) : *An E. coli cell of volume 10^{-12} cm³ contains 60 molecules of lac repressor. The repressor has a binding affinity (K_d) of 10^{-8} M and 10^{-9} M with and without lactose respectively, in the medium.*

16. The molar concentration of the repressor in the cell is

(a) 1.1 nM

(b) 1 nM

(c) 10 nM

(d) 100 nM

17. Therefore the lac-operon is

(a) repressed and can only be induced with lactose.

(b) repressed and cannot be induced with lactose.

(c) not repressed.

(d) expressed only when glucose and lactose are present.

2010

18. Identify the enzyme that catalyzes the following reaction

$$\alpha\text{-Ketoglutarate} + NADH + NH_4^+ + H^+ \xrightarrow{?} \text{Glutamate} + NAD^+ + H_2O$$

(a) Glutamate synthetase

(b) Glutamate oxoglutarate aminotransferase

(c) Glutamate dehydrogenase

(d) α-ketoglutarate deaminase

19. The degree of inhibition for an enzyme catalyzed reaction at a particular inhibitor concentration is independent of initial substrate concentration. The inhibition follows

(a) competitive inhibition

(b) mixed inhibition

(c) un-competitive inhibition

(d) non-competitive inhibition

20. An immobilized enzyme being used in a continuous plug flow reactor exhibits an effectiveness factor (η) of 1.2. The value of η being greater than 1.0 could be apparently due to

(a) substrate inhibited with internal pore diffusion limitation

(b) external pore diffusion limitation

(c) sigmoidal kinetics

(d) unstability of the enzyme

21. The turnover numbers for the enzymes, E1 and E2 are 150 s^{-1} and 15 s^{-1} respectively. This means

(a) E1 binds to its substrate with higher affinity than E2

(b) The velocity of reactions catalyzed by E1 and E2 at their respective saturating substrate concentrations could be equal, if concentration of E2 used is 10 times that of E1

(c) The velocity of E1 catalyzed reaction is always greater than that of E2

(d) The velocity of E1 catalyzed reaction at a particular enzyme concentration and saturating substrate concentration is lower than that of E2 catalyzed reaction under the same conditions

22. The dissociation constant K_d for ligand binding to the receptor is 10^{-7} M. The concentration of ligand required for occupying 10% of receptors is

(a) 10^{-6} M (b) 10^{-7} M

(c) 10^{-8} M (d) 10^{-9} M

23. Receptor R is over expressed in CHO cells and analysed for expression. 6×10^7 cells were incubated with its radioactive ligand (specific activity 100 counts per picomole). If the total counts present in cell pellet was 1000 cpm, the average number of receptors R per cell is (assume complete saturation of receptors with ligand and one ligand binds to one receptor)

(a) 10^4 (b) 10^5

(c) 10^6 (d) 10^7

2009

24. One unit of glucoamylase enzyme activity is defined as the amount of enzyme required to produce (imol of glucose per min in a 4% solution of Lintner starch at pH 4.5 and 60°C. If in a reaction mixture with 1ml of the crude enzyme preparation containing 8 mg protein and 9 ml of 4.44% starch, 0.6 μmc of glucose/ml-min is produced, what will be the specific activity of the crude enzyme preparation?

(a) 1 unit/mg protein

(b) 1.5 units/mg protein

(c) 0.25 units/mg protein

(d) 0.75 units/mg protein

2008

25. Which of the following techniques is best suited for immobilizing an affinity ligand?

(a) Physical adsorption

(b) Gel entrapment

(c) Cross-linking with a polymer

(d) Covalent linkage to a spacer arm

26. A DNA fragment of 4500 bp has to be tailed with dT residues by using dTTP and the enzyme 'terminal transferase'. The stock solution of dTTP that is used as a substrate has a concentration of 150 M. Ten l of this stock solution is added to a total volume of 200 l reaction. What will be the concentration of dTTP in the reaction?

(a) 7.5 M (b) 75 M

(c) 0.75 M (d) 0.075 M

27. Evaluate the Michaelis constant for the following lipase catalyzed trans-esterification reaction for the production of biodiesel

$$\text{Vegetable oil} + \text{Lipase} \underset{k_2}{\overset{k_1}{\rightleftharpoons}} \text{Oil-lipase complex} \xrightarrow{k_3} \text{Biodiesel} + \text{Glycerol}$$

where, $k_1 = 3 \times 10^8\ M^{-1} s^{-1}$; $k = 4 \times 10^3\ s^{-1}$ and $k_2 = 2 \times 10^3\ s^{-1}$

(a) 4.2×10^{-3} M (b) 14.0×10^{-4} M

(c) 6.4×10^{-6} M (d) 1.4×10^{-4} M

28. In a chemostat, evaluate the dilution rate at the cell wash-out condition by applying Monod's model with the given set of data: $_{max}$ = $1h^{-1}$; $Y_{x/s} = 0.5\ g\ g^{-1}$; $K_s = 0.2\ g\ L^{-1}$; $S_0 = 10\ g\ L^{-1}$

(a) 1.00 h^{-1} (b) 0.49 h^{-1}

(c) 0.98 h^{-1} (d) 1.02 h^{-1}

Common Data Questions

Common Data for Questions 29 and 30:

An enzyme (24000 Da) undergoes first-order deactivation kinetics while catalyzing a reaction according to Michaelis-Menten kinetics ($K_m = 10^{-4}$ M). The enzyme has a turnover number if 10^4 molecules substrate/min-(molecule enzyme) and a deactivation constant (k_d) of 0.1 min^{-1} at the reaction conditions. The reaction mixture initially contains 0.6 mg/l of active enzyme and 0.02 M of the substrate.

29. The time required to convert 10% of the substrate will be rapproximately

(a) 16 min (b) 24 min

(c) 32 min (d) 8 min

30. The maximum possible conversion for the enzymatic reaction will be

(a) 100% (b) 50%

(c) 25% (d) 12.5%

Statement for Linked Answer Questions 31 and 32:

A Nick Translation reaction in a final volume of 100 l was carried out by using 25 Ci of labeled [α-^{32}P]–dCTP for labeling a 1.2 Kb λ-Interferon DNA fragment.

31. After completion of Nick translation reaction, 10 l of reaction was spotted on a glass-fiber filter that upon counting resulted into 4.2×10^4 cpm in reaction. Another 10 l was processed for TCA precipitation to determine redioisotope incorporation. The TCA precipitated sample gave 2.94×10^4 cpm. What is the percent of [α-^{32}P]-dCTP incorporation into the DNA sample?

(a) 40% (b) 50%

(c) 60% (d) 70%

32. If 2.94×10^4 cpm of TCA precipitable counts of the 10 l sample were taken from 1/10 dilutioa of the 100 l Nick Translation reaction containing 1 g of γ-Interferon DNA, what is the specific activity of the labeled product?

(a) 1.47×10^6 cpm/ g

(b) 1.47×10^7 cpm/ g

(c) 2.94×10^6 cpm/ g

(d) 2.94×10^7 cpm/ g

Statement for Linked Answer Questions 33 and 34:

A double reciprocal plot was created from the specific growth rate and limiting-substrate concentration data obtained from a chemostat experiment. A linear regression gave values of 1.25 hr and 100 mg-hr-l^{-1} for the intercept and slope, respectively.

33. The respective values of the Monod kinetic constants $_m$ (hr^{-1}) and K_s (mg/l) are as follows:

(a) 0.08, 8 (b) 0.8, 0.8

(c) 0.8, 80 (d) 8, 8

34. The same culture (with the $_m$ and K_s values as computed above) is cultivated in a 10-litre chemostat being operated with a 50 ml/min sterile feed containing 50 g/l of substrate. Assuming an overall yield coefficient of 0.3 g-DCW/g-substrate, the respective values of the outlet biomass and substrate concentration are

(a) 15 g/l, 48 mg/l (b) 15 g/l, 0.48 g/l

(c) 48 g/l, 15 g/l (b) 4.8 g/l, 4.8 g/l

2007

35. Determine the correctness or otherwise of the following Assertion [a] and Reason [r]

Assertion : Enzymatic method of tissue dispersion is milder than chemical and mechanical methods.

Reason : Enzymes work at optimal temperature and pH

(a) Both [a] and [r] are true and [r] is the correct reason for [a]

(b) Both [a] and [r] are true but [r] is not the correct reason for [a]

(c) [a] is true but [r] is false

(d) [a] is false but [r] is true

Common Data for Questions 36 and 37 :

Enzyme X converts substrates S_1 and S_2 (which are similar but not identical) to products P_1 and P_2 respectively

36. K_m values of enzyme X for substrate S_1 and S_2 are 0.1 mM and 0.01 mM, respectively This suggest that

P. enzyme X has more affinity towards S_1

Q. enzyme X has low affinity towards S_1

R. enzyme X has more affinity towards S_2

S. enzyme X has low affinity towards S_2

(a) P, Q

(b) R, S

(c) Q, S

(d) Q, R

37. What would happen if enzyme X is incubated with a mixture of 0.1 mM of S_1 and S_2?

(a) Products P_1 and P_2 are produced at equal concentrations

(b) Only product P_2 is produced

(c) More P_2 and less P_1 are produced

(d) More P_1 and less P_2 are produced

2006

38. Immobilization of enzymes using entrapment method requires

(P) photosensitive polyethylene glycol dimethacrylate

(Q) CNBr activation of sepharose

(R) polyfunctional reagent like hexamethylene diisocyanate

(S) radiation of polyvinyl alcohol

(a) P, Q

(b) R, S

(c) P, S

(d) Q, S

39. Immobilization of enzymes

(P) increases the specificity of the enzyme for its reactants

(Q) facilitates reuse of the enzyme in batch reactions

(R) makes it unsuitable for its use in a continuous reactor system

(S) decreases the operational cost of the industrial process

(a) Q, S

(b) Q, R

(c) R, S

(d) P, Q

NUMERICAL TYPE QUESTIONS

2016

1. In an assay of the type II dehydroquinase of molecular mass 18 kDa, it is found that the V_{max} of the enzyme is 0.0134 $\mu mol.min^{-1}$ when 1.8 g enzyme is added to the assay mixture. If the K_m for the substrate is 25 μM, the k_{cat}/K_m ratio will be _____ $10^4\ M^{-1}.s^{-1}$.

2. The activity of lactate dehydrogenase can be measured by monitoring the following reaction:

Pyruvate + NADH → Lactate + NAD^+

The molar extinction coefficient of NADH at 340 nm is 6220 $M^{-1}.cm^{-1}$. NAD^+ does not absorb at this wavelength. In an assay, 25 µL of a sample of enzyme (containing 5 µg protein per mL) was added to a mixture of pyruvate and NADH to give a total volume of 3 mL in a cuvette of 1 cm pathlength. The rate of decrease in absorbance at 340 nm was 0.14 min^{-1}. The specific activity of the enzyme will be _____ $\mu mol.min^{-1}.mg^{-1}$.

3. An enzyme is immobilized on the surface of a non-porous spherical particle of 2 mm diameter. The immobilized enzyme is suspended in a solution having bulk substrate concentration of 10 mM. The enzyme follows first order kinetics with rate constant 10 s^{-1} and the external mass transfer coefficient is 1 $cm.s^{-1}$. Assume steady state condition wherein rate of enzyme reaction ($mmol.L^{-1}.s^{-1}$) at the surface is equal to mass transfer rate ($mmol.L^{-1}.s^{-1}$). The substrate concentration at the surface of the immobilized particle will be _____ mM.

2015

4. A single subunit enzyme converts 420 µmoles of substrate to product in one minute. The activity of the enzyme is _____ $\times 10^{-6}$ Katal.

5. The K_i of novel competitive inhibitor designed against an enzyme is 2.5 μM. The enzyme was assayed in the absence or presence of the inhibitor (5 μM) under identical conditions. The K_m in the presence of the inhibitor was found to be 30 μM. The K_m in the absence of the inhibitor is ___ μM.

2014

6. The degree of reduction of ethanol is _____

7. Triose phosphate isomerase converts dihydroxy acetone phosphate (DHAP) to glyceraldehyde-3-phosphate (G-3-P) in a reversible reaction. At 298 K and pH 7.0, the equilibrium mixture contains 40 mM DHAP and 4 mM G-3-P. Assume that the reaction started with 44 mM DHAP and no G-3-P. The standard free-energy change in kJ/mol for the formation of G-3-P [R = 8.315 J/mol.K] is _____.

ANSWERS

EXERCISE – I

MCQ Type Questions

1. (d)	2. (b)	3. (a)	4. (a)	5. (c)	6. (c)	7. (d)	8. (d)	9. (a)	10. (d)
11. (b)	12. (b)	13. (c)	14. (c)	15. (c)	16. (d)	17. (d)	18. (c)	19. (d)	20. (a)
21. (c)	22. (d)	23. (c)	24. (b)	25. (c)	26. (c)	27. (a)	28. (c)	29. (b)	30. (c)
31. (c)	32. (a)	33. (c)	34. (a)	35. (c)	36. (e)	37. (e)	38. (e)	39. (a)	40. (d)
41. (c)	42. (c)	43. (c)	44. (b)	45. (d)	46. (a)	47. (b)	48. (a)	49. (d)	50. (a)
51. (c)	52. (d)	53. (a)	54. (b)	55. (a)	56. (a)	57. (c)	58. (a)	59. (b)	60. (b)
61. (d)	62. (a)	63. (b)	64. (c)	65. (a)	66. (b)	67. (b)	68. (a)	69. (b)	70. (b)
71. (a)	72. (a)	73. (b)	74. (a)	75. (a)	76. (c)	77. (a)	78. (c)	79. (d)	80. (b)
81. (a)	82. (c)	83. (a)	84. (d)	85. (a)	86. (c)	87. (a)			

Numerical Type Questions

1. more than 2000 enzymes
2. 1 : 10,00,000

EXERCISE – II

MCQ Type questions

1. (c)	2. (b)	3. (b)	4. (d)	5. (a)	6. (b)	7. (c)	8. (d)	9. (d)	10. (a)
11. (c)	12. (c)	13. (a)	14. (c)	15. (d)	16. (d)	17. (b)	18. (c)	19. (d)	20. (*)
21. (b)	22. (c)	23. (b)	24. (*)	25. (d)	26. (a)	27. (d)	28. (c)	29. (*)	30. (*)
31. (d)	32. (d)	33. (c)	34. (a)	35. (b)	36. (d)	37. (c)	38. (c)	39. (a)	

Numerical Type Questions

1. (8.6 : 9.4)
2. (525.0 : 555.0)
3. (7.5 : 7.5)
4. (7)
5. (90 or 10)
6. (6 to 6)
7. (2.46 to 2.50)

EXPLANATIONS

EXERCISE – I

MCQ TYPE QUESTIONS

1. Zymase is complex of enzyme. It was obtained from yeast cells by Edward Bucher
2. All enzymes are proteins but all proteins are not enzymes
3. Arber and Nathans got Nobel Prize in 1978 for the discovery of restriction endonuclease.
4. First time fermentative enzymes were discovered from yeast.
5. In all living cells of the plant body, because in all living things biochemical reaction takes place.
6. Enzymes are the polymers of amino acids because enzymes are made up of proteins which are polymers of amino acids.
7. Viruses are acellular organisms. So enzymes are absent in viruses.
8. Enzymes work as organic catalyst. They are not used up in biochemical reactions.
9. Potassium acts as an activator of several enzymes including DNA polymerase.
10. Peptidase is the enzyme that is not used in making detergents.
11. Enzymes are polymer of amino acid
12. Zn acts as a co-factor.
13. Coenzymes are loosely attached complex non-protein.
14. The proteinaceous part of an enzyme is called apo enzyme.
15. Permeases are found in plasma membrane of protein. So they are carrier proteins.
16. FAD and FAM contains vitamin B2(riboflavin).
17. Ag is non- essential part of plants.
18. The substance which causes change in allosteric sites are known as modulators. Hexokinase is the example of inhibitor modulator.
19. Molybdenum play important role in nitrogen fixation because it is an activator of nitrate reductase.
20. In each and every molecule energy of activation is found, in which a few have more while other s have less energy.
21. Km is a Michaelis Menten equation constant, which indicates the substrate concentration at which the chemical reaction catalyzed by an enzyme attains half its maximum velocity.
22. Amylase digests the stored food material of germinated cotton seed.
23. The enzyme which converts glucose into alcohol is Zymase.
24. Rate of enzyme actively initially increases and then decreases.

25. Enzymes are active on definite pH, a few are active in acid medium and few are in alkaline medium. This correlation is called as pH sensitivity.
26. Zymogens are enzyme precursors which in need can be converted in enzymes.
27. Isoenzymes: A few enzymes are isomerically equal to others and are only differing in their molecular structure.
28. More than 100 enzymes are known to have isoenzymes, isoenzymes in humans which catalyse change of pyruvate to lactate.
29. **Reason:** J.B. Sumner (1926) prepared a pure crystalline form of urease enzyme.
30. **Reason:** Each enzyme has its own different favourable pH value but same temperature optima.
31. **Reason:** Biochemical reactions of living beings are different from non-living world because in non-living beings, catalysts of reactions are mostly inorganic metals.
32. **Reason:** One gene determines the synthesis of one enzyme.
33. **Reason:** Because a particular enzyme can catalyse only a particular type of reaction.
34. **Reason:** Some enzymes are normally present in cell but their synthesis is ceased when the concentration of their end product become high, such enzymes are called **repressible enzymes** whereas the end product is called **co repressor.**

    A regulator gene produces the apc repressor which unites with co repressor to form a functional repressor molecule. This repressor molecule inhibits mRNA synthesis by all genes specifying **enzymes.**
35. Enzyme has specific site for substrates called as active sites and substrate has reactive sites. These active and reactive sites help in making of substrates enzyme complex.
36. The enzymes substrate complex is short lived. The substrate is changed into products. The products remain complexed with the active site of the enzyme for a brief period. They soon separate and the active site is free to perform another catalytic act. Enzymes activity depends upon their affinity of substrates. If turnover number of substrate is higher, then enzymes show high affinity towards substrate. The number of substrate molecules changed per minute by a molecule or enzyme is called turn over number.
37. Desmolysing enzymes are those which catalyse reactions by the other methods other than hydrolysis, **e.g.,**aldolases, dehydrogenases, oxidases, etc. Digestive enzymes function by catalysing hydrolysis. Larger molecules are broken into smaller ones. They are grouped into three types - proteolytic (breaks protein molecule), amylolytic (breaks sugar molecule) and lipolytic (breaks lipid molecule).
38. Cofactor may be inorganic or organic in nature. Organic cofactors are of two types, coenzymes and prosthetic groups. Coenzymes are easily separable non protein organic cofactors. Prosthetic groups are non protein organic cofactors firmly attached to apo enzymes (protein part of enzyme).
39. We know that all biological reactions are catalysed by special catalysts called enzyme, thus enzymes are defined as biological proteins. We also know that enzymes are small organic molecules which are weakly held to the protein and can be easily separated by dialysis. Therefore chemically all enzymes are globular proteins.
40. We know that DNA molecules are found primarily in the nucleus of the cell but RNA molecules are found outside the nucleus. By heating, its special structural arrangement is changed irreversibly, this result in the conversion of enzyme into a fibrous or insoluble form. Due to this irreversible change, enzymes lose their specific activity when heated.
41. The number of substrate molecules changed per minute by a molecule or enzyme is called turn over number. The higher the turn-over number, the more efficient an enzyme is. It depends upon the number of active sites present over an enzyme.
42. Feedback inhibition is a type of reversible inhibition found in allosteric enzymes. The inhibitor is non competitive and is usually a low molecular intermediate or product of metabolic pathway having a chain of reactions involving a number of enzymes.
43. Enzyme becomes inactive below minimum temperature. Low temperature preserves the enzymes- in the inactive.state. High temperature" destroys enzymes by causing their denaturation.
44. Activation energy is an external supply of energy which is needed for the initiation of the chemical reaction. Activation energy required for such a large number of reactions cannot be provided by living systems. Enzymes lower the activation energy required for a reaction. Enzymes are generally specific for their substrates.
45. According to Holum, the cofactor my be:
    1. A coenzyme - a non-protein organic substance which is dialyzable, thermostable and loosely attached to the protein part.
    2. A prosthetic group - an organic substance which is dialyzable and thermostable which is firmly attached to the protein or apoenzyme portion.
    3. A metal-ion-activator - these include K+, Fe++, Fe+++, Cu++, Co++, Zn++, Mn++, Mg++, Ca++, and Mo+++.

46. Absolute specificity - the enzyme will catalyze only one reaction.
47. A RING domain is an example of Ubiquitin ligase
48. Transaminase is involved in transfer of radical
49. Cholesterol side-chain cleavage enzyme, also called 20, 22-Desmolase
50. Zero order reaction rate is independent of substrate concentration
51. In the second order reaction rate is proportional to the first power of each of two reactants
52. Proofreading mechanisms are also found in RNA polymerase, aminoacyl tRNA synthetases and ribosomes.
53. Some enzymes do not need any additional components to show full activity. However, others require non-protein molecules called cofactors to be bound for activity. Cofactors can be either inorganic (e.g., metal ions and iron-sulfer clusters) or organic compounds (e.g., flavin and heme)
54. Louis Pasteur recognized in 1860 that enzymes were essential to fermentation but assumed that their catalytic action was inextricably linked with the structure and life of the yeast cell. Not until 1897 was it shown by German chemist Edward Buchner that cell-free extracts of yeast could ferment sugars to alcohol and carbon dioxide;
55. The qualitative test for tyrosine is millons test
56. Reverse transcriptase is also called as RNA directed DNA polymerase
57. One gene and one enzyme hypothesis was conducted on Neurospora
58. Alpha amylase is also called as ptyalin
59. Steady state refers to a time period of the enzymatic reaction during which the rate of formation of the ES complex is exactly matched by its rate of decay to free enzyme and products.
60. The catalytic efficiency of an enzyme is best defined by the ratio of the kinetic constants, kcat/Km
61. For an enzyme in solution, the rate-determining step in catalysis will be either k1, the rate of ES formation, or one of the multiple steps contributing to kcat.
62. Number of molecules of substrate converted to product by one enzyme site per second is the turn over number
63. Y = (Vmax * X)/(Km + X)

    In the above reaction X represents substrate
64. Vmax is measured as moles of product produced per minute
65. The value of slope formed in lineweaver - Burk plot taking 1/V on y-axis and 1/S on x-axis is km/Vmax
66. In Lineweaver-Burk plot the most accurate values of KM and Vmax are obtained by nonlinear regression
67. The term which is used for binding of substrate to one binding site increases the activity of neighboring sites is positive cooperativity
68. The unit used to measure kcat is sec-1
69. The overall free energy change for a reaction is related to the equilibrium constant
70. Small km value indicates tight binding
71. Reversible inhibitors interact with an enzyme via non covalent association
72. Turn over number is formulated as

    kcat = vmax/[E]t
73. Chymotrypsin suitable pH is at small intestine
74. The main difference between ribozymes and enzymes is that RNA catalysts are composed of nucleotides, whereas enzymes are composed of amino acids.
75. An inhibitor binding at an allosteric site can be noncompetitive, because it can bind to both the free enzyme and the enzyme-substrate complex.
76. Inhibitor that chemically react at a substrate binding site irreversibly inactivate the enzyme and are also very effective inhibitors. These may be called active-site-directed or suicide inactivators.
77. In biochemistry, eicosanoids are signaling molecules made by oxidation of twenty-carbon essential fatty acids, (EFAs). They exert complex control over many bodily systems, mainly in inflammation or immunity, and as messengers in the central nervous system. The networks of controls that depend upon eicosanoids are among the most complex in the human body.
78. Eicosanoids derive from either omega-3 or omega-6 EFAs.
79. Covalently binding inhibitors are grouped into mechanism-based inhibitors, affinity labels, and pseudoirreversible
80. Inhibition of the enzyme catechol-O-methyltransferase (COMT) is an important approach in the treatment of Parkinson's disease
81. The slope for a graph drawn between Ki on y-axis with various concentration of inhibitor on x-axis is Km/Ki
82. An inhibitor that binds covalently to irreversibly inactivate the enzyme is called an irreversible inhibitor or inactivator
83. If an inhibitor binds reversibly to the same site as the substrate, the inhibition will be competitive. Competitive inhibitors are common in nature.

84. Reversible inhibitors bind to enzymes with non-covalent interaction such as hydrogen bonds, hydrophobic interactions and ionic bonds. Multiple weak bonds between the inhibitor and the active site combine to produce strong and specific binding. In contrast to substrates and irreversible inhibitors, reversible inhibitors generally do not undergo chemical reactions when bound to the enzyme and can be easily removed by dilution or dialysis.
85. In uncompetitive inhibition, the inhibitor binds only to the substrate-enzyme complex; it should not be confused with non-competitive inhibitors. Both Vmax and Km decrease
86. Slow-tight inhibition occurs when the initial enzyme-inhibitor complex EI undergoes isomerisation to a second more tightly held complex, EI*, but the overall inhibition process is reversible
87. Some protein kinase inhibitors have chemical structures that are similar to adenosine triphosphate.

NUMERICAL TYPE QUESTIONS

1. More than 2000 enzymes are known to occur in an average living cell.
2. **Reason:** Enzymes show reversible reactions and act by lowering energy of activation by more than 50%. They work in milliseconds and rate of enzyme to substrate is as high as 1 : 1000000.

EXERCISE – II

MCQ TYPE QUESTIONS

1. Enzyme inhibitors which resembled the transition state structure would bind more tightly to the enzyme than the actual substrate. Transition state analogs can be used as inhibitors in enzyme-catalyzed reactions by blocking the active site of the enzyme. Eg: Saquinavir.
2. Higher cell densities can be achieved using microcarriers which increases the surface area for cell growth and also mediates the culture of different type of cells. They are used for anchorage dependant cells. Eg: Cytodex I. Absence of surface charge doesn't help in anchorage.
3. Trypsin cleaves at carboxy side of R. So R is written on the C termini. On the other hand, Trypsin digested fragment includes RAV And RSY. Any of the fragment can be placed depending on the chymotrypsin digestion which is performed after trypsinisation. Chymotripsinisation produced ARVY and SR after cleaving C termini of another R. So the resultant amino acid sequence will be AVRYSR.
4. Scatchard analysis is plot of ratio of concentration of bound ligand to unbound ligand verses the bound ligand concentration. The slope = affinity for ligand binding and Y intercept = total concentration of antibody.

$$\frac{[Ab-Ag]}{[Ag]} = -\frac{[Ab-Ag]}{[Ag]}.ka + nka$$

Then compare with equation $y = mx + c$

5. Graph A represent uncompetitive inhibition
6. Enzymes or 'biocatalysts' are remarkable discovery in the field of bioprocess technology. Biocatalysis has been widely accepted in diverse sectors owing to their ease of production, substrate specificity and green chemistry. However, for large extent commercialization of these bio-derived catalysts, their reusability factor becomes mandatory, failing which they would no longer be economic. Consequently, immobilized enzymes with functional efficiency and enhanced reproducibility are used as alternatives in spite of their expensiveness. Immobilized biocatalysts can either be enzymes or whole cells. Enzyme immobilization is confinement of enzyme to a phase (matrix/support) different from the one for substrates and products. Inert polymers and inorganic materials are usually used as carrier matrices.
7. The ratio k(cat)/K(M)—often referred to as the "specificity constant"—is a useful index for comparing the relative rates of an enzyme acting on alternative, competing substrates. However, an alternative description, "catalytic efficiency", is frequently used to compare the reactivity of two enzymes acting on the same substrate.

$$E + S \underset{K_1}{\overset{K_1}{\rightleftharpoons}} ES \xrightarrow{K_2} E + P$$

K_2 here is also known as k_{cat}/k_m, the catalytic efficiency of enzyme.

$$k_{cat}/k_m = k_2 = V_{Max}/[E]_0$$

8. If a reversible inhibitor can bind to the enzyme active site in place of the substrate, it is described as a "competitive inhibitor." In pure competitive inhibition, the inhibitor is assumed to bind to the free enzyme but not to the enzyme-substrate (ES) complex. At the macroscopic level, the effect of competitive inhibition is to increase the substrate concentration required to achieve a given reaction velocity; in other words, to raise the Km. Here, with increase in the concentration of the inhibitor [I], the rate of the reaction changes and so does the slope of the plot. Hence it is a competitive inhibition.

9. From the graph we can see that the value of r is same for all the receptors. 'r' is the moles of bound ligand per moles of receptor. So, both R1 and R3 have the same number of sites for ligand binding. Now, in the graph, r = 2. Therefore the number of ligand binding sites is 2.
10. Analysing the graph it can be seen that r/c value is the maximum for Recpetor1. Here, r is the moles of bound ligand per moles of receptor and c is concentration of free ligand. So, more the value of r/c, more will be the efficiency of binding the ligand by a receptor. Since r/c for Recpetor1 is 4 which is the highest, so Receptor1 is the most efficient binder of the ligand.
11. One katak is 6×10^7 IU units
12. Enzymes accelerate the rate of the reverse reaction as well as the forward reaction, it would be helpful to ignore any back reaction by which E1P might form ES. The velocity of this back reaction would be given by $v = k_{-2}[E][P]$.
18. Glutamate dehydrogenase (GLDH) is an enzyme, present in most microbes and the mitochondria of eukaryotes, as are some of the other enzymes required for urea synthesis, that converts glutamate to α-ketoglutarate, and vice versa. The α-ketoglutarate to glutamate reaction does not occur in mammals as glutamate dehydrogenase equilibrium favours the production of ammonia and α-ketoglutarate. NAD^+ or $NADP^+$ is a cofactor for the glutamate dehydrogenase reaction, producing α-ketoglutarate and ammonium as a byproduct.
19. A non- competitive inhibitor can combine with an enzyme molecule to produce a dead- end complex, regardless of whether a substrate molecule is bound or not. Hence, inhibitor binds at a different site from the substrate. Km (Michelis Menton) constant remains unchanged but the value of Vmax (maximum velocity) is altered.
20. The degree of effectiveness factor is determined by Hill equation for Non Michaelis-Menten kinetics. The Hill equation accounts for allosteric binding at sites other than the active site. n is the "Hill coefficient." When n < 1, there is negative effectiveness factor; When n = 1, there is no effect; When n > 1, there is positive effectively.
21. Turnover number is expressed in molecules of product / active site / second. It can be defined as the number of molecules of a substrate acted upon in a period of 1 minute by a single enzyme molecule; with the enzyme working at a maximum rate.here the statement shows the dP/dt of the two enzymes at unit time. Again dP/dt is the velocity of the reaction i.e. the E1 has greater velocity than of E2.
22. Given,
    Dissociation constant, $K_d = 10^{-7}M$
    The reaction is: L + R → LR
    Thus, K_d for the reaction = [L][R]/[LR]
    or [LR]/[LR][R] = 10/100
    Therefore, [L] = 1/9 * 10^{-7}M = 10^{-8}M
23. Number of ligands = total radioactivity/ radioactivity in each ligand
    = 1000 cpm/ 100 cpm/pmole
    = 10 pmole
    = 10 * 10^{-12} * 6.023 *10^{-12} * 6.023 * 10^{23} ligands
    Thus, total ligands= 6.023 * 10^{12}
    Average number of receptors, R per cell= total receptor/number of cells
    = 6.023 * 10^{12}/ 6 * 10^7 = 10^5
25. Affinity chromatography utilizes the specific interactions between two molecules for the purification of a target molecule. A ligand having affinity for a target molecule is covalently attached to an insoluble support and functions as bait for capturing the target from complex solutions. The affinity ligand can be any molecule that will bind the target without also binding other molecules in the solution.
26. Stock solution= 150 M
    Dilution factor= 10/200 l = 1/20 l
    Concentration of d TTP= (1/20)*150 M= 7.5 M
27. $K_M = \frac{K_{-1}+K_2}{K_1}$
    $= \frac{4\times10^4\,S^{-1}+2\times10^3\,S^{-1}}{3\times10^8\,M^{-1}S^{-1}}$
    $= \frac{42000}{3\times10^8}$
    $= 1.4\times10^{-4}M$
28. We know, from Monod's equation
    $\mu = \mu_{max}[S_0/(K_S + S_0)]$
    $\Rightarrow \mu = 1 * [10/(0.2+10)] = 10/10.2 = 0.98$
    At wash out, Dilution rate D = μ
    Therefore,
    the required dilution rate = 0.98 hr^{-1}
31. % incorporation is calculated by incorporation into precipitated form. (The polymerized form will precipitate).
    $= \frac{\text{precipitate form cpm}}{\text{total cpm}}$
    $= \frac{2.94\times10^4\text{ cpm}}{4.2\times10^4\text{ cpm}}\times100 \cong 70\%$

32. 2.94×10^5 cpm is from 0.1 dilution of 10 µl sample. Thus, 10 µl sample will have 2.94×10^6 cpm but, this 10 µl sample is taken from 100 µl which contain 1 µg of λ-interferon. So, 100 µl of 1 µg interferon sample will have 2.94×10^7 cmp/µg.

35. A : Enzymatic method is mild. Chemical and mechanical methods may tend to destroy the tissue.

R : Enzymes show their catalytic activity at a particular optimum p H and temperature.

36. The Michaelis constant (Km) is a means of characterizing an enzyme's affinity for a substrate. The Km in an enzymatic reaction is the substrate concentration at which the reaction rate is half its maximum speed. Thus, a low Km value means that the enzyme has a high affinity for the substrate (as a "little" substrate is enough to run the reaction at half its max speed).

37. Because this enzyme has a high KM, less substrate is converted into product.

38. Photosensitive Polyethylene glycol dimethylacrylate is used in free radical copolymer crosslinking reactions which cross links enzyme with matrix and hence used in immobilzation of enzymes. Polyvinyl alcohol is used for formation of spherical structured beads in immobilisation process but after the process is over it needs to be degraded and for this purpose gamma radiation induced degradation of polyvinyl alcohol is carried out.

39 The immobilized enzyme is easily removed from the reaction making it easy to recycle the biocatalyst. Immobilized enzymes typically have greater thermal and operational stability than the soluble form of the enzyme. This reduces the operational cost.

NUMERICAL TYPE QUESTIONS

1. $V_{max} = 0.0134\ \mu mol\ min^{-1}$

$= 02.233 \times 10^{-4}\ \mu mol\ s^{-1}$

$= 2.233 \times 10^{-10}\ mol\ s^{-1}$, $K_m = 25\ \mu m.$

$$E_t = \frac{1.8}{18} = 0.1\ \mu mol = 0.1 \times 10^{-6}\ mol.$$

Now, $$K_{cat} = \frac{V_{max}}{E_t} = \frac{2.233 \times 10^{-10}}{0.1 \times 10^{-6}} = 2.233 \times 10^{-3}$$

$$\therefore \quad \frac{K_{cat}}{K_m} = \frac{2.233 \times 10^{-3}}{25 \times 10^{-6}} = 0.0893 \times 10^3$$

$= 8.93 \times 10^1\ M^{-1}s^{-1}$

2. Specific Activity of enzyme

$$= \left(\frac{\text{Change in OD. Per min} \times 1000 \times \text{total reaction mix in ml}}{\text{Extinction coefficient} \times \text{Vol. of enzyme in sample in ml} \times \text{concentration of protein in g/ml}} \right)$$

$$= \frac{0.14 \times 1000 \times 3}{(6220 \times 25 \times 10^{-3} \times 5 \times 10^{-3})} = 540$$

3. $$V_1 = \frac{V_{max} A[s]}{K_m + [s]}, \quad V_2 = K_L A([s_0] - [s])$$

Reaction rate Mass transfer rate

The given condition is

$$V_1 = V_2 \text{ at steady state.}$$

Further $V_{max} = K_L s_0 = 1 \times 10 = 10.$

$$\therefore \quad \frac{10[s]}{10 + [s]} = [10 - s] \Rightarrow s^2 + 10s - 100 = 0$$

Solving for s

Using $$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$s = 6.180 \text{ or } -16.180$$

$$\therefore \quad s = 6.180$$

4. 1 Katal = The one mole of substrate converted by enzymes in 1 second.

$$= \frac{420}{60} \frac{\mu \text{moles}}{\text{seconds}} = 7 \times 10^{-6} \text{katal}$$

5. $$\frac{1}{V} = \frac{K_m}{[S]} \frac{1}{V_{max}} + \frac{1}{V_{max}} \quad \ldots(1)$$

$$\frac{1}{V} = \frac{K_{m_1}}{V_{max}} \left[1 + \frac{I}{KI} \right] \frac{1}{[S]} + \frac{1}{V_{min}} \quad \ldots(2)$$

Then equation (1) – (2) and by simplification

$$K_m = 30 \left[1 + \frac{5}{2.5} \right] = 90$$

Note : If we take K_m in the absence of the inhibitor 30 µM then K_m in the presence of the inhibitor can be calculated as per above formulae 10 µM.

6. In ethanol, there are 6 hydrogen atoms which are present in it. So, the degree of reduction of ethanol is 6 as there are six hydrogen atoms gained by the ethanol molecule. Hence, degree of reduction for ethanol is 6.

7. The standard Gibb's free energy change for the formation of G-3-P is given by:

$$\Delta G^0 = -RT \ln K$$

$$\Rightarrow \quad \Delta G^0 = -8.315 \times 298 \times \ln 0.4$$

$$\Rightarrow \quad \Delta G^0 = 2{,}270.449\ J/mol = 2.27\ kJ/mol$$

■■

11 CHAPTER

Biochemical Engineering and Bioprocess Technology

Microbial Growth Kinetics

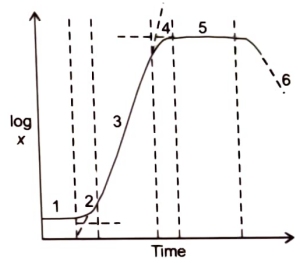
Batch Culture

Batch cultivation represents growth in closed system since no fresh medium is added to the culture.

When a suitable growth supporting medium is inoculated with cells a characteristic sequence of events termed the growth cycle takes place which can be depicted by ploting a graph of cell dry weight, (X), $(g.l^{-1})$ versus period of incubation (h) hours.

Growth phases

1. lag, **2.** acceleration, **3.** exponential, **4.** deceleration, **5.** stationary, **6.** decline.



Batch growth cycle

(i) The Lag Phase

Immediately after incoculation no apparent growth may occur for a period of time. The length of the lag phase is variable and dependent on the previous growth history of the cells. It represents a period of adaptation to grow in a new environment, and involves the synthesis of enzymes required for growth in this enviornment.

(ii) The Exponential or Log Phase

This is frequently spoken of as the phase of 'balanced growth' where the synthesis of all cell constituents increases at a constant rate so that the cell population doubles, and continues to double at regular intervals. This is not a true phase of balanced or steady-state growth since nutrients are continuously being consumed (substrate concentration varies with time), and end-products of metabolism are accumulating consequently the environment is continously changing.

(iii) The Stationary Phase

This is characterized by no net growth. In fact, growth may be occuring but it is balanced by the rate of cell death or cell lysis. The cells may remain viable for long period since this phase with endogenous metabolism, the oxidation of storage polymers, proteins, etc., occurring. The population usually enters the stationary phase as the result of the depletion of some essential nutrient, the build-up of toxic products, or a change in the physical environment.

(iv) Decline Phase

During this phase the death rate may become higher than the growth rate in this case the cell density falls.

(v) Acceleration and Deceleration Phases

These are occasionally referred in the literature. The deceleration phase is important since growth is 'unbalanced' and the growth rate varies as a function of residual substrate concentration in rate-limited cultures.

It is important to realize that the growth cycle is not an inherent property of the organism but the inevitable consequence of the interaction of the organism with the environment in a closed system.

Growth Kinetics in Batch Culture

Unicellular microbial growth is autocatalytic such that the rate of growth is proportional to the cell concentration already present. During exponential growth the cell dry weight (or number) increases as follows

$$X_0 \rightarrow 2X_0 \rightarrow 4X_0 \rightarrow 8X_0 \ldots\ldots\ldots\ldots nX_0$$

where X is the cell dry weight ($g.l^{-1}$) and the interval is constant and termed the doubling time (t_d).

Consequently the number of doublings, n, after time t, is given by

$$n = \frac{t}{t_d}$$

It follows that the cell density (X_t) after time, t, will be related to the density orignally present (X_0) according to

$$X_t = X_0\, 2^n = X_0 2^{t/td}$$

Rearranging and taking logarithms yields

$$\ln (X_t/X_0) = \ln 2^{(t/td)} = \frac{t}{t_d} \ln 2$$

Thus,
$$\frac{\ln X_t - \ln X_0}{t} = \frac{\ln 2}{t_d} = \frac{0.693}{t_d} \qquad \ldots(i)$$

This is more commonly expressed in terms of

$$2.303 \frac{(\log X_t - \log X_0)}{t} = \frac{0.693}{t_d}$$

A plot of log X versus time yields a straight line with a slope of $\frac{0.693}{t_d}$. Above equation can be write in differential equation form as -

From equation
$$\frac{d(\ln X)}{dt} = \frac{0.693}{t_d}$$

However,
$$\frac{d(\ln X)}{di} = \frac{d(\ln X)}{dX} \times \frac{dX}{dt}$$

and since
$$\frac{d(\ln X)}{dt} = \frac{1}{X}\frac{dX}{dt}$$

we can write
$$\frac{1}{X}\frac{dX}{dt} = \frac{0.693}{t_d} = \mu$$

or
$$\frac{dX}{dt} = X$$

Equation is termed as the growth equation where is the specific growth rate defined as the rate of increase in cell concentration per unit of cell concentration (h^{-1}).

Equation is the differential form of the growth equation and can be used to describe growth at any stage during the growth cycle, whilet equation (i) is the integral form which only applies during exponential growth where is constant.

The doubling time (t_d) is usually equivalent to the mean generation time (mgt), the average period for one complete cell division cycle. Exceptions occur where a fraction of the cell population is non-viable.

In the special case where $\frac{X_t}{X_0} = 2$, i.e. the cell population doubles, equation simplifies to:

$$= \frac{2.303 \log 2}{t_d} = \frac{0.693}{t_d}$$

The Substrate Utilization Rate

Substrates are consumed not only to provide the necessary carbon, energy, and structural components for cell growth but for the maintenance of cell viability (cell repair mechanisms, substrate transport process, maintenance of membrane potential, etc.) and for product formation. Thus the rate of a single substrate utilization can be determined from :

Substrate accumulation = substrate feed – substrate consued for growth – substrate consumed for product synthesis – Substrate consumed for maintenance – substrate removel

$$-\frac{dS}{dt} = \frac{F}{V}.S - \frac{\mu X}{Y_{X/S}} - \frac{q_p X}{Y_{P/S}} - mX - \frac{F'}{V'}.S'$$

where F and F′ are the flow rates of medium into and out of the bioreactor respectively ($1\ h^{-1}$), V is the volume of the culture (l); S and S′ are the substrate concentrations going in and out of the culture (gl^{-1}); $Y_{X/S}$ is the biomass yield coefficient (g cell dry weight formed per g substrate consumed): $Y_{P/S}$ is the product yield coefficient (g product formed per g substrate consumed): and q_p are the specific growth and product formation rates (h^{-1}) respectively; and m is the maintenance coefficient (g substrate consumed per g cell dry weight per hour).

With no substrate feed and substrate removal becomes :

$$-\frac{dS}{dt} = \frac{-X}{Y_{X/S}} - \frac{q_p X}{Y_{P/S}} - mX$$

Frequently especially with aerobic cultures, the maintenance coefficient is very small compared with the term $X/Y_{X/S}$, and if no products are formed, then

$$\frac{dS}{dt} = \frac{X}{Y_{X/S}}$$

The specific rate of substrate utilization $q_S(h^{-1})$ can be determined by

$$q_S = \frac{1}{X}.\frac{dS}{dt} = \frac{\mu}{Y_{X/S}} + \frac{q_P}{Y_{P/S}} + m$$

Product Formation Rate

If products are formed during the growth of the microbial cells, the rate of formation is described by :

product accumulation = product synthesis – production removal – product denaturation

$$\frac{dP}{dt} = q_p X - \frac{F}{V}.P - \beta P$$

where P is the product concentration (gl^{-1}) and β the rate of product denaturation (h^{-1}). If the product is stable and not removed from the bioreactor, then

$$\frac{dP}{dt} = q_p X$$

and the specific rate of product formation q_P is given by:

$$q_P = \frac{1}{X}.\frac{dP}{dt}$$

Growth rate, substrate consumption rate, and product formation rate can be expressed either in terms of volumetric or specific rates. Thus the volumetric rate ($gl^{-1}h^{-1}$) is dependent on cell concentration; it describes the rate of synthesis of the desired product, or rate of substrate utilization, per unit volume of the bioreactor.

Biomass and Product Yields

The biomass and product yields. $Y_{X/S}$ and $Y_{P/S}$ are extremely important parameters since they represent the efficiency of conversion of the substrate into biomass or product. They are defined as the mass of biomass or product formed per mass of substrate consumed.

Thus $$\frac{dX}{dt} = Y_{X/S} \cdot \frac{dS}{dt} \quad \text{and} \quad \frac{dP}{dt} = Y_{P/S} \cdot \frac{dS}{dt}$$

These yields are normally determined either by measuring the total amount of biomass or product formed and substrate consumed over a period.

$$Y_{X/S} = \frac{\Delta X}{\Delta S} \quad \text{and} \quad Y_{P/S} = \frac{\Delta P}{\Delta S}$$

or they can be determined from the rates of biomass or product formation and substrate consumption rates :

$$Y_{X/S} = \frac{\mu}{q_s} \quad \text{and} \quad Y_{P/S} = \frac{q_P}{q_s}$$

Pirt Equation

The yeild coefficient may not be constant throughout the growth curve mainly because of the consumption of substrate for maintenance requirements, consequently Pirt (1975) derived an expression which relates the effect of maintenance on the measured biomass yield $Y_{X/S(M)}$:

$$Y_{X/S(m)} = \frac{\Delta X}{\Delta S_G + \Delta S_m}$$

where ΔS_G is true substrate conusmed and ΔS_m is substrate consumed for maintenance. The true growth yield $Y_{X/S(G)}$ will be

$$Y_{X/S(G)} = \frac{\Delta X}{\Delta S_G}$$

In batch culture the rate of substrate utilization

$$\frac{dS}{dt} = \left(\frac{dS}{dt}\right)_m + \left(\frac{dS}{dt}\right)_G$$

where $$\frac{dX}{dt} = X \quad \text{and} \quad \frac{dS}{dt} = \frac{\mu X}{Y_{X/S(m)}}$$

However, $$\left(\frac{dS}{dt}\right)_G = \frac{\mu X}{Y_{X/S(G)}} \quad \text{and} \quad \left(\frac{dS}{dt}\right)_m = mX$$

where m is the maintenance coefficient. Then

$$\frac{dS}{dt} = mX + \frac{\mu X}{Y_{X/S(G)}}$$

$$\frac{dS}{dt} \cdot \frac{1}{X} = m + \frac{\mu}{Y_{X/S(G)}}$$

$$\frac{1}{Y_{X/S(m)}} = \frac{m}{\mu} + \frac{1}{Y_{X/S(G)}}$$

The maintenance coefficient is determined by plotting $1/Y_{X/S(m)}$ versus $1/$ to give a straight line with a slope of m and intercept $1/Y_{X/S(G)}$. Maintenance coefficient determinations are usually made by using chemostat techniques since it is difficult to vary the value of in batch culture. Where $1/Y_m$ is measured or calculated growth yield and $1/Y_G$ is true growth yield.

Productivity

Volumetric productivity is measured in terms of grams product formed per litre per hour.

The effect of substrate concentration on the specific growth rate ( )

The effect of substrate concentration, S, on can be described by the Monod equation.

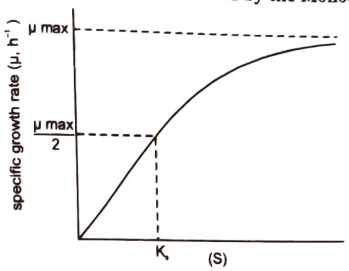


Fig. The relation ship between the specific growth rate ( ) and the substrate concentration (S).

$$\mu = \mu_m \frac{S}{K_s + S}$$

where is the specific growth rate (h^{-1}). $_m$ is the maximum specific growth rate (h^{-1}) which is a constant for a particular substrate. K_S is the saturation or Monod constant (gl^{-1}) and S is the substrate concentration (gl^{-1}).

If $S > K_S$ then equation simplifies to : $= {}_m \frac{S}{S} = {}_m$

Alternatively, if $K_S = S$, then: $= \frac{\mu_m S}{2S}$ therefore $= {}_m/2$.

Thus the saturation constant K_S is numerically equivalent to the substrate concentration at which the growth rate is half the maximum $_m$. K_S expresses the affinity of the organism for the substrate S. A *high* value of K_S represents a *low* affinity for the substrate and vice versa.

Accurate determinations of $_m$ and K_S may be obtained by linearizing equation by taking reciprocals of both the left- and right-hand side and rearranging to give :

$$\frac{1}{\mu} = \frac{1}{S} \cdot \frac{K_S}{\mu_m} + \frac{1}{\mu_m}$$

Thus when $1/S = 0$, $1/\mu = 1/{}_m$, and when $1/ = 0$, $1/S = -1/K_S$, consequently a plot of $1/$ versus $1/s$, the so-called Lineweaver-Burk plot yields a straight line with a slope of $K_S/{}_{max}$.

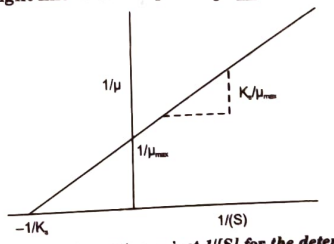


Fig.: A Lineweaver-Burk plot of 1/ against 1/[S] for the determination of the saturation constant K_S(mg.l⁻¹) and $_{max}$(h^{-1}).

Effect of pH on the growth rate

The pH of the culture medium can have an important influence on the growth and product formation of an organism. In general the maximum growth rate may be maintained over 2-5 units. As a rule of thumb. bacteria usually grow in the rang of pH 4–8, yeasts 3–6, moulds 3–7 and higher eukaryotic cells 6.5–7.5.

Continuous Culture

Continuous culture is a system for growing cells in an open system where the microbial population is maintained in a continuous state of balancded growth by continuously removing some of the culture and replacing it with fresh medium at the same rate.

There are two types of continuous culture system :

(1) the *chemostat* which is operated by supplying an essential growth-limiting nutrient at a constant rate with the result that the culture density and growth rate adjust themselves to the supply, and

(2) the *turbidostat* which is operated by maintaining a constant cell density by supplying fresh medium as required.

A basic requirement for both system is a method for controlling a constant bioreactor volume. Many methods can be used to maintain constant volume. The simplest method uses an overflow tube or weir set to a constant height within the bioreactor vessel so that as fresh medium is pumped into the bioreactor an equal volume of culture enters the overflow tube and passes to the collection vessel gravimetrically.

Chemostat theory

In a Chemostat the growth rate is determined by the rate of supply of an essential nutrient which can be carbon, nitrogen, phosphorus, or a trace element or vitamin. All other essential nutrients are present in excess. The ability to vary the growth rate by changing the rate of supply of fresh nutrient is the major advantage of continuous over batch culture, and explains the self-regulatory property of the chemostate. By maintaining the supply rate constant the culture can be maintained in a steady state of growth indefinitely.

The relationship between dilution rate and cell concentration

The self-regulatory property of chemostat and steady state growth can be expressed by a series of equations which relate the cell and limiting nutrient concentrations to the primary independent operating variable – the medium supply rate. This is done by developing material balances on the cells and limiting substrate , in the bioreactor. Thus for cells :

cell accumulation = cell in – cell out + growth – death

$$\frac{dX}{dt} = \frac{FX_0}{V} - \frac{FX}{V} + \mu X - \alpha X$$

where F = flow rate of medium supply ($1h^{-1}$)

V = constant working volume of bioreactor (l)

X_0 = cell concentration in medium supply (gl^{-1})

X = cell concentration in bioreactor (gl^{-1})

= specific growth rate (h^{-1})

α = specific death rate (h^{-1})

With a single-stage chemostat the medium supply is usually sterile (assuming no cell recycle) and $\mu >> \alpha$ consequently expression can be simplified to:

$$\frac{dX}{dt} = -\frac{FX}{V} + \mu X$$

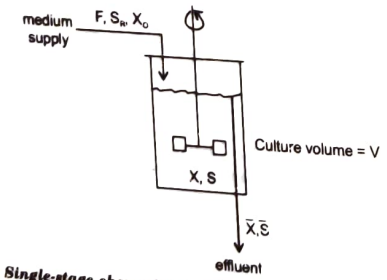


Fig. : Single-stage chemostat showing nutrient and biomass flows.

The term F/V is termed as *dilution rate*, that is the number of culture volumes passing through the reactor per hour, consequently expression can be rewritten as:

$$\frac{dX}{dt} = -DX + \mu X = X(\mu - D)$$

where D is the dilution rate.

During steady state growth $dX/dt = 0$ consequently $\mu = D$. This expression shows that, simply by varying the medium supply rate the growth rate can be varied. This holds until $D > \mu_m$ (the maximum specific growth rate). Under these conditions the nutrient is no longer limiting, and the expression $(\mu - D)$ becomes negative. The result is a fall in cell concentration and wash-out of the cutlure. The critical dilution rate, D_c, is defined as the lowest dilution rate at which wash-out occurs. In practice D_c is approximately equal to μ_m.

At dilution rates approaching D_c the chemostate becomes less stable since slight fluctuations in the flow rate due to pumping errors etc., cause large changes in X. Under these conditions, unstable steady states exist which can rapidly lead to wash out. Thus the major drawback of chemostats is that they work best at lower diluution rates where the changes in X and S are small and any fluctuations in flow rate have little effect on the steady state.

The turbidostat, on the other hand maintains a constant cell density by controlling the flow rate of medium input. Consequently, it is very sensitive to small changes in X and it works best a high dilution rates. Chemostats and turbidostats are therefore complementary.

The relationship between dilution rate and substrate concentration

This relationship can be examined by making a material balance around the chemostat on the limiting nutrient :

Nutrient accumulation =nutrient in – nutrient out – consumption – maintenance – product formation

$$-\frac{dS}{dt} = \frac{F S_R}{V} - \frac{F S}{V} - \frac{\mu X}{Y_{X/S}} - mX - \frac{q_p X}{Y_{P/S}}$$

where S_R = limiting substrate concentration in medium reservoir (gl^{-1})

$Y_{X/S}$ = biomass yield coefficient (g cell dry weight formed per g substrate consumed)

$Y_{P/S}$ = product yield coefficient (g product formed per g substrate consumed)

S = limiting substrate concentration in chemostat (gl^{-1})

m = maintenance requirement (g substrate consumed per g cell dry weight per hour)

= specific growth rate (h^{-1})

q_p = specific rate of product formation (g product formed per litre per hour)

In general, $mx << X/Y_{X/S}$ and can be neglected. Thus taking the case where no produucts are formed, expression becomes:

$$-\frac{dS}{dt} = \frac{FS_R}{V} - \frac{FS}{V} - \frac{\mu X}{Y_{X/S}}$$

At a steady state F/V= D and $dS/dt = 0$, and also $S = \bar{S}$ & $X = \bar{X}$

then, $\frac{\mu \bar{X}}{Y_{X/S}} = D(S_R - \bar{S})$

but μ = D at steady state, therefore rearranging gives the equation as -

$$\bar{X} = Y_{X/S}(S_R - \bar{S})$$

Note that at steady state, X and S are labelled $\bar{X}$ and $\bar{S}$ by convention.

If is assumed that the absolute value of x is independent of all nutrients except the growth-limiting nutrient.

The fundamental equations of continuous culture

It is also necessary to develop a model which relates X and S to the dilution rate, D. This is done by substituting values of from the Monod equation:

$$= \mu_m \frac{S}{K_s + S}.$$

(*i*) *Non-steady state equations*

For biomass : $\frac{dX}{dt} = X(\ - D)$

substituting value, then $\frac{dX}{dt} = X\left(\mu_m \frac{S}{K_s + S} - D\right)$

For limiting substrate : $\frac{dS}{dt} = D(S_R - S) - \frac{\mu X}{Y_{X/S}}$

substituting the vlaue of $\frac{dS}{dt} = D(S_R - S) - \frac{\mu X}{Y_{X/S}} \mu_m \frac{S}{K_s + S}$

(*ii*) *Steady state equations*

At steady state $dX/dt = 0$ and $dS/dt = 0$, so X and S become constant and are represented by $\bar{X}$ and $\bar{S}$ and = D. Consequently the Monod equation becomes :

$$D = \ _m \frac{\bar{S}}{K_s + \bar{S}}$$

solving for $\bar{S}$: $\bar{S} = \frac{K_s D}{\mu_m - D}$

now $\bar{X} = Y_{X/S}(S_R - \bar{S})$

therefore substituting for $\bar{S}$ in equation yields :

$$\bar{X} = Y_{X/S}\left(S_R - \frac{K_S D}{\mu_m - D}\right)$$

These fundamental equations describing steady state growth show that the steady state substrate concentration, $\bar{S}$, varies as a function of D alone, whilst biomass concentration $\bar{X}$ varies as a function of D and S_R.

The critical dilution rate

The critical dilution rate, D_c, is the lowest value of D at which wash-out occurs. In general, D_c is approximately eqaul to $_m$. However, $D_c =$ when $\bar{S} = S_R$. This is clear that since at wash-out there are no cells remaining in the chemostat, therefore no substrate is consumed. Consequently we can write :

$$D_c = \mu_m \frac{S_R}{K_s + S_R}$$

It is clear that D_c varies as a function of S_R. Thus if the value of S_R is high relattive to K_S.

$$D_c = \mu_m \frac{S_R}{S_R} = \mu_m$$

However, as S_R is reduced, the term K_S is no longer negligible, and D_c will be less than $_m$.

Productivity

In continuous culture the biomass productivity $P_x (g/l/h)$ is defined as:

$$P_x = D\bar{X}$$

substituting $P_x = DY_{X/S}\left(S_R - \frac{K_s D}{\mu_m - D}\right)$

Productivity as a function of dilution rate. The dilution rate giving rise to maximum productivity. D_m, can be calculated by setting the first derivative of equation to zero.

$$D_m = \left[1 - \left(\frac{K_s}{K_s + \bar{S}}\right)^{1/2}\right]$$

Usually, the dilution rate giving maximum productivity is frequently not the same as that giving maximum yield or conversion of substrate to cells. Consequently a compromise must frequently be made between productivity and residual substrate in the effluent.

The productivity of growth-realted products (P_p) can be described in a similar way:

$$P_p = D\bar{p}$$

where $\bar{p}$ represent the steady state product concentration (gl^{-1}).

Determination of Maximum Growth Rate

Since it is difficult to maintain stable steady states close to $_m$, the simplest method of determining $_m$ is to use the Monod equation and solve for K_s:

$$D = \mu_m \frac{\bar{S}}{K_s + \bar{S}}$$

and $K_s = \bar{S}\frac{(\mu_m - D)}{D}$

By operating the chemostat under identical conditions at a minimum of two dilution rates and determining $\bar{S}$ for each $_m$ can be found by solving the following equation:

$$\frac{\bar{S}_1(\mu_m - D_1)}{D_1} = \frac{\bar{S}_2(\mu_m - D_2)}{D_2}$$

Having found $_m$, K_s can be determined by substituting the value of $_m$ into equation

$$D = \frac{\mu_m \bar{S}}{K_s + \bar{S}}$$

The biomass yield, $Y_{X/S}$, can be determined at each steady state from measurements of $\bar{S}$ and $\bar{X}$ from equation

$$Y_{X/S} = \frac{\bar{X}}{S_R - \bar{S}}$$

Modifications to the Basic Chemostat System

1. Cell recycle system

It is possible to improve the efficiency of chemostat operation by recycling a proportion of the cells in the outlet stream back into the fermenter, or by using a number of chemostats in series.

Cell recycle is an important method of continually inoculating the chemostat to stablize the system by minimizing process perturbations and enabling the chemostat to be operated at dilution rates greater than $_m$. A further advantage of cell recycle is the ability to operate the chemostat close to D_m, the dilution rate giving maximum cell productivity without excessive substrate being left in the effluent unutilized. Normally, at D_m substrate remains in the effluent. By recycling cells, a higher steady state biomass consentration can be maintained which results in higher substrate consumption.

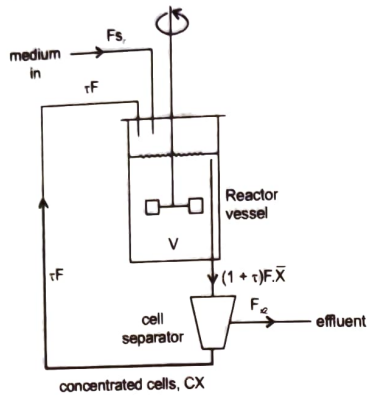


Fig. Schmatic representation of a chemostat with cell recycle. C, the concentration factor : τ, the recylce ratio

The theory for chemostat operation with cell recycling is similar to basic chemostat theory except that it includes two extra, terms, τ the recycle ratio and C the factor by which the outlet stream from the chemostat is concentrated berfore returning to the chemostat. Writing a material balance for cell mass and substrate around the fermenter the cell concetration yields :

For biomass :

cell accumulation = cell in medium supply + cell in recycle stream + cell growth – cells out

$$\frac{dx}{dt} = \frac{FX_0}{V} + \frac{FCX}{V} + \mu X - \frac{(1+\tau)FX}{V}$$

Cell maintenance and death are considered negligible.

Under steady state conditions $dX/dt = 0$ and $D = F/V$ and equation can be simplified to:

$$= D(1 + \tau - \tau CV)$$

Equation shows that the dilution rate no longer exceeds the growth rate because the expression $(1 + \tau - \tau C)$ is always less than one due to C being greater than 1.

For substrate :

substrate accumulation = substrate in medium supply + substrate in recycle stream – consumption – substrate out

$$\frac{dS}{dt} = \frac{FS_R}{V} + \frac{FS}{V} - \frac{\mu X}{Y_{X/S}} - \frac{(1+\tau)F_s}{V}$$

Under steady state condition $dS/dt = 0$ and $D = F/V$, and equation can be simplified to:

$$\bar{X} = \frac{D}{\mu} Y_{X/S}\left(S_R - \bar{S}\right)$$

Substituting with

$$\bar{X} = \frac{Y_{X/S}\left(S_R - \bar{S}\right)}{(1+\tau-\tau C)}$$

Equation shows that X will always be higher with recycle by a factor of $1/(1 + \tau - \tau C)$.

An expression for growth rate as a function of substrate concentration can be developed by substituting the Monod equation

$$\bar{S} = K_s \frac{\mu}{\mu_m - \mu} = K_s \frac{D(1+\tau-\tau C)}{\mu_m - D(1+\tau-\tau C)}$$

and

$$\bar{X} = \frac{Y_{X/S}}{(1+\tau-\tau C)}\left[S_R - K_s \frac{D(1+\tau-\tau C)}{\mu_m - D(1+\tau-\tau C)}\right]$$

2. Multistage System

In multistage continuous culture the outlet from the first stage acts as the inlet stream to the second stage. Such a system allows a great degree of flexibility since the first stage outlet continuously inoculates the second stage. The second stage can also be operated with a substrate supply so that D can be greater than μ_m.

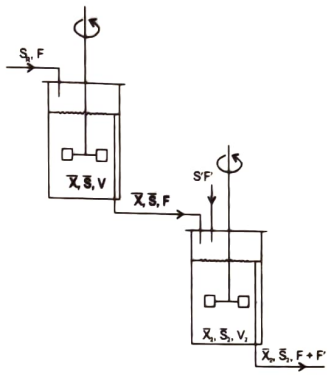


Fig. — A two-stage chemostate

A further possibility is to have different media in each stage. An example would be to maximize biomass production in the first stage and product formation in the second or subsequent stages. If desired, a cell recycle stream could be added to either stage.

Material balances on substrate and biomass can be developed around each stage of a two stage system without cell recycle and medium supply to the second stage. Balances arround the first stage have already been discussed.

For biomass (2nd stage) :

cell accumulation = Cells in – cells out + cell growth

$$\frac{dX_2}{dt} = \frac{FX}{V_2} - \frac{FX_2}{V} + \mu_2 X_2$$

Under steady state conditions $dX_2/dt = 0$ and $D_2 = F/V_2$

$$_2 = D_2\left(1 - \frac{\bar{X}_2}{\bar{X}}\right)$$

For substrate :

substrate accumulation = substrate in – substrate out – consumption

$$\frac{dS_2}{dt} = \frac{FS}{V_2} - \frac{FS_2}{V} - \frac{\mu_2 X_2}{Y_{X/S}}$$

Under steady state conditions: $\bar{X}_2 = \frac{D_2}{\mu_2} Y_{X/S}\left(\bar{S} - \bar{S}_2\right)$

From these equations it is clear that if no nutrient supply is added to the second stage the growth rate is less than the dilution rate, and the net growth in the second stage will be small. This system allows the study of non-growth-related events.

Addition of a nutrient supply to the second stage results in a higher biomass density at steady state and growth rate higher than $_m$

$$_2 = D_2 - \frac{F}{V_2}\frac{\bar{X}}{\bar{X}_2} \quad ; \quad \bar{X}_2 = \frac{Y_{X/S}}{\mu_2}\left(\frac{F}{V_2}\bar{S} + \frac{F'}{V_2}S' - D_2\bar{S}_2\right)$$

where F′ and S′ are the flow rate (lh^{-1}) and limiting substrate concentration (gl^{-1}) in the second stage medium supply.

Media Formulations

Micro-organisms need nutrients, a source of energy and certain environmental conditions in order to grow and reproduce. In the environment microbes have adapted to the habitats most suitable for their needs. In the labortary, however, these requirements must be met by a culture medium. This is basically an aqueous solution to which all the necessary nutrients have been added. A good media would meet the following criteria

1. It will produce the maximum yield of product or biomass per gram of substrate used.
2. It will produce the maximum concentration of product or biomass.
3. It will permit the maximum rate of product formation.
4. There will be the minimum yield of undesired products.
5. It will be consistent quality.

Depending on the type and combination of nutrients, different categories of media can be made.

Complex Media

Complex media are rich in nutrients they contain water soluble extracts of plant or animal tissue (e.g. enzymatically digested animal proteins such as peptone and tryptone). Usually a sugar often glucose is added to serve as the main carbon and energy source. The combination of extracts and sugar creates a medium which is rich in minerals and organic nutrients but since the exact composition is unknown, the medium is called complex. It contains undefined constituents like soyabeen meal, molasses, corn steep liquor etc.

Synthetic Media

It is also called as chemically defined media. These media are composed of pure ingredients in carefully measured concentrations, dissolved in double distilled water *i.e.* the exact chemical composition of the medium is known. Typically they contian a simple sugar as the carbon source, an inorganic nitrogen source, various mineral salts and if necessary growth factors (purified aminoacids, vitamins, purines and pyrimidines).

Sclective/Differential Media

These are media based on either of the two categories above suppremented with growth promoting or growth inhibiting additives. The additives may be species or organism selective (e.g. a specific substrate, or an inhibitor such as cyclo hexamide (articlone) which inhibit all eucaryotic growth and is typically used to prevent fungal growth in mixed cultures).

Media	Purpose
Complex	Grow most heterotrophic organisms.
Defined	Grow specific heterotrophs and are often mandatory for chemoautotrophs, photoautotrophs and for microbiological assays.
Selective	Suppress unwanted microbes or encourage desired microbes.
Differential	Distinguish colonies of specific microbes from others.
Enrichment	Similar to selective media but designed to increase the numbers of desired micro-organisms to a detectable level without stimulating the rest of the bacterial population.
Reducing	Growth of obligate anaerobes.

The mixture of necessary nutrients can be used as a liquid medium, or a solidifying agent can be added. "Agar-Agar" is a natural polysaccharide produced by marine algae and is the most commmonly used as a solidfying agent added to media (conc. 1.5% w/v)

Nutrient Media : Composition and Prepartion

Carbon Sources

Organisms that are photosynthetic, or obtain energy from the oxidation of inorganic compounds, typically use carbon-dioxide as the sole source of cellular carbon, and this fixation of carbon dioxide requires energy which is obtained from sunlight or inorganic compounds. All other organisms obtain their carbon from organic nutrients which serve a dual role both as a source of energy and carbon. Micro-organisms are extraordinarily diverse with respect to both the kind and number of organic compounds that they can use as a principal source of carbon and energy.

(*i*) **Xylose :** This has only been used as a nutrient in the preparation of glucose isomerase enzyme, although laboratory scale investigations are in progress using xylose for the production of ethanol using *Kluyveromyces marxianus.*

(*ii*) **Glucose :** This is used frequently in fermentations intended for highly purified products of high value. Glucose used is formed by acid or enzymatic hydrolysis of corn or potato starch and can be supplied as a powder, paste or syrup. It is used for the production of antibiotics, steroids, amino-acids, butyric, tartaric acids, xanthan gum, and hetero-polysaccharides.

(*iii*) **Sucrose :** Sucrose is the main substrate for citric acid fermentation, but sucrose is mainly supplied as molasses either from sugarbeet or sugarcane, Beet molasses is a syrupy brownish liquid containing approximately 50% sucrose but its composition can vary greatly. In addition to sugars, molasses contain some 30% dry matter consisting of nitrogenous compounds which can be of value. Molasses is the major media component in the production of bakers yeast, citric acid, acetone, butanol, organic acids and amino-acids. Sugarcane molasses are supplied as refiners or raw cane sugar molasses (blackstrap), contain high levels of invert sugar and biotin. This has been used as a carbon source for the production of ethanol, feed yeasts, acetone and butanol.

(*iv*) **Lactose :** Pure lactose is not often used as a carbon source except for some antibiotics, but whey produced during cheese manufacture contains high levels of lactose (75%) which is used for ethanol, SCP, and citric acid production.

(*v*) **Maltose :** It is present in malt, malt extract and brewer's wort. It is seldom used outside the brewing industry except for laboratory work.

(*vi*) **Starch :** Starch is a principal reserve carbohydrate in plants and is used pure or as a component from potatoes, wheat, oats, barley, sorghum, corn, tapioca and sago. The starch can be used directly after gelatinization for acetone or butanol production or after saccharification with acid or enzymes for a range of products.

(*vii*) Dextrin : Dextrin is the α-amylase enzyme degradation preduct of starch breakdown and has been used for some antibiotic production.

(*viii*) **Inulin :** This is the reserve carbohydrate in Jerusalem artichokes, and chicory representing some 25% of the dry weight. It's enzymatic breakdown to levulin and fructose has allowed its use as a substrate.

(*ix*) **Cellulose :** Cellulose, alongwith lignin, hemicellulose, and pectin are the main components of plant cell walls and as such represents an enormous carbohydrate resource. Sulphite waste liquors, a by-product form wood pulping, has been used as a substrate for ethanol and single-cell protein production.

(*x*) **Methanol :** This has been used as a carbon source for the production of single-cell protein.

(*xi*) **Ethanol :** Traditionally used for vinegar production it has also been used for single-cell protein production, but is now more likely to be used as a fuel in cars.

(*xii*) **Glycerol :** This is often used for antibiotic or steroid fermentations.

(*xiii*) **Carboxylic acids :** Acetic acid has recently been used as an inexpensive carbon source, and succinic acid is used in the production of pectinases.

(*xiv*) **Fats :** Fats and oils are often employed as antifoam agents, but not as sole carbon sources.

(*xv*) **Hydrocarbons :** Methane, *n*-butane and *n*-pentane have been used as a carbon source for single-cell protein production, but only *n*-paraffins have been extensively employed as carbon sources. Commercial media containing $C_{10} - C_{20}$ paraffins have been used for the production of biomass, citric acid and other organic acids.

Nitrogen Source

Next to the carbon source a supply of nitrogen is vital. In laboratory media the nitrogen is normally supplied as ammonium or nitrate salts, but for industrial processes other sources are used.

(*i*) **Urea :** This has been used in the growth of *Candida utilis* but is not stable during sterilization and is thus of limited value.

(*ii*) **Meals :** Soybean meal, cottonseed meal, rapeseed meal, cornmeal, and fish meal are all used as nitrogen sources.

(*iii*) **Cornsteep liquor :** This is a by-product of corn starch processing and is used frequently as a nitrogen source as it contains amino-acids, vitamins, and mineral salts.

(*iv*) **Yeast extracts :** Yeast autolystate are frequently used in laboratory media.

Table : Nitrogen source in fermentation

Nitrogen source	Comments
Salts	
Ammonium	Mainly laboratory use
Nitrate	Mainly laboratory use
Organic	
Urea	Not stable on sterilization
Meals	
Soybean meal	Frequently used
Cottonseed meal	
Repeseed meal	Because of toxicity cannot be used as animal feed, hence used in fermentations
Corn meal	
Corn glutin meal	By product of corn wet milling process
Fish meal	Rarely used
Cornsteep liquior	Great variety in composition
Yeast extracts	laboratory uses
Protein hydrolysates	Mainly used for pharmaceuticals
Distillation residues	Often used for livestock

Inorganic Minerals

It includes macronutrients (e.g.) nitrogen, phosphorous, potassium, calcium, magnesium, sulphur) in the form of salts in large amounts and microelements (e.g. boron, molybdenum copper, zinc, manganess, iron and chloride). For example addition of calcium indirectly improved the yield of streptomycin. The organic phosphate concentration also influences production of bacitracins, citric acid, ergot, monomycin etc. The conentration of manganese, iron and zinc are most critical in secondary metabolism.

Oils and Fats

Oils were first used as carriers for antifoams in antibiotic processes. Vegetable oils (olive, maize, cotton seed, linseed, soyabeen etc.) may also be used as carbon source particularly for their content of fatty acids, oleic, linoleic etc.

Growth Factors

Some common growth factors required are vitamins, amino acids, fatty acids or sterols. The simple glucose ammonium-salts medium will support a surprizing number of organisms.

Amino acids are available individually in technical gardes or can be added as mixed hydrolysates such as casein. Vitamins are often required in very small quantities as they are related to co-enzyme function. For eg. calcium pantothenate has been used in one medium formulation for vinegar production. In process used for production of glutamic acid, limited concetration of biotin must be present in medium.

Buffers

The control of pH may be extremely important it optimal productivity is to be achieved A compound may be added to the medium to serve specifically as a buffer, or may be used a nutrient source. Many media are buffered at about pH 7.0 by the incorporation of calcium carbonate (as chalk). If the pH decreases the carbonate is decomposed. Obviously, phosphates which are part of many media also play an important role in buffering.

The balanced use of the carbon and nitrogen sources will also form a basis for pH control as buffering capacity can be provided by the proteins, peptides and amino acids, such as in corn-steep liquor. The pH may also be controlled externally by addition of ammonia or sodium hydroxide and sulphuric acid.

Inducers

The majority of enzymes which are of industrial interest are inducible. Induced enzymes are synthesized only in response to the presence in the environment of an inducer. Inducers are often substrates such as starch or dextrins for amylases, maltose for pullulanase and pectin for pectinases. Some inducers are very potent, such as isovaleronitrile inducing nitralase.

Specific and general inhibitors used in fermentations

Product	Inhibitor	Main effect	Micro-organism
Glycerol	Sodium bisulphate	Acetaldehyde production repressed	*Saccharomyces cerivisiae*
Tetracycline	Bromide	Chlortetracycline formation repressed	*Streptomyces aureofaciens*
Glutamic acid	Penicillin	Cell wall permeability	*Micrococcus glutamicus*
Citric acid	Alkali metal/phosphate, pH below 2.0	Oxalic acid repressed	*Aspergillus niger*
Valine	Various inhibitors	Various effects with different inhibitors	*Brevibacterium roseum*
Rifamycin B	Di-ethyl barbiturate	Other rifamycins Inhibited	*Nocardia mediterranei*
7-Chloro-6-de methyltetracycline	Ethionine	Affects one-carbon transfer reaction	*Sterptomyces aureofaciens*

Some examples of of industrially important enzyme inducers

Enzyme	Inducer	Micro-organims
α-Amylase	Starch Maltose	*Aspergillus spp, Bacillus subtillis*
Pullulanase	Maltose	*Aerobacter aerogenes*
α-Mannosidase	Yeast mannans	*Streptomyces griseus*
Penicillin acylase	Phenylacetic acid	*Escherichia coli*
Proteases	Various proteins	*Bacillus spp. Streptococcus spp. Streptomyces spp. Streptomyces spp. Asperigillus spp. Mucor spp.*
Cellulose	Cellulose	*Trichoderma viride*
Pectinases	Pectin (beet pulp, apple pomace, citrus peel)	*Aspergillus spp.*
Nitralase	Isovaleronitrile	*Rhodococcus rhodochrous*

Antifoams

In most microbiological processes, foaming is a problem. It may be due to a component in the medium or some factor produced by the micro-organism. The most common cause of foaming is due to proteins in the medium, such as cornsteep liquor, pharma media, peanut meal, soybean meal, yeast extract or meat extract.

The foaming can cause removal of cells from the medium which will lead to autolysis and the further release of microbial cell proteins will probably increases the stability of the foam. If uncontrolled, then numerous changes may occur and physical and biological problems may be created.

Antifoams are surface active agents, reducing the surface tension in the foams and destabilizing proteins films by (a) hydrophobic bridges between two surfaces, (b) displacement of the absorbed protein, and (c) rapid spreading on the surface of the film.

The following compounds, which meet most of these requirements have been found to be most suitable in different fermentations processes :

1. Alcohols.
2. Esters.
3. Fatty acids and derivatives, particularly glycerides, which include cottonseed oil, soy-bean oil, olive oil, castor oil, sunflower oil, rapeseed oil and cod liver oil.
4. Silicones.
5. Sulphonates.

Sterilization

Generally, culturing of micro-organisms needs to be carried out in an aseptic manner. Sterilization is the elimination by removal or killing of all micro-organisms, and the inactivation of viruses in or on a product.

(*a*) Media Sterilization is the destruction or removal of all life forms from the medium.

(*b*) Air Sterilization involves the continuous removal of life forms from the air stream being fed into the fermentor during the course of fermentation.

Media Sterilization

Thermal treatment, in general, is suitable because of its relatively low cost, its effectiveness in achieving an accptable level of sterility, and ease of validation.

Filtration, on the other hand, being comparatively expensive, has the advantage of minimum adverse effects on the medium quality.

Thermal Death Kinetics

The relationship between non-survival rate (death rate) and number of viable cells is

$$-\frac{dN}{dt} = k_d N$$

$$\Rightarrow \qquad -\int_{N_0}^{N} \frac{dN}{N} = \int_0^t k_d dt$$

where, N = number of viable organisms at any given time t

t = time from the start of treatment

k_d = specific death constant

Above equation can be re-written as

$$\log \frac{N_0}{N} = \frac{k_d}{2.303} t$$

or

$$N_0 = N.e^{-kdt}$$

where, N_0 = number of organism at the start of treatment.

Media Sterilization by Removal

A large number of methods have been used in the bioprocess industry for the removal of particulate matter from fluids.

Following table summarizes those most commonly used.

Methods	Type
(*a*) Filtration	Absolute and depth
(*b*) Adsorption	Ion-exchange, electrostatic, charcoal, Affinity,
(*c*) Sedimentation	Centrifugation, Flocculation, Flotation.

Filtration is the only "effective" way of physically removing micro-organisms from fluids. Filtration is particularly useful for sterilizing heat-labile materials.

Thus, sterilization by filtration is quite common when it is essential to process nutrients or products is liquid form. Examples include sterilization of complex organic media such as those used in animal cell culture and final product sterilization.

Media Sterilization By Inactivation

(*i*) Methods of Inactivation

An alternative to physical removal is inactivation of cells by thermal, chemical, or physical means. The goal is to simply kill or Inactivate without necssarily disintegrating the cells.

(*a*) Heat : Heat is the most widely used lethal agent but, steam is by for the best method. Aqueous solution must be sterilized using steam. Small laboratory bioreactors, shake flasks containing medium,

and othher small glass and plastic articles are normally treated with steam in an autoclave. These are metal vessels which can filled with steam at pressure by generation internally or by injection. The steam is normally at a pressure above atmospheric (15 lb) which gives a temperatuure of 121° C.

These methods raise the temperature for a sufficient time to kill all possible contaminants. In general, 15 – 40 min at 121°C (15 *l*b) is sufficient for shake flasks and small bioreactors, but for larger bioreactors some understanding of the conditions are required.

In studying, the effect of temperature on sterilization decimal reduction time (D) is used; D being the time required to reduce the cell population by 10 – fold, i.e.

$$D = \frac{N}{N_0} = \frac{1}{10}$$

Substituting in equation $\left[\log \frac{N_0}{N} = \frac{kt}{2.303}\right]$

$$\Rightarrow \qquad \log 10 = \frac{kD}{2.303} \qquad \text{or} \quad D = \frac{2.303}{k_d}$$

The time required to kill all the cells in a suspension at specific temperature starting at a known cell number, refered as F value:

$$F = nD$$

where n = number of log cycles between start and finish.

For example, in many bioreactors a probability of contamination of 10^{-6} is often used as a baseline. If the initial count of contaminants was 10^{11} and D at 120°C is 1.5 min for the most resistant spores of *Bacillus stearothermophiles*.

then $n = 11 + 6 = 17$

$$F_{120°C} = 17 \times 1.5$$

$$= 25.5 \text{ min.}$$

(*b*) Dry heat : Dry heat is used principally to sterilize glassware or other heat-stable solid materials. Dry heat requires a much longer exposure as heat conduction is less rapid in air than in steam and both spores and vegetative cells are more heat resistant in the absence of moisture.

(*c*) Pasteurization and Tyndallization : Pasteurization is not strictly a method of sterilization but a method making food and beverage safe for consumption. The technique involved heating the substance for a short period at a temperature below those used for sterilization which is sufficient to kill many organisms. For example, the most widespread used in the pasteurization of milk where it is heated at 62° C for 30 minutes.

(*d*) Tyndallization : Involves heating the material at atmospheric pressure to 100°C on successive days. It is useful where a medium is affected by temperatures of above 100°C or where spores are present. Those spores which survive the first heat treatment germinate into vegetative cells which are destroyed by the second heat treatment.

(*e*) Chemicals : Although not used for preparative medium sterilization, a variety of chemicals have been used to disinfect apparatus, glassware, etc. Examples of disinfectants are formaldehyde, phenolics such as lysol or hypochlorite (bleach) are often used to place infected pipettes and glasswares in as means of restricting contamination.

(*f*) Ionizing Radiation : Ultra-violet rays, high-energy electrons and gamma rays have been suggested for media sterilization. Radiation is used to sterilize plastic labware, petri dishes and other similar products on an increasing scale. Ultra-violet lamps are often installed in culture or clean rooms as these are effective in killing airborne micro-organisms but can only be operated in absence of workers.

Del factor

The term $\ln N_0/N$ has also been described as the design criterion or Del or Nabla factor and is represented by ∇. It is a measure of the reduction in organisms at a certain temperature.

$$\nabla = \ln \frac{N_0}{N}$$

The ∇_{total} represents the total Del factor for a whole process, and can be calculated as follows.

If the initial concentration of contaminants N_0 was 10^6 ml^{-1}; the vessel volume was 10^7 ml (10000) and the final level of contaminating organisms should be 1 in 1000 vessel then N should be 1:1000 or 10^{-3}.

$$\nabla_{total} = \ln \frac{N_0}{N} = \ln \frac{10^6 \times 10^7}{10^{-3}} = \ln 10^{16} = 36.8$$

However, any batch sterilization will involve the heating up and cooling down in addition to holding the bioreactor or medium at the required temperature. The heating and cooling of a bioreactor are dependent on equipment design, steam temperature, cool water temperature, and flow rate. The temperature profile can be measured experimentally.

Thus the total sterilization may be broken into three components:

$$\nabla_{total} = \nabla_{heating} + \nabla_{holding} = \nabla_{cooling}$$

Using the Arrhenius relationship, it is possible to calculate the cell death on heating and cooling from the temperature profile. In general the holding time is usually much smaller than the heating and cooling times, but because the holding time is at the maximum temperature the major sterilization occurs at this stage.

Air Sterilization

Large quantities of air are required to pass through a fermentor during the course of a fermentation. This is due to the low solubility of oxygen in water. Typically 0.5 to 1.5 vvm (volume of air per volume of liquid per minute) are used in aerobic fermentations.

Air is a source of potential contamination and must be sterilized before it enters the fermentor. These may also be a need to sterilize exhaust gas from the fermentor in order to maintain containment.

Although heat is sometimes used as means to sterilize the air, filtration is by far the most common method. In contrast to heat sterilization of media where sterilization is based thermal destruction of the most heat sensitue organism (the bacterial spore), air sterilization by filtration is usually based on the removal of the entire population rather than a resistant subset of a population.

(a) Depth Filtration Using Fibrous Materials

There are four primary mechanisms for depth filtration *i.e.*, inertial impaction, interception, diffusion, and sedimentation.

Difficulties Associated With Fibrous Depth Filters

While packed beds of fibrous filters where the standard method of air sterilization earlier, they are used much frequently today. Some of the problems are due to the use of **glass fibers** as the primary medium of columns packing the glass fibers would retain moisture after sterilization and look considerable time to dry; for this reason they often were not sterilized after every fermentation batch. If wet, the pressure drop across the filter was high. Hence not only did the cost of air compression increase, but also the air flow to the fermentor decreased. In addition, if the fibrous filters were wet, contaminants would accumulate in the filter. This would allow passage of bacteria into the fermentor, and thus contamination would follow. Moisture retention can be prevented by keeping the filter hot, but this adds additional cost.

Two of the important consideration in setting up a fibrous filtration system are the issues of **channelling and internal movement** of fibers. Both must be minimized. The integrity of the bed is usually inspected visually during plant shut-down. Some or all of the beds would need to be replaced to remove captured dirt and reduce pressure drop. Glass fibre material is hazardous; hence considerable care is necessary in its handling.

(b) Cartridge Air Filtration

Cartridge air filters have become the standard method of filtration today, and essentially all new facilities install pleated cartridge filters. These consist of preformed filter media is an easy-to handle form and provide highly predictable performance. They features high particle-collection efficiency and good capacity to hold solids.

Cartridge filters can be either

(a) Fibrous media-based depth filters, or

(b) Membrane-based absolute filters.

- Fibrous media-based cartridge filters can be either **wound or pleated.**

  Wound filters create the same effect as a stack of woven cloth, but the yarn is wound around a mandrel rather being interwoven as a cloth.

  These fibers can be made of staple fibers (wool or cotton) or mono filaments.

  Pleated fibers, on the other hand, are fabricated from sheets of porous non-woven fabrics that have been pleated to provide additional surface area and strength. The efficiency of pleated filters is more sensitive to particle size.

- Membrane-based cartridge filters on the other hand, are typically produced by the controlled precipitation of a continuous polymeric sheet from a solvent phase.

- Cartridge filters have a number of **advantage** over conventional fibrous filters.

  (a) They typically have a longer life and are often used for more then 50 sterilization cycles.

  (b) In contrast to most fibrous materials, membranes are typically hydrophobic in anture and reject moisture.

  (c) They are safe to use and are not fiber releasing.

  (d) Their operation cost can be lower, because they do not require heat tracing to keep the filter medium dry.

  (e) An important additonal advantages of cartridge filter is their ease of validation.

Theory of Depth Filter

If it is assumed that if a particle touches a fibre it remains attached to it, and that there is a uniform concentration of particles at any given depth in the filter, then each layer of a unit thickness of the filter should reduce the population entering it by the same proportion; which may be expressed mathematically as :

$$d\,N/\,dx = -KN$$

where N is the concentration of particles in the air at a depth, x in the filter and K is a constant.

On integrating equation over the length of the filter it becomes :

$$\int_{N_0}^{N} \frac{dN}{N} = -\int k dx$$

$$\Rightarrow \quad N/\,N_0 = e^{-Kx}$$

where N_0 is the number of particles entering the filter

and N is the number of particles leaving the filter.

On taking natural logarithms, equation becomes :

$$\ln (N/N_0) = -Kx$$

Equation is termed as the log penetration relationship. The efficiency of the filter is given by the ratio of the number of particle removed to the original number present, thus:

$$E = (N_0 - N)/N_0 \qquad (5.16)$$

where E is the efficiency of the filter.

But : $$(N_0 - N)/N_0 = 1 - (N - N_0) \qquad (5.17)$$

Substituting $$N/N_0 = e^{-Kx}$$

Thus : $$(N_0 - N)/N_0 = 1 - e^{-Kx} \qquad (5.18)$$

and $$E = 1 - e^{-Kx}$$

The log penetration relationship has been used by Humphrey and Gaden (1995) in filter design, by using the concept X_{90}, the depth of filter required to remove 90% of the total number of particles entering the filter; thus :

If N_0 is 10 and x is X_{90} then N would be 1 :

$$\ln (1/10) = -KX_{90}$$

or

$$2.303 \log_{10} (1/10) = -KX_{90}$$

$$2.303 (-1) = -KX_{90}$$

therefore

$$X_{90} = 2.303/K$$

The design of depth filters

It is required to provide a 20-m^3 fermenter with air at a rate of 10 m^3 min^{-1} for a fermentation lasting 100 hours. From an investigation of the filter material to be used, the optimum linear air velocity was shown to be 0.15 m sec^{-1}, at which the value of K was 1.535 cm^{-1}. The dimensions of the filter may be calculated as followed:

The log penetration relationship states that:

$$\ln (N/N_0) = -Kx$$

The air in the fermentation plant contained approximately 200 micro-organisms m^{-3}

Therefore,

N_0 = total amount of air provided × 200

$$N_0 = 10 \times 60 \times 100 \times 200 = 12 \times 10^6 \text{ organisms.}$$

The acceptable degree of contamination is one in a thousand,

Therefore,

$$N = 10^{-3}$$

$$\ln \{10^{-3}/(12 \times 10^6)\} = -Kx,$$

$$\ln 8.33 \times 10^{-11} = -Kx,$$

$$\ln 8.33 \times 10^{-11} = -1.535x,$$

$$x = -23.21/-1.535 = 15.12 \text{ cm.}$$

Therefore, the filter to be used should be 15.12 cm long.

The cross-sectional area of the filter is given by volumetric air flow rate divided by the linear air velocity :

$$\pi r^2 = 10/0.15 \times 60$$

where r is the radius of the filter

$$r = 0.59 \text{ m.}$$

Thus the filter to be employed should be 15.12 cm long and 0.59 m radius.

However, as Humphrey (1960) pointed out, the efficient operation of the filter is dependent on the supply of air at the optimum linear velocity. If the air velocity increases or decreases the value of K will decrease, resulting in a loss of filtration efficiency. Considering the example calcuulation, if the linear air velocity were to drop to 0.03 m sec^{-1}, then the value of K would decline to 0.2 cm^{-1}. The number of organisms which would enter the fermentation in 1 minute at this reduced air-flow rate would be as calculated below :

$$\ln (N/N_0) = -Kx.$$

At a linear air velocity of 0.03 m sec^{-1}, in 1 minute 0.03 × 60 × the cross-sectional area of the filter m^{-3} of air would enter the filter, i.e. 1.98 m^3. At a microbial contamination level of 200 organisms m^{-3} this means that 396 organisms would enter the filter in 1 minute.

Thus :

$$\ln (N/396) = -0.2 \times 15.12,$$

$$N = 19.24.$$

Therefore, 19.24 organisms would have entered the fermenter in 1 minute at the decreased air-flow rate. If the filter had been designed to meet this contingency, then the length would have been :

$$\ln (10^{-3}/396) = -0.2x$$

$$x = 64.4$$

Thus a filter length of 64.4 cm would have been required to have maintained the same probability of contamination over the 1 minute of reduced air flow.

Design of a Fermenter

The main function of a fermenter is to provide a controlled environment for the growth of microorganisms, animal cells or plant cells to obtain a desired product.

Two basis types of fermenters are used:

1. A glass vessel with a round or flat bottom and a top flanged carrying plate. The large glass containers originally used were borosilicate battery jars. All vessels of this type have to be sterilized by autoclaving.
2. A glass cylinder with stainless-steel top and bottom plates. These fermenters may be sterilized in situ, but 30 cm diameter is the upper size limit to safety withstand working pressures.

Basic Functions of a Fermenter

Temperature Control

Normally in the design and construction of a fermenter there must be adequate provision for temperature control which will affect the design of the vessel body. Heat will be produced by microbial activity and mechanical agitation and if the heat generated by these two processes is not ideal for the particular manufacturing process then heat may have to be added to, removed from, the system.

Aeration and Agitation

primary purpose of aeration is to provide microorganisms in submerged culture with sufficient oxygen for metabolic requirements, while agitation should ensure that a uniform suspension of microbial cells is achieved in a homogeneous nutrient medium.

The structural components of the fermenter involved in aeration and agitation are:

(*a*) The agitator (impeller),

(*b*) Stirrer glands and bearings,

(*c*) Baffles,

(*d*) The aeration system (sparger).

(*a*) **The Agitator (Impeller) :** The agitator is required to achieve a number of mixing objectives, e.g. bulk fluid and gas-phase mixing, air dispersion, oxygen transfer, heat transfer, suspension of solid particles and maintaining a uniform environment throughout the vessel contents. Agitators may be classified as disc turbines, vaned discs, open turbines of variables pitch and propellers.

(*b*) **Stirrer Glands and Bearings :** The satisfactory sealing of the stirrer shaft assembly has been one of the most difficult problems to overcome in the construction of fermentation equipment, which can be operated aseptically far long period the stirrer shaft can enter the vessel from the top, or bottom of the vessel. Top entry is most commonly used, but bottom entry may be advantageous if more space is needed on the top plate for entry ports, and the shorter shaft permits higher stirrer speeds to be used by eliminating the problem of the shaft whipping at high speeds.

(*c*) **Baffles :** Four baffles are normally incorporated into agitated vessels of all sizes to prevent a vortex and to improve aeration efficiency. Baffles minimizing microbial growth on the impeller and the fermenter walls.

(*d*) **The Aeration System (Sparger) :** A sparger may be defined as a device for introducing air into the liquid in a fermenter. Three basic types of sparger have been used and may be described as the porous sparger, the orifice sparger (a perfortated pipe) and the nozzle sparger. A combined sparger-agitator may be used in laboratory fermenters.

(*e*) **Foam Control :** In any fermentation is it important to minimize foaming. When foaming becomes excessive, ther is a danger that filter become wet resulting in contamination. There is also the possibility that siphoning will develop, leadint to the loss of all or part of the contents of the fermenter. In certain circumstances antifoams may cause problems with aeration or downstream processing. Foam breakers which break down foam by an impact mechanism created by some type of rotating mechanisms.

(*f*) **pH control :** The metabolism of most micro-organisms results in a change in the pH of the culture medium with a concomitant change in the physiology of the microbe. It is usual practice to continuously monitor the pH of the fermentation broth with a pH probe in situ, and to maintain a fixed pH, by the addition of acid or alkali.

(*g*) **Temperature control :** Microbes produce heat as they metabolize substrates. Exchange coils inside the vessel and/or a vessel jacket are used to counter this metabolic heat. As the fermentation proceeds, the cell density increases, as does the heat output, and cooling water is circulated through the coils or jacket. Coling water is used in an identical way to reduce the temperature of the fresh medium after it has been sterilized.

Different Types of Bioreators

A. Stirred Tank Reactors

Stirred tank bioreactor consist of a cylindrical vessel with a motor-driven central shaft that support one or more impellers. The shaft may alter through the top or the bottom of the reactor vessel. A typical stirred tank reactor in microbial culture vessels are generally provided with four baffles projecting into the vessel from the walls to prevent swirling and vortexing of the fluid. The aspect ratio (i.e. height-to-diameter ratio of the vessel is 3-5. Except in animal cell culture application where aspect ratio do not normally exceed 2. Often, the animal cell culture vessels are unbaffled (especially small-scale reactors) to reduce turbulence that may damage the cells. The number of impeller depends on the aspect ratio. The bottom impeller is located at a distance about 1/3 of the tank diameter above the bottom of the tank. Additional impellers are spaced approximately 1.2 impeller diameter distance apart, the impeller diameter is about 1/3 of the vessel diameter for gas dispersion impellers such as Rushton disc turbines and concave bladed impellers. Larger hydrofoil impellers with diameters of 0.5 to 0.6 times the tank diameter are especially effective bulk mixers and are used in fermenters for highly viscous mycelia broths.

Animal cell culture vessels typically employ a single, large diameter, low-shear impeller such as a marine propeller. Gas is sparged into the reactor liquid below the bottom impeller using a perforated pipe ring sparger with a ring diameter that is slightly smaller than that of the impeller. Alternatively, a single hole sparger may be used.

B. Bubble Column Bioreactor

In bubble column bioreactor usually, the column is cylindrical with an aspect ratio of 4-6 (height-to-diameter). Gas is sparged at the base of the column through perforated pipes, perforated plates, or sintered glass or metal micro-porous spargers. O_2 transfer, mixing and other performance factors are influenced mainly by the gas flow rate and the rhelogical properties of the fluid. Internal such as horizontal perforated plates, vertical baffles may be placed in the vessel to improve mass transfer and modify the basic design.

C. Airlift Bioreactors

In airlift bioreactor, the fluid volume of the vessel is divided into two interconnected zones by means of a baffle or draft tube. Only one of the two zones is sparged with air or other gas. The sparged zone is known as the riser; the zone that receives no gas is the downcomer. The bulk density of the gas-liquid dispersion in the gas-sparged riser tends to be lower than the bulk density in the downcomer, consequently the dispersion are two separate vertical pipes that are interconnected at the top and the bottom to form an external circulation loop. For optimal gas-liquid mass transfer performance, the riser-to-downcomer cross-sectional area ratio should be between 1.8 and 4.3. External loop configuration may be either a concentric draft tube device or a split cylinder.

Airlift bioreactors are highly energy-efficient relative to stirred fermenters, yet the productivities of both types are comparable. Being especially suited to shear-sensitive cultures, airlift devices are often employed in large-scale manufacture of bio pharmaceutical proteins obtained from fragile animal cells. Heat and mass transfer capabilities of airlift reactor are at least as good as those of oter systems, and airlift reactors are more effective in suspending solids than are bubble columns.

D. Enzyme Reactors

A vessel employed to carry out the desired conversion using an enzyme is called ***enzyme reactor.*** Several different types of reactors are available the choice of reactor type depending on the form of enzyme (free or immobilized) to be used, kinetics of reaction, etc., and the scale of operation. The different types of reactors used for enzyme-mediated conversion are as follows

(*i*) stirred tank reactor (STR), (*ii*) membrane reactors, (*iii*) continuous flow reactors, *e.g.*

(*a*) continuous flow stirred tank reactors (CFSTR), (*b*) packed bed reactors (PBR) and (*c*) fluidized bed reactors.

Stirred Tank Reactors : These reactors are simple and consist of a tank containing a stirrer and, usually, fixed baffles to improve mixing. These reactors are used in batch mode, and free enzymes can be employed. Efficient arrangements are provided to maintain temperature, pH, etc. of the reaction mix.

Membrane Reactors : A membrane reactor uses a membrane, *e.g.*, dialysis membrane, to certain the enzyme in a chamber into which the substrate moves and the product moves out. Generally, membrane reactors use a hollow fiber of 200 m diameter with 50 m thick membrane. Each reactor contains hundreds of such fibers into which the enzyme is retained; usually these reactors use soluble enzymes. The substrate is kept in the main chamber of the reactor. These reactors can also be used in a continuous mode, and the substrate flow rate is adjusted to achieve the desired level of reaction.

These reactors are

(*i*) easy to establish, (*ii*) permit the use of more than one enzyme to catalyze a chain of reactions, (*iii*) allow easy replacement of enzymes, and (*iv*)are useful in producing small scale (g to kg) quantites of the product.

The chief limitations of these systems are as follows:

(*i*) regular replacement of membranes adds to cost, and (*ii*) the need for substrate diffusion through the membrne often limits its applications.

Continuous Flow Reactors

Continuous flow reactors are based on immobilized enzymes. They offer the following advantages :

(*i*) greater productivity per unit amount of enzyme, (*ii*) can be used for substrates having low solubility, and (*iii*) uniform product quality, i.e., lack of batch to batch variation.

These reactors may be of the following three types :

(*i*) continuous flow stirred tank reactor, (*ii*) packed bed reactor and (*iii*) fluidized bed reactor.

Continuous Flow Stirred Tank Reactor : The main features of these reactors are as follows :

(*i*) efficient stirring of the tank, (*ii*) use of immobilized enzymes (when immobilized enzymes are not available, enzymes in solution may be used; in such a case an ultrafiltration membrane is used to filter the reactor output to retain the enzyme within the reactor), (*iii*) continuous pumping in of the substrate, which is, in an idealized situation, completely mixed with the reactor contents, and (*iv*) continuous removal of the product (without the enzyme).

The reactor volume is usually 5-10 times the volume of immobilized enzyme. The matrix used for immobilization must be strong and should not disintegrate due to strring. If the matrix is sufficiently dense, particles are small as 10 m in diameter may be used.

Packed Bed Reactor : These reactors contain a settled bed of immobilized enzyme particles; the reaction mix continuously enters from one end and the product moves out from the other end of the reactor. These reactors are like columns, and the degree of reaction, for a fixed flow rate, is proportional to the length of reactor column. A turbulent flow of reaction mix through the reactor column is preferred as it improves mixing and heat transfer. These reactors are preferred in cases of process involving product inhibition, substrate activation, substrate activation, and reaction reversibility. Generally, immobilized enzyme pellets of 1-3 mm diameter are used; the matrix used for immobilization is usually fairly rigid.

Packed bed reactors may become clogged up by colloids or precipitates present in the reaction mix. Temperature of pH is not easily regulated espically in reactors of >15 cm diameter. Channels may form in the reactor bed due to excessive pressure drop, irregular packing or uneven flow stream. It is desirable to use pellets of uniform size and an upward flow of substrate stream.

Fluidized Bed Reactor : In these reactors, the immobilized enzyme particles are fluidized, *i.e.*, the particles become suspended in the substrate stream, by the flow of the substrate stream. Usually, the minimum flow rate needed for fluidization in about 0.2-1 cm/second. If needed, a stream of, preferably, gas

may be used for fluidization. The immobilized enzyme particles are usually quite small, *e.g.*, 20-40 m in diameter, if their density is sufficiently high; otherwise, larger particles have to be used to prevent them from being flown out of the reactor.

These reactors have kinetic properties between continuous flow stirred tank reactors are packed bed reactors. Therefor, they are used for processes that require these intermediate characteristics, *e.g.*, where high conversion is needed, but the substrate stream is colloidal, or the reaction leads to substantial changes in pH and temperature. They are especially suited for reactions, which either use or liberate a gas. Initial fluidization of the bed requires a large power input, and such reactors are difficult to scale up. These reactors are tapered at their upper end to allow a range of flow rates, but very high flow rates cannot be used.

Mass Transfer

(1) Molecular Diffusion

Molecular diffusion is the movement of component molecules in a mixture under the influence of a concentration difference in the system. Diffusion of molecules occurs in the direction required to destroy the concentration gradient. If the gradient is maintained by constantly supplying material to the region of high concentration and removing it from the region of low concentration, diffusion will be continuous.

(a) Diffusion Theory

Consider a system containing molecular component A and B initially, the concentration of A in the system is not uniform; as indicated in Figure 1, concetratin C_A varies from C_{A1} to C_{A2} as a function of distance y, in response to this concentration gradient, molecules of A will diffuse from the region of high concentration until eventually the whole system acquires uniform composition. If there is no large-scale fluid motion in the system, e.g. due to stirring, mixing occurs solely by random molecular movement.

Assume that mass transfer of A occurs across area a perpendicular to the direction of diffusion. In single-phase systems, the rate of mass transfer due to molecular diffusion is given by *Fick's law of diffusion*, which states that mass flux is proportional to the concentration gradient:

$$J_A = \frac{N_A}{a} = -D_{AB}\frac{dC_A}{dy} \qquad ...(1)$$

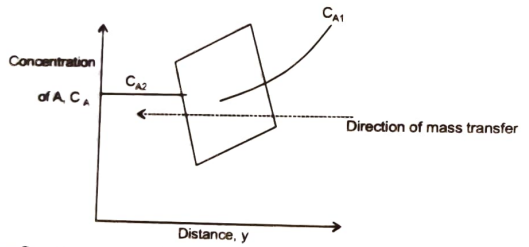


Figure Concentration gradient of component A inducing transfer across area a.

J_A is the mass flux of component A, N_A is the rate of mass transfer of component A, a is the area across which mass transfer occurs, D_{AB} is the binary diffusion coefficient or diffusivity of component A in a mixture of A and B, C_A is the concentration of component A, and *y* is distance, dC_A / dy is the concentration gradient, or change in concentration of A with distance. As indicated in Equation (1) mass flux is defined as the rate of mass transfer per unit area perpendicular to the direction of movement; J_A has units of, g mol s^{-1} m^{-2}.

Corresponding units for N_A is g mol s^{-1}, for C_A g mol m^{-3}, and for D_{AB} m^2 S^{-1}.

(b) Film Theory

The two-Film theory is a useful model for mass transfer between phase. Mass transfer of solute from one phase to another involves transport form the bulk of one phase to the phase boundary or interface, and then from the interface to the bulk of the second phase. The film theory is based on tthe idea that a fluid film or mass-transfer boundary layer forms whereever there is contact between two phases.

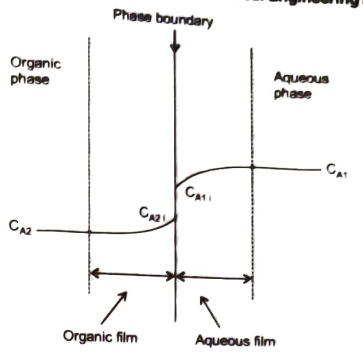


Figure : Film resistance to mass transfer between two immiscible liquids

Let us consider mass transfer of component A across the phase interface represented in Figure Assume that the phases are two immiscible liquids such as water and chloroform, and that A is initially at higher concentration in the aqueous phase then in the organic phase. Each phase is well mixed and in turbulent flow. The concentration of A in the bulk aqueous phase is C_{A1}, the concentration of A in the bulk organic phase is C_{A2}.

According to the film theory, turbulence in each fluid dies out at the phase boundary. A thin film of relatively stagnant fluid exists on either side of the interface; mass transfer through this film is effected solely by molecular diffusion. The concentration of A changes near the interface as indicated in Figure C_{A1i} is the interfacial concentration of A in the aqueous phase; C_{A2i} is the interfacial concentration of A in the organic phase, Most of the resistance of mass transfer resides in the liquid films rather than in the bulk liquid.

(2)Convective Mass Transfer

The term convective mass transfer refers to mass transfer occurring in the presence of bulk fluid motion. Molecular diffusion will occur whenever there is a concentration gradient; however if the bulk fluid is also moving, the overall rate of mass transfer will be higher due to the contribution of convective currents. Analysis of mass transfer is most important in multi-phase systems where interfacial boundary layers provide significant mass-transfer resistance.

Rate of mass transfer is directly proportional to the driving force for transfer, and the area available for the transfer process to take place. This can be expressed as:

$$\text{Transfer rate} \propto (\text{transfer area}) \times (\text{driving force})$$

The proportionality coefficient in this equation is called the mass transfer coefficient, so that :

$$\text{Transfer rate} = (\text{mass-transfer coefficient}) \times (\text{transfer area}) \times (\text{driving force})$$

For each fluid on either side of a phase boundary, the driving force for mass transfer can be expressed in terms of a concentration difference. Therefore, rate of mass transfer to a phase boundary is given by the equation :

$$N_A = ka\,\Delta C_A = ka\,(C_{A0} - C_{Ai}) \qquad (2)$$

Where N_A is the rate of mass transfer of component A, *k* is the mass-transfer coefficient, *a* is the area availlable for mass transfer, C_{A0} is the bulk concentration of component A away from the phase boundary, and C_{Ai} is the concentration of A at the interface. Equation (2) is usually used to represent the volumetric rate of mass transfer, so units of N_A are, for example, g mol m^{-3} s^{-2}. Consistent with this representation, *a* is the interfacial area per unit volume with dimensions L^{-1} and units of, for example, m^2m^{-3} or m^{-1}. The dimensions of the mass transfer coefficient are LT^{-1} Sl units are m s^{-1}. By analogy for heat transfer, Equation (2) can also be written in the form :

$$N_A = \frac{\Delta C_A}{R_m} \text{ and } R_m = \frac{1}{Ka} \qquad (3)$$

Where R_m is the resistance to mass transfer:

Three mass-transfer situations which occur in bio-processing are ***liquid-solid mass transfer, liquid-liquid mass transfer*** between immiscible solvents and ***gas-liquid mass transfer.***

(a) Liquid-Solid Mass Transfer

Let us assume that component A is required for reaction at the surface of a solid. The situation at the interface between flowing liquid containing A and the solid is illustrated in figure Near the interface, the fluid velocity is reduced and a boundary layer develops. As A is consumed by reaction, the local concentration of A at the surface decrease and a concentration gradient is established through the film. The concentration difference between the bulk liquid and the phase interface drives mass-transfrer of A from the liquid to the solid, allowing the reaction to continue. If the solid is non-porous, A does not penetrate further than the surface. The concentration of A at the phase boundary is C_{A_i} the concentration of A in the bulk liquid outside the film is C_{A0}. If a is the liquid-solid interfacial area per unit volume, the volumetric rate of mass transfer can be determined from Equation (2) as :

$$N_A = k_L a(C_{A0} - C_{A_i}) \tag{4}$$

Where k_L is the *liquid-phase mass-transfer coefficient.*

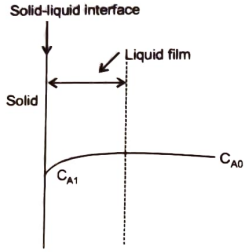


Figure : Concentration gradient for liquid-solid mass transfer.

(b) Liquid-Liquid Mass Transfer

Liquid-liquid mass transfer between immiscible solvents is most often encounterd in the product-recovery stages of bio processing. Organic solvents are used to isolate antibiotics, steroids and alkaloids from fermentation broths; two-phase aqueous systems are used in protein purification. Liquid-liquid mass transfer is also important when hydrocarbons are used as substrates in fermentation, e.g. in production of microbial biomass for single-cell protein.

The situation at the interface between two immiscible liquids is shown in figure Component A is present at bulk concentration C_{A1} in one liquid phase; this concentration falls to C_{A1i} at the interface. In the other liquid, the concentration of A falls from to C_{A2i} at the interface to C_{A2} in the bulk. The rate of mass transfer N_A in each liquid phase can be obtained from Equation (2)

$$N_{A1} = k_{L1} a(C_{A1} - C_{A1i}) \tag{5}$$

and

$$N_{A2} = k_{L2} a(C_{A2} - C_{A2i}) \tag{6}$$

Where, k_L is the liquid-phase mass-transfer coefficient, and subscripts 1 and 2 refer to the two liquid phases.

(c) Gas-Liquid Mass transfer

Gas-liquid mass transfer is paramount importance in bio-processing because of the requirement for oxygen in aerobic fermentations. Transfer of a solute as oxygen from gas to liquid is analysed in a similar way to liquid-liquid and liquid-solid mass transfer.

Figure 4 shows the situation at an interface between gas and liquid phases containing component A. Let us assume that A is transferred from the gas phase into the liquid. The concentration of A in the liquid is C_{AL} in the bullk and C_{AL_i} at the interface. In the gas, the concentration is C_{AG} in the bulk and C_{AG_i} at the interface.

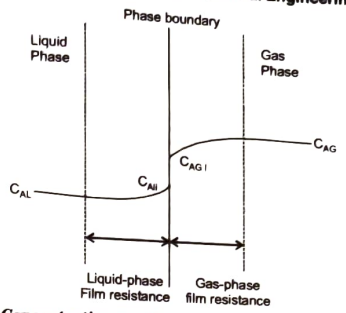


Figure : Concentration gradients for gas-liquid mass transfer.

From Equation (3), the rate of mass transfer of A through the gas boundary layer is:

$$N_{AG} = k_G a(C_{AG} - C_{AG_i}) \tag{7}$$

And the rate of mass transfer of A through the liquid boundary layer is:

$$N_{AL} = k_L a(C_{AG_i} - C_{AG}) \tag{8}$$

Where k_G is the gas-phase mass-transfer coefficient and K_L is the liquid-phase mass-transfer coefficient.

Factors Affecting Cellular Oxygen Demand

The rate at which oxygen is consumed by cells in fermenters determines the rate at which it must be transferred from gas to liquid. Many factors influence oxygen demand; the most important of these are cell species, culture growth phase, and nature of the carbon source in the medium. In batch culture, rate of oxygen uptake varies with time. The reasons for this are two fold. First, the concentration of cells increaes during the course of batch culture and the total rate of oxygen uptake is proportional to the number of cells present. In addition, the rate of oxygen consumption per cell, known as the specific oxygen uptake rate, also varies.

If Q_0 is the oxygen uptake rate per volume of broth and q_0 is the specific oxygen uptake rate :

$$Q_0 = q_0 x \tag{9}$$

Where, x is cell concentration. Typical units for q_0 are g g^{-1} s^{-1}, and for Q_0 g l^{-1} s^{-1}.

Choice of substrate for the fermentation can also significantly affect oxygen demand. Because glucose is generally consumed more rapidly than other sugars or carbons-containing substrates, rates of oxygen demand are higher when glucose is used. For example, maximum oxygen-consumption rates of 5.5, 6.1 and 12 mmol $l^{-1}h^{-1}$ have been observed for Penicillum mould growing on lactose, sucrose and glucose, respectively.

Oxygen Uptake in Cell Cultures/Oxygen transfer from gas bubble to cell

In aerobic fermentation, oxygen molecules must overcome a series of transport resistance before being utilized by the cells. Eight mass-transfer steps involved in transport of oxygen from the interior of gas bubbles to the site of intracellular reaction. They are.

(*i*) Transfer from the interior of the bubble to the gas-liquid interface.

(*ii*) Movement across the gas-liquid interface.

(*iii*) Diffusion through the relatively stagnant liquid film surrounding the bubble.

(*iv*) Transport through the bulk liquid.

(*v*) Diffusion through the relatively stagnant liquid film surrounding the cells.

(*vi*) Movement across the liquid-cell interface.

(*vii*) If the cells are in a floc, clump or solid particle, diffusion through the solid to the individual cell.

(*viii*) Transport through the cytoplasm to the site of reaction.

Note that resistance due to the gas boundary layer on the inside of the bubble has been neglected; becauuse of the low solubility of oxygen in aqueous solutions, we can assume that the liquid-film resistance dominates gas-liquid mass transfer. If the cells are individually suspended in liquid rather than in a clump, step (*vii*) disappears.

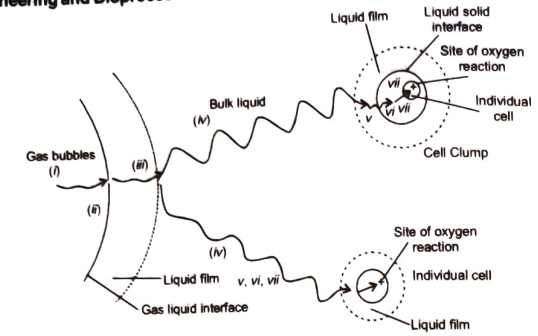


Figure: Step for transfrer of oxygen from gas bubble to cell

For most bioreactors the following analysis is valid :

(*i*) Transfer through the bulk gas phase in the bubble is relatively fast.

(*ii*) The gas-liquid interface itself contributes negligible resistance.

(*iii*) The liquid film arouund the bubbles is a major resistance to oxygen transfer.

(*iv*) In a well-mixed fermenter, concentration gradients in the bulk liquid are minimized and mass-transfer resistance in this region is small. However, rapid mixing can be difficult to achieve in viscous fermentation broths; if this is the case, oxygen-transfer resistance in the bulk liquid may be important.

(*v*) Because single cells are much smaller than gas bubbles, the liquid film surrounding each cell is much thinner than that around the bubbles and its effect on mass transfer can generally be neglected. On the other hand, if the cells form large clumps, liquid-film resistance can be significant.

(*vi*) Resistance at the cell-liquid interface is generally neglected.

(*vii*) When the cells are in clumps, intra-particle resistance is likely to be significant, as oxygen has to diffuse through the solid pellet to reach the interior cells. The magnitude of this resistance depends on the size of the clumps.

(*viii*) Intracellular oxygen-transfer resistance is negligible because of the small distance involved.

When cells are dispersed in the liquid and the bulk fermentation broth is well mixed, the major resistance to oxygen transfer is the liquid film surrounding the gas bubbles. Transport through this film becomes the rate-limiting step in the complete process, and controls the overall mass-transfer rate.

At steady state there can be no accumulation of oxygen at any location in the fermenter; therefore, the rate of oxygen transfer from the bubbles must be equal to the rate of oxygen cosumption by the cells.

$$K_L a(C^*_{AL} - C_{AL}) = q_0 x \qquad (10)$$

$K_L a$ is used to characterize the oxygen mass-transfer capability of fermenters. If $K_L a$ for a particular system is small, the ability of the reactor to deliver oxygen to the cells is limited. We can predict the response of the fermenter to changes in mass-transfer operating conditions using equation (10). For example if the rate of cell metabolism remains unchanged but $K_L a$ is increased, e.g. by raising the stirrer speed to reduce the thickness of the boundary layer around the bubbles, the dissolved-oxygen concentration C_{AL} must rise in order for the left-hand side of equation (10) to remain equal to the right-hand side. Similarly, if the rate of oxygen consumption by the cells accelerates while $K_L a$ is unaffected, C_{AL} must decrease.

$$x_{max} = \frac{K_L a C^*_{AL}}{q_0} \qquad (11)$$

Aeration and Agitation

Aerobic microbial culture must be supplied with oxygen during growth at a rate sufficient to satisfy the organisms demand. The oxygen demand of an industrial fermentation process is normally satisfied by aerating and agitating the fermentation broth. However, the productivity of many fermentations is limited by oxygen availability and, therefore, is important to consider the factors which affect a fermenter's efficiency in supplying microbial cells with oxygen.

Oxygen Requirements

Maximum biomass production may be achieved by satisfying the organism's maximum specific oxygen demand by maintaining the dissolved oxygen concentration greater than the critical level. If the dissolved oxygen concentration were to fall below the critical level then the cells may be metabolically disturbed. However, it must be remembered that it is frequently the objective of the fermentation technologist to produce a product of the microorganism rather than the organism itself and that metabolic disturbance of the cell by oxygen starvation may be advantageous to the formation of certain products.

Equally, the provision of a dissolved oxygen concentration far greater than the critical level may have no influence on biomass production, but may stimulate product formation. Thus, the aeration conditions necessary for the optimum production of a product may be different from those favoring biomass productions.

Oxygen Supply

Oxygen normally supplied to microbial cultures in the form of air, this being the cheapest available source of the gas. The method for aeration of a culture with a supply of air varies with the scale of the process :

(*i*) Laboratory-scale cultures may be aerated by means of the shake-flask technique where the culture (50 to 100 cm^3) is grown in a conical flask (250 to 500 cm^3) shaken on a platform contained in a controlled environment chamber.

(*ii*) Pilot-and industrial-scale fermentations are normally carried out in stirred, aerated vessels, termed fermenters. It is often advantageous to culture relatively small volumes (1 dm^3) in a sttirred, aerated vessel as this enables the cultural conditions to be better monitored and controlled, and facilitates the addition of supplements and the removal of samples.

Bartholomew et al. (1950) represented the transfer of oxygen from air to the cell, during fermentation, as occurring in number of steps :

(*i*) The transfer of oxygen from an air bubble into solution.

(*ii*) The transfer of the dissolved oxygen through the fermentation medium to the microbial cell.

(*iii*) The uptake of the dissolved oxygen by the cell.

These workers demonstrated that the limiting step in the transfer of oxygen from air to the cell in *Streptomyces griseus* fermentation was the transfer of oxygen into solution.

The rate of oxygen transfer from air bubble to the liquid phase may be described by the equation :

$$d\,C_L/dt = K_L a(C^* - CL) \qquad (1)$$

Here, C_L is the concentration of dissolved oxygen in the fermentation broth (m moles dm^3), t is time (hours), dC_L/dt is the change in oxygen concentration over a time period, i.e., the oxygen transfer rate (m moles O_2 dm^{-3} h^{-1}),

K_L is the mass transfer coefficient (cm h^{-1}),

a, is the gas/liquid interface area per liquid volume (cm^2 cm^{-3}),

C^* is the saturated dissolved oxygen concentration (m moles dm^{-3})

K_L may be considered as the sum of the reciprocals of the resistance to the transfer of oxygen from gas to liquid and ($C^* - C_L$) may be considered as the 'driving force' across the resistances. It is extremely difficult to measure both K_L and 'a' in a fermentation seperately and, therefore, the two terms are generally combined in the term $K_L a$, the volumetric mass-transfer coefficient, the units of which are reciprocal time (h^{-1}). The volumetric mass-transfer coefficient is used as a measure of the aeration capacity of a fermenter. The larger the $K_L a$, the higher the aeration capacity of the system, the $K_L a$, value will depend upon the design and operating conditions of the fermenter and will be affected by such variables as aeration rate, agitation rate and impeller design.

Determination of $K_L a$ Values

The determination of the $K_L a$ of a fermenter is essential in order to establish its aeration efficiency and to quantify the effects of operating variables on the provision of oxygen. The section considers the merits and limitations of the methods available for the determination of $K_L a$ values. It is important to remember at this stage that dissolved oxygen is usually monitored using a dissolved oxygen electrode, which records dissolved oxygen activity or dissolved oxygen tension (DOT) whilst the equation describing oxygen transfer are based on dissolved oxygen concentration. The solubility of oxygen is affected by dissolved solutes so that pure water and a fermentation medium saturated with oxygen would have different dissolved oxygen concetration yet have the same DOT, i.e. and oxygen electrode would record 100% for both. Thus, to translate DOT into concentration the solubility of oxygen in the fermentation medium must be known and this can present difficulties.

(a) The Sulphite Oxidation Technique

Copper et al. (1944) were the first to describe the determination of oxygen-transfer rates in aerated vessels by the oxidation of sodium sulphite solution. This technique does not require the measurement of dissolved oxygen concentration but relies on the rate of conversion of a 0.5 M solution of sodium sulphite to sodium sulphate in the presence of a copper or cobalt catalyst: $Na_2SO_3 + 0.5O_2 = Na_2SO_4$

The rate of reaction is such that as oxygen enters solution it is immediately consumed in the oxidation of sulphite, so that the sulphite oxidation rate is equivalent to the oxygen-transfer rate. The dissolved oxygen concentration, for all practical purposes, will be zero and the $K_L a$ may then be calculated from the equation :

$$OTR = K_L a.C^* \qquad (2)$$

(where OTR is the oxygen transfer rate)

(b) Gassing-out Techniques

The estimation of the $K_L a$ of a fermentation system by gassing-out techniques depends upon monitoring the increases in dissolved oxygen concentration of solution during aeration and agitation. The oxygen transfer rate will decrease during the period of aeration as C_L approaches C^* due to the decline in the driving force $(C^* - C_L)$. The oxygen transfer rate, at any one time, will be equal to the slope of the tangent to the curve of values of dissolved oxygen concentration against time of aeration.

(i) The static method of gassing out

In this technique, the oxygen concentration of the solution is lowered by gassing the liquid out with nitrogen gas, so that the solution is 'scrubbed' free of oxyygen. The deoxygenated liquid is then aerated and agitated and the increases in dissolved oxygen monitored using some form of dissolved oxygen probe. The increase in dissolved oxygen concentration has already been described by equation (1), i.e.: $\frac{d_{CL}}{dt} = k_L a\left(C^* - C_L\right)$

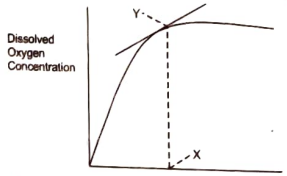
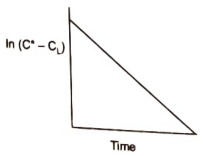


Figure: The increase in dissolved oxygen concentration of a solution over a period of aeration. The oxygen transfer rate at time X is equal to the slope of the tangent at point Y.

Integration of equation (1) yield :

$$\ln (C^* - C_L) = - K_L at \qquad (3)$$



A plot of the $\ln (C^* - C_L)$ against time of aeration, the slope of which equals $- K_L a$.

Thus, a plot of $\ln (C^* - C_L)$ against time will yield a straight line of slope $-K_L a$, as shown in figure. This technique has the advantage over the sulphite oxidation method in that it is very rapid (normally taking up 15 minutes)

(ii) The dynamic method of gassing out

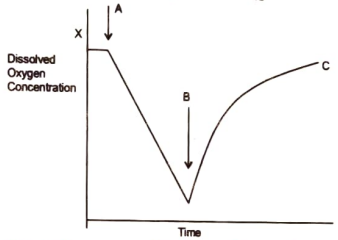
The respiratory activity of a growing culture in the fermenter to lower the oxygen level prior to aeration. Therefore, the estimation has the advantage of being carried out during a fermentation, which should give a more realistic assessment of the fermenter's efficiency. Because of the complex nature of fermentation broths the probe used to monitor the change in dissolved oxygen concentration must be of the membrane-covered type, the procedure involves stopping the supply of air to the fermentation, which results in a linear decline in the dissolved oxygen concentration due to the respiration of the culture. The slope of the line AB in figure is a measure of the respiration rate of the culture. At point B the aeration is resumed and the dissolved oxygen concentration is the difference between the transfer of oxygen into solution and the uptake of oxygen by the respiring culture as expressed by the equation :

$$dC_L/dt = K_L a(C^* - C_L) - xQ_{o2} \qquad (4)$$

Where, x is the concetration of biomass and

xQ_{o2} is the specific respiration rate (m moles of oxygen g^{-1} biomass h^{-1})

$$C_L = 1 / K_L a(dC_L/dt) + xQ_{02} + C^* \qquad (5)$$



Thus, from equation (5), a plot of C_L versus $dC_L/dt + xQ_{o2}$ will yield a straight line, the slope of which will equal-1 / $K_L a$, as shown in Figure. This technique is convenient in that the equations may be applied using DOT rather than concentration because it is the rates of transfer and uptake that are being monitored so that the percentage saturation readings generated by the electrode be used directly.

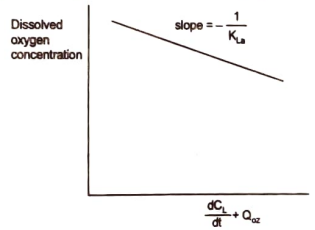


Figure : The dynamic method for determination of $K_L a$ values.

Theory of Mixing

The object of mixing in a gas/liquid/solid system are to:

(a) Create small bubbles and to increase the interfacial area.
(b) Disperse the bubbles throughout the liquid.
(c) Keep the bubbles in the liquid by recirculation for a sufficient time.
(d) Maintain uniform conditions in the vessel and provide turbulent eddies to feed liquid to and from the interfaces.
(e) Maintain the solid in suspension and in particular prevent sedimentation on the bottom of the vessel.
(f) Maintain a uniform temperature in the vessel.

Characterization of Agitation

The following treatment of agitation is restricted to fluids that approximate to Newtonian liquids.

As mixing is extermely complex, the variables involved are lumped together into dimensionless groups to obtain correlations that describes the system. The **Reynolds number** is to characterize the flow

$$Re = \frac{D_i^2 N\rho}{\mu}$$

where D_i = impeller diameter, m

N = impeller speed, Hz

ρ = fluid density, kg/m^3

= fluid viscosity, kg/m/s.

Fully turbulent flow exists above a Reynolds number of 10^4, while fully laminar flow exists below 100, and in between is the transition region. Another group which is used to characterize mixing in a vessel which takes into account gravitational effect is the **Froude number** :

$$Fr = \frac{N^2 D_i}{g}$$

where g = gravitational acceleration, m/s^2.

A third group which relates to the energy required by the agitator is the **Power number** :

$$P_o = \frac{P}{N^3 D_i^5 \rho}$$

Where P = agitator power

The power number has been correlated with Reynold's number for several types of agitator. The power refers to the shaft power and so allowances must be made for electrical and friction losses when calculating motor power input.

In general, the laminar region, where K_1 is a constant

$$P_o = K_1 Re^{-1}$$

and in the turbulent region the power number is independent of the Reynolds number. Under these conditions

$$P_o = K_2 N^3 D_i^5 \rho$$

and the power is independent of the viscosity.

It is also useful to define another number which relates to the flow rate generated by the impeller. The action of an agitator blade can be likened to a caseless pump. The power input can be written as

$$P = \rho Q g H'$$

where Q = flow rate created by the impeller

H' = the head which is dissipated on circulation through the vessel.

With low-viscosity fluids, the head is equivalent to the turbulence generated, which is most intense in the region of the impeller and decreases considerably in the outer area of the vessel.

Substituting the power term into the expression for the power number,

$$P_o = \left(\frac{Q}{ND_i^3}\right)\left(\frac{gH'}{N^2 D_i^2}\right)$$

The resulting dimensionless groups are $\frac{gH'}{N^2 D_i^2}$, the **head number**, and $\frac{Q}{ND_i^3}$, the **flow number, Fl**.

In an aerated system, the effective density of the mixture is reduced, which affects the power requirement, and so it is useful to define a **modified flow number** as follows,

$$Fl_g = \frac{Q_g}{ND_i^3}$$

Where Q_g is the gas flow in the reactor, m^{3}s^{-1}.

Fluid Rheology

Fluids may be described as Newtonian or non-Newtonian depending on whether their rheology (flow) characteristics obey Newton's law of viscous flow. Consider a fluid contained between two parallel area A and distance x apart. If the lower plate is moved in one direction at a constant velocity, the fluid adjacent to the moving plate will move in the same direction and impart some of its momentum to the 'layer' of liquid directly above it causing it, also, to move in the same direction at a slightly lower velocity.

Newton's law of viscous flow states that the viscous force, F, opposing motion at the interface between the two liquid layers, flowing with a velocity gradient of dv/dx, is given by the equation :

$$F = A(dv/dx)$$

Where is the fluid viscosity, which may be considered as the resistance of the fluid to flow.

then $$F/A = (dv/dx)$$

F/A is termed the shear stress (τ) and is the applied force per unit area, dv/dx is termed the shear rate (γ) and is the velocity gradient. Thus :

$$\tau = \gamma$$

Equation conforms to the general relationship

$$\tau = K\gamma^n$$

Where, K is the consistency coefficient and n is the flow behavior index or power law index.

For a Newtonian fluid n is 1 and the consistency coefficient is the viscosity, which is the ratio of shear stress to shear rate. Thus, a plot of shear stress against shear rate, for a Newtonian fluid, would produce a straight line, the slope of which would equal the viscosity. Such a plot is termed a rheogram.

Bingham Plastic Rheology

Bingham plastics are similar to Newtonian liquids apart from the fact that shear rate will not increases until a threshold shear stress is exceeded. The threshold shear stress is termed the yield stress or yield value τ_0. A linear relationship of shear stress to shear rate is given once the yield stress is exceeded and the slope of this line is termed as the coefficient of rigidity or the plastic viscosity. Thus, the flow of a Bingham plastic is described by the equation :

$$\tau = \tau_0 + n\gamma$$

Where n is the coefficient of rigidity and τ_0 is the yield stress.

Pseudoplastic Rheology

The apparent viscosity of a pseudoplastic liquid decreases with increasing shear rate. Most polymer solutions behave as pseudoplastics. The decrease in apparent viscosity is by the long chain molecules tending to align with each other at high shear rates resulting in easier flow. The flow of a pseudoplastic liquid may be described by the power law model, i.e.:

$$\tau = K(\gamma)^n$$

K has the same units as viscosity and may be taken as the apparent viscosity. The flow behaviour index is less than unity for a pseudoplastic liquid, smaller the value of n, greater the flow characteristic of the liquid deviate from those of a Newtonian fluid. Equation may be converted to the logarithmic form as:

$$\log \tau = \log K + n \log\gamma$$

Thus, a plot of log shear stress against log shear rate will produce a straight line, the slope of which will equal the flow-behaviour index and intercept on the shear axis stress axis will be equal to the logarithm of the consistency coefficient.

Dilatants Rheology

The apparents viscosity of a dilatants liquid increases with increasing shear rate. The flow of a dilatants liquid may also be described by equation as pseudoplastic but in this case the value of the flow-behaviour index is greater than 1, greater the value greater the flow characteristics deviate from those of a Newtonian fluid. Thus, the values of K and n may be obtained from a plot of log shear stress against log shear rate.

Casson Body Rheology

Casson described a type of non-Newtonian fluid, termed as Casson body, which behaved as a pseudoplastic in that the apparent viscosity decreased with increasing shear rate but displayed a yield stress, and therefore, also resembled a Bingham plastic. The flow characteristics of a casson body may be described by the following equation :

$$\sqrt{\tau} = \sqrt{\tau_0} + K_c \sqrt{\gamma}$$

Wherer K_c is the casson viscosity.

A plot of $\sqrt{\tau}$ against $\sqrt{\gamma}$ will give a straight line, the slop of which will equal to the Casson viscosity and the intercept of the $\sqrt{\tau}$ against $\sqrt{\tau_0}$.

Role et al. claimed that the rheology of a penicillin broth could be best described in terms of a Casson body.

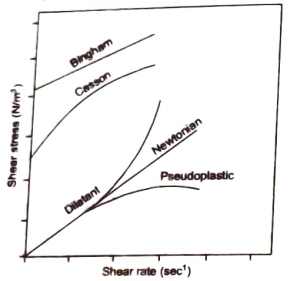


Fig. Rheological Characteristics of Fluids

Heat Transfer

Thermal energy is related to the temperature of matter. For a given material and mass, the higher the temperature, the greater its thermal energy. Heat transfer is a study of the exchange of thermal energy through a body or between bodies which occurs when there is a temperature difference. Heat always transfers from hot to cold.

Modes of Heat Transfer

There are three modes of heat transfer: conduction, convection, and radiation. Any energy exchange between bodies occurs through one of these modes or a combination of them.

1. Conduction

Conduction is heat transfer through solids or stationery fluids. Two mechanisms explain how heat is transferred by conduction : lattice vibration and particle collision.

In solids, atoms are bound to each other by a series of bonds, analogous to springs. When there is a temperature difference in the solid, the hot side of the solid experiences more vigorous atomic movements. The vibrations are transmitted through the springs to the cooler side of the solid. Eventually, they reach an equilibrium, where all the atoms are vibrating with the same energy.

Solids, especially metals, have free electrons, which are not bound to any particular atom and can freely move about the solid. The electrons in the hot side of the solid move faster than those on the cooler side. As the electrons undergo a series of collisions, the faster electrons give off some of their energy to the slower elctrons. Eventually, through a series of random collisions, an equilibrium is reached, where the electrons are moving at the same average velocity. Conduction through electron collision is more effective than through lattice vibration; this is why metals generally are better heat conductors than ceramic materials, which do not have many free electrons.

In fluids, conduction occurs through collisions between freely moving molecules. The mechanism is identical to the electron collisions in metals.

The effectiveness by which heat is transferred through a material is measured by the thermal conductivity, k. A good conductor, such as copper, has a high conductivity; a poor conductor, or a insulator, has a low conductivity. Conductivity is measured in watts per meter per Kelvin (W/mK). The rate of heat transfer by conduction is given by:

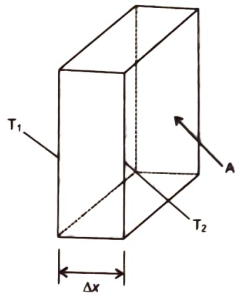


Figure : Heating curve of water

$$q_{conduction} = -kA\frac{\Delta T}{\Delta x}$$

where A is the cross-sectional area through which the heat is conducting, T is the temperature difference between the two surfaces separated by a distance Δx. In heat transfer, a positive q means that heat is flowing into the body, and a negative q represents heat leaving the body. This is also known as Fourier's law.

2. Convection

Convection uses the motion of fluids to transfer heat. In a typical convective heat transfer, a hot surface heats the surrounding fluid, which is then carried away by fluid movement such as wind. The warm fluid is replaced by cooler fluid, which can draw more heat away from the surface. Since the heated fluid is constantly replaced by cooler fluid, the rate of heat transfer is enhanced.

Natural convection (or free convection) refers to a case where the fluid movement is created by the warm fluid itself. The density of fluid decrease as it is heated; thus, hot fluids are lighter than cool fluids. Warm fluid surrounding a hot object rises, and is replaced by cooler fluid. The result is a circulation of air above the warm surface.

Convection coefficient, h, is the measured in W/m²K, and is determined by factors such as the fluid density, viscosity, and velocity.

$$q_{convection} = -hA(T_{surface} - T_\infty)$$

where A is the surface area of the object, $T_{surface}$ is the surface temperature, and T_∞ is the ambient or fluid temperature.

3. Radiation

Radiative heat transfer does not require a medium to pass through; thus, it is the only form of heat transfer present in vacuum. It uses electromagnetic radiation (photons), which travels at the speed of light and is emitted by any matter with temperature above 0 degrees Kelvin (–273 °C). Radiative heat transfer occurs when the emitted radiation strikes another body and is absorbed.

The amount of radiation emitted by an object is given by:

$$q_{emitted} = \varepsilon\sigma AT^4$$

where A is the surface area, T is the temperature of the body, σ is a constant called *stefan Boltzmann constant*, equal to $5.67 \cdot 10^{-8}$ W/m²K⁴, and ε is a material property called *emissivity*. The emissivity has a value between zero and 1, and is a measuure of how efficiently a surface emits radiation. It is the ratio of the radiation emitted by a surface to the radiation emitted by a perfect emitter at the same temperature.

Heat Transfer Coefficient

The **heat transfer coefficient,** in thermodynamics is used in calculating the heat transfer, typically by convection or phase change between a fluid and a solid :

$$h = \frac{q}{A.\Delta T}$$

where q = heat flow in input or lost heat flow, J/s = W,

h = heat transfer coefficient, W/m^2K,

A = heat transfer surface area, m^2,

ΔT = difference in temperature between the solid surface and surrounding fluid area, °K.

From the above equation, the heat transfer coefficient is the proportional coefficient between the heat flux that is a heat flow per unit area, q/A, and the thermodynamic driving force for the flow of heat (i.e., the temperature difference, ΔT).

The heat transfer coefficient has SI units in watts per meter square-kelvin: $W/(m^2K)$. Heat transfer coefficient is the inverse of thermal insulance.

Combining heat transfer coefficients

For two or more heat transfer processes acting in parallel, heat transfer coefficients simply add :

$$h = h_1 + h_2 + \ldots$$

For two or more heat transfer processes connected in series, heat transfer coefficients add inversely.

$$\frac{1}{h} = \frac{1}{h_1} + \frac{1}{h_2} + \cdots$$

For example, consider a pipe with a fluid flowing inside. The rate of heat transfer between the bulk of the fluid inside the pipe and the pipe external surface is:

$$q = \left(\frac{1}{\frac{1}{h} + \frac{t}{k}}\right).A.\Delta T$$

where q = heat transfer rate (W),

h = heat transfer coefficient ($W/(m^2.K)$),

t = wall thickness (m),

k = wall thermal conductivity (W/m.K),

A = area (m^2), ΔT = difference in temperature.

Overall Heat Transfer Coefficient

The overall heat transfer coefficient U is a measure of the overall ability of a series of conductive and convective barriers to transfer heat. It is commonly applied to the calculation of heat transfer in heat exchangers, but can be applied equally well to other problems.

For the case of a heat exchanger, U can be used to determine the total heat transfer between the two streams in the heat exchanger by the following relationship :

$$q = UA\Delta T_{LM}$$

where q = heat transfer rate (W),

U = overall heat transfer coefficient ($W/m^2.K$)),

A = heat transfer surface area (m^2),

ΔT_{LM} = log mean temperature difference (K).

The overall heat transfer coefficient takes into account the individual heat transfer coefficients of each stream and the resistance of the pipe material. It can be calculated as the reciprocal of the sum of a series of thermal resistances (but more complex relationships exist, for example when heat transfer takes place by different routes in parallel) :

$$\frac{1}{UA} = \Sigma\frac{1}{hA} + \Sigma R$$

where R = Resistance(s) to heat flow in pipe wall (K/W)

The heat transfer coefficient is the heat transferred per unit area per kelvin. Thus area is included in the equation as it represents the area over which the transfer of heat takes place. The areas for each flow will be different as they represent the contact area for each fluid side.

The thermal resistance due to the pipe wall is calculated by the following relationship:

$$R = \frac{x}{k.A}$$

where x = the wall thickness (m),

k = the thermal conductivity of the material (W/(m.K)),

A = the total area of the heat exchanger (m^2).

This represents the heat transfer by conduction in the pipe. The thermal conductivity is a characteristic of the particular material. The *convection heat transfer coefficent* for each stream depends on the type of fluid, flow properties and temperature properties.

Scale-up

Generally, fermenters maintain a height-to-diameter ratio of 2 to 1 or 3 to 1. If the height-to-diameter ratio remains constant, then the surface-to-volume ratio decreases dramatically during scale-up. This change decreases the relative contribution of surface aeration to oxygen supply and dissolved-carbon-dioxide removal in comparison to the contribution from sparging.

Scale-up based on constant power input (P_g/V), constant liquid circulation rate inside the vessel (pumping rate of impeller per unit volume, Q/V), constant shear at impeller tip (ND_i), and constant Reynolds number ($ND_i^2\rho/$) can be calculated. Note that $P \propto N^3D_i^5$, $V \propto D_i^3$, $Q \propto ND_i^3$, $P/V \propto N^3D_i^2$ and $Q/V \propto N$. Thus, fixing N and D_i fixes all the quantities. Since these quantitites have different dependencies on N and D_i, a change of scale must result in changes in the physcial environment that the cells experience. Thus, different scale-up rules (constant P/V implies constant OTR, constant R_e implies geometrically similar flow patterns, constant N to give constant mixing times, and constant tip speed to give constant shear) can give very different results.

Scale-up Example

Consider the scale-up of a fermentation from a 10 1 to 10,000 1 vessel. The small fermenter has a height-to-diameter ratio of 3. The impeller diameter is 30% of the tank diameter. Agitator speed is 500 rpm and three Rushton impellers are used. Determine the dimensions of the large fermenter and agitator speed for:

(*a*) Constant P/V

(*b*) Constant impeller tip speed

(*c*) Constant Reynolds number

Assume geometric similarity

Solution : Assume the vessel is a cylinder.

Thus $V = (\pi/4)D_i^2H.$

but $H = 3\ D_i,$

So $V = (\pi/4)\ D_i^3 = 10{,}000\ cm^3$

Solving for D_i, H, and $D_i = 0.3\ D_i$ gives:

$D_i = 16.2$ cm

$H = 48.6$ cm

$D_i = 4.8$ cm

The scale-up factor is the cube root of the ratio of tank volumes, or

$$(10{,}0001/10\ 1)^{1/3} = 10$$

To maintain geometric simularity the larger vessel will have its dimensions increased by a factor of 10. That is:

$D_i = 1.62$ m

$H = 4.86$ m

$D_i = 0.48$ m

(*a*) For constant P/V to apply $N^3D_t^2$ must be the same in both vessels. Let subscript 1 refer to the small vessel and 2 to the large vessel.

$$N_2 = N_1\left(\frac{D_{t1}}{D_{t2}}\right)^{2/3} = 500\,\text{rpm}(1/10)^{2/3} = 107\,\text{rpm}$$

(*b*) For constant tip speed to apply ND_t must be the same in both vessels.

$$N_2 = N_1\left(\frac{D_{t1}}{D_{t2}}\right) = 500\,\text{rpm}(1/10) = 50\,\text{rpm}$$

(*c*) For constant Re to apply ND_t^2 must be the same in both vessels.

$$N_2 = N_1\left(\frac{D_{t1}}{D_{t2}}\right)^2 = 500\,\text{rpm}(1/10)^2 = 5\,\text{rpm}$$

Scale-up on the basis of constant P/V would be the most likely choice unless the culture was unusually shear sensitive. Scale-up based on constant Re would not be used.

Downstream Processing

The various processes used for the actual recovery of useful products from a fermentation or any other inductrial process together constitute ***downstream processing.*** The cost of downstream processing (DSP) is often more than 50% of the manufacturing cost, and there is product loss at each step of DSP. In addition, the product is either present in the cells, in the medium or both. In either case, the concentration of product is usually rather low, and it is generally mixed with other molecules from which it has to be separated. Therefore, the DSP should be efficient, involve as few steps as possible to avoid product loss, and be cost-effective. The various steps in DSP are as follows :

(*i*) separation of particles,

(*ii*) disintegration of cells,

(*iii*) extraction,

(*iv*) concentration,

(*v*) purification, and (*vi*) drying.

Separation of Particles

The first step in DSP is the separation of solids, usually cells, from the liquid medium. This is generally achieved as follow :

Filtratiion : It is used for the separation of filamentous fungi and filamentous bacteria, *e.g.*, *Streptomycetes*, and often for yeast flocks. Filtration uses pressure created by overpressure or vacuum, and its rate depends on filter area, fluid viscosituy and the resistance generated by filter-cake, which increaes with time. The various techniques of filtration employed are as follows: surface filtration, depth filtration, centrifugal filtration, cross-flow filtration, rotary drum vacuum filtration. ***Rotary drum vaccum filters*** are the most commonly used.

Centrifugation : It may be used to separate bacteria and, usually, protein precipitates. But difficulties arise due to small differences in the densities of the particles and the medium. In addition, equipment cost, power consumption, temperature, etc. are the other disadvantages.

Flocculation and Floatation : Small bacterial cells, which are difficult to separate even by centrifugation, can be recovered as follows. ***Flocculation,*** *i.e.*, sticking together of cells, can be induced by inorganic salts, mineral hydrocolloids and organic polyelectrolytes. Since sedimentation rate of a particle increases with size, flocculated cells can be recovered by centrifugation. In cases, where flocculation is not effective, very fine gas bubbles can be created by sparging, release of overpressure or electrolysis. In any case, the gas bubbles adsorb to and surround the cells, raising them to the surface of medium in form of foam (***floatation***); long chain fatty acids or amines promote stable foam formation. The cells collected in the foam are readily recovered. Flocculation and floatation are used for the most efficient recovery of microbial biomass in some single cell protein production systems.

Disintegration of Cells

Disruption of microbial cells is usually difficult due to their small size, strong cell wall and high osmotic pressure inside cells. Generally, cell disruption is achieved by mechanical means, lysis or drying. The method of cell disruption must not damage the product of interest; the suitability of various methods is usually assessed in terms of recovery of a cellular enzyme activity following cell disruption.

Mechanical Cell Disruption : This approach uses shear, *e.g.*, grinding in a ball mill, colloid mill, etc., pressure and pressure release, *e.g.*, homogenizer, and ultrasound. A widely used method is as follows: the cell suspension is forced through a fine nozzle; the cells disintegrate due to hydrodynamic shear and caviation.

Drying : The cells may be dried, *e.g.*, by adding the cells into a large excess of cold acetone and subsequently extracted using buffer or salt solutions. Drying induces changes in cell wall structure, which facilitate extraction. This method is widely used.

Lysis : Microbial cells may be lysed by chemical means, *e.g.*, salts or surfactants, osmotic shock, freezing, or by lytic enzymes, *e.g.*, lysozyme, etc.

In general, recovery of enzyme as product is the best following cell disruption using enzymes or ultrasound, followed by thermal and osmotic methods, while mechanical methods are the least desirable. However, ultrasound method is confined to laboratory mainly due to difficulties in heat removal on a large scale.

Extraction

The process of recovering a compound or a group of compound from a mixture or from cells into a solvent phase is called ***extraction.*** Extraction usually achieves both separation as well as concentration of the product. It is especially useful for the recovery of lipophilic substances, and in antibiotic recovery; it is often an early step after cell separation.

Liquid-Liquid Extration : It employs two immiscible liquids into which the product is differentially soluble. Usually, successively smaller volumes of the solvent are used for repeated extration of a given sample; back-extraction also tends to increase the selectivity of extraction. The extraction may be performed in a single step, by multi-stage parallel-flow extraction, or by counter-current extraction (most complex but most effective).

Whole Broth (Medium + cells) Extraction : It should be used wherever possible since it reduces the number of steps as well as product loss. The effectiveness of extraction may, however, be reduced due to the presence of cells.

Aqueous Multiphase Extraction : It is used for separation of enzymes from cells/cell debris. The enzymes are extracted in an aqueous polyethylene glycol-dextran mixture, which form separate phase. Recovery of enzymes from these phases is rather easy and free from some of the dificulties encountered in centrifugation.

Concentration

Some concentration of the product may occur during the extraction step. Further concentration may be achieved by the following :

(*i*) evaporation

(*ii*) membrane filtration

(*iii*) ion exchange methods and

(*iv*) adsorption methods.

Evaporation : It is generally used in cases of solvent extration using various devices, *e.g.*, continuous flow evaporators, falling film evaporators, thin film evaporators, centrifugal thin film evaporators and spray-dryers. Efficient arrangements must be made for recovery of the evaporated solvent to reduce costs. For low grade products, often evaporation of the whole broth is undertaken using a spray-drier.

Membrane Filtration : It generally achieves both concentration and separation of the products, usually, bases on the size of molecules. The different processes of membrane filtration are as follows: microfiltration, ultrafiltration, reverse osmosis, and electrodialysis. Micro- and ultrafiltrations work as sieves and separate molecules of different sizes, but reverse osmosis can separate molecules of similar size. Microfiltration can be used for cell separation as well.

Ion Exchange Resins : These are polymers having firmly attached ionizable groups (anions or cations), which ionize under a suitable environment. These may be solid, *e.g.*, dextran, cellulose, polyamine, acrylate, etc., or liquid, e.g., a solvent carrying a functional group like phosphoric acid mono-or diester, etc. Solid ion exchangers may be used in two ways as follows: (*i*) they may be packed in columns (*ii*) they may be added to the extract and removed by decantation. Liquid ion exchangers dissolve only in nonaqueous solvent carriers and the separation is similar to liquid-liquid extration. Some antibiotics are recovered directly from the whole broth using ion exchange resins. The product is recovered from the ion exchangers by ion displacement; this also regenerates the ion exchanger.

Adsorption Resins : These are porous polymers without ionization. Most compounds are adsorbed to the resins in non-ionized state. The porosity of the resin determines the surface available for adsorption. These resins may be apolar (*e.g.*, styrene-divinyl beneze), polar (*e.g.*, sulfoxide, amide, etc.), or semipolar (*e.g.*, acrylic ester). The products are recovered from such resins by solvent (organic) extraction, changed pH, etc.

Purification

The final step in the recovery of a product is ***purification*** which aims at obtaining the product in a highly purified state. The earlier steps will have achieved variable degrees of purification, which may determine the degree of resolution necessary during the purification step. The degree of resolution will mainly depend on the similarities of the metabolite with other molecules present in the concentrate, and the degree of purity required in the final product. Purification is achieved by the following procedures :

(*i*) crystallization and

(*ii*) chromatographic procedures.

crystallization : It is maily used for purification of low molecular weight compounds like antibiotics, e.g., penicillin G is usually extracted from the fermentation broth in butyl acetate and cystallized by the addition of potassium acetate in ethanolic solution. Crystallization is the final stage in purification of products like citric acid, sodium glutamate, etc.

Chromatographic Methods : These are used for the purification of low molecular weight compounds from mixtures of similar molecules, *e.g.*, homologous antibiotics, and or macromolecules, especially enzymes, which are similar in properties. The different chromatographic procedures are as follows:

(*i*) adsorption, (*ii*) ion exchange, (*iii*) gel filtration, (*iv*) hydrophobic, (*v*) affinity, (*vi*) covalcnt and (*vii*) partition chromatography.

Adsorption chromatography separates molecules due to their differential affinities for the surface of a solid matrix, e.g., silica gel, alumina, hydroxyapatite (all inorganic) or and organic polymer. In case of ***ion exchange chromatography***, resins or polysaccharides, *e.g.*, cellulose, sepharose, having attached ionized functional groups are used for a high resolution separation of macromolecules, *e.g.*, protiens. ***Gel filtration*** uses molecular sieves, composed of neutral cross-linked carriers e.g., polymers like agarose, dextrans, of different pore sizes. Molecules smaller than the pore size enter the carrier and are retained; they are later eluted (in order of molecule size) and collected. Gel filtration is used in aqueous systems. ***Hydrophobic carriers*** are used for purification of hydrophobic molecules, *e.g.*, many enzymes and other proteins.

Affinity chromatography uses molecules, called ***effectors*** or **ligand** to which the product has high and specific affinity, e.g., using an antibody (effector) for purification of the antigen to which it is specific. The effector is immobilized on a water insoluble carrier, which is packed in a column through which the mixture is passed. The effector binds only to the molecules for which it is specific and retains it in the columns; it is later recovered by elution using a buffer solution of a specified pH. For example, human leucocyte interferon is recovered in high yield and in high purity by affinity chromatography using monoclonal antibody imobilized on a sepharose column. ***Group specific affinity chromatography*** and ***covalent chromatography*** are based on chemical interactions between the carrier and the product molecules.

Drying

Drying makes the products suitable for handling and storage. It should be accomplished with a minimum rise in temperature due to heat sensitivity of most products. Addition of sugars or other stabilizers improves the heat tolerance of some products like enzymes and pharmaceutical preparations. The most common approaches to drying are as follows :

(*i*) vacuum drying,

(*ii*) spray drying and

(*iii*) freeze drying.

In ***spray drying,*** the solution or slurry to be dried is atomized by a nozzle or a rotating disc. A current of hot air (150-250°C) is passed; the drying is so rapid that the temperature of particles remains very low. Spray drying is used for enzymes, antibiotics and food products. ***Vacuum drying*** uses both heat and vacuum for drying; it can be applied both in batch mode, e.g., chamber dryers, or in continuous mode e.g. rotating drum vacuum dryers.

In ***freeze drying,*** the liquor to be dried is first frozen and the water is sublimed from the frozen mass. A very low pressure (partial vacuum) is maintained to promote sublimation of water. The energy needed for sublimation is provided by heated plates and radiation onto the suurface. The temperature of solid is regulated by regulating the pressure in the drying chamber. This is the most gentle method of drying, and is used for many pharmaceutical products, *e.g.*, viruses, vaccines, plasma fractions, enzymes, etc., and in food industries.

In Situ Recovery of Products

In many cases, the fermentation products are inhibitory to the product formation. In all such cases, removal of the product during the fermentation process, *i.e.*, **in situ *recovery of the products,*** greatly increases product yields. A number of methods have been successfully used for this purpose, which can be grouped into the following three broad categories: (*i*) use of vacuum, (*ii*) two-phase systems, and (*iii*) dialysis for product removal.

Use of Vaccum

Vacuum has been used to remove volatile products like ethanol from the fermentation broth. Either the fermentation vessel itself is kept under vacuum (impractical due to high energy needs) or, alternatively, the broth is circulated through a vacuum chamber while the fermenter is kept at atmospheric pressure. Both approaches have been used to enhance ethanol production.

Two-Phase Systems

In this approach, a water immiscible material (*the second phase*) is used to remove the product from the aqueous broth (*the first phase*). The second phase may be either present in the fermentation broth or located in a separate chamber through which the broth is passed for removal of the product. The second phase may either be a liquid, *e.g.*, *n*-hexane, *n*-octane, iso-octane, kerosene, toluene, isoamyl alcohol, etc., or a solid. The latter may be of the following two types :

(*i*) adsorbants, *e.g.*, activated charcoal, polymeric resins, etc., and

(*ii*) ion exchangers, *e.g.*, ion exchange resins.

1. ***Liquid second phase*** has been used for the separation of products like butyric acid, valeric acid, caproic acid from suppressed methane anaerobic fermentation.
2. ***Adsorbants*** like activated charcoal, etc. have been employed for the *in situ* recovery of a variety of products.
3. ***Anion exchange resin*** (added into the broth) has been used to remove salicylic acid during fermentatiion of naphthalene by *Pseudomonas aeruginosa;* this greatly increases salicylic acid yields.

Dialysis

The fermenter may contain a chamber separated by a selectively permeable membrane. This membrane allows the product to diffuse from the fermentation chamber into the dialysis chamber only (one-way diffusion) from where the product is recovered while the fermentation continues. This approach has been used for the removal of lactic acid during fermentation by *Lactobacillus delbrueckii*.

Material and Energy Balance

One of the simplest concepts in process engineering is the material and energy balance. Because mass in biological system is conserved at all times, the law of conservation of mass provides the theoretical framework for material balances. Energy effects are important because biological catalysts are very sensitive to heat and changes in temperature.

(*a*) System

A specified part of the universe with real or imaginary boundaries on which studies of thermodynamics etc. are to be made.

(*b*) Surrounding

The rest part of the universe, adjacent to real or imaginary boundries of system.

(c) Type of System

System can be classified as:

1. **Open System**

   Those in which exchange of mass and energy is possible with surrounding.

2. **Closed System**

   Those in which exchange of energy (not mass) take place with surrounding.

(d) Process

A process cause change in the system or surrounding. Sevral term used to describe process

(i) Batch Process

It is operate in a closed system. All material are added to the system at the start of the process, the system is then closed and product removed only when the process is complete.

(ii) Semi-batch Process

It allow either input or output of mass but not both.

(iii) Fed-batch Process

It allow input of material to the system but not output.

(iv) Countinuous Process

It allow matter to flow in and out of the system. If rate of mass input and output are equal continuous process can be operated indefinitely.

(v) Steady State Process

If all properties of a system, such as temperature, pressure, concentration, volume, mass etc. do not vary with time. The process is said be at a steady state.

(vi) Transient or Unsteady–State Process

The mass of the system is either increasing or decreasing with time during fed-batch and Semi-batch process. In batch process, even though the total mass is constant, changes occuring inside the system, cause the system properties to vary with time. Such process are called transisent or unsteady state process.

Steady state is often confused with another thermodynamic term equilibrium. A system at equilibrium is one in which all opposing forces are excatly counter balance. So that the properties of the system do not change with time.

The Nature of Energy

Energy is the capacity to transfer heat on to do work.

Type of Work

There are two type of work:

(a) Shaft Work is a work done by a moving part witin the system or Impeller, mixing a fermentation broth.

(b) Flow Work (w_f) is the energy required to push matter into the system. Flow work is given by the expression

$$W_f = PV,$$

where P = Pressure, V = Volume

Type of Energy

Energy takes three basic forms

(a) Kinetic energy (E_k) is the energy of motion and its magnitude depends upon mass of an object and its velocity.

(b) Potential energy (E_p) an object can also passes energy by virtue of its position relative to other. This energy is knwon Potential energy.

(c) Internal energy (U) is the sum of all molecular atomic and subatomic energies of matter, internal energy cann't measured directly or known its absolute term.

Unit of Energy

SI unit of energy is Joule (J), 1 J = 1 Newton metre (Nm)

Traditionaly, energy changes accompanying chemical reaction are expressed in calorie. One calorie (cal) is the amount of energy required to raise the temperature of 1g water by 1°C from 14.5°C to 15.5°C.

$$1 \text{ cal} = 4.1868 \text{ J}$$

Also,

Material and Energy Balance Equation

In this universe two things are always conserved. One is the energy and another is the matter *i.e;* material

Law of Conservation of Mass

The law of conservation of mass can be written as

$$\begin{bmatrix}\text{mass in}\\ \text{through system}\\ \text{boundary}\end{bmatrix} - \begin{bmatrix}\text{mass out}\\ \text{through system}\\ \text{boundary}\end{bmatrix} + \begin{bmatrix}\text{mass}\\ \text{generated}\\ \text{within system}\end{bmatrix} - \begin{bmatrix}\text{mass consumed}\\ \text{within system}\end{bmatrix} = \begin{bmatrix}\text{mass accumulate}\\ \text{within system}\end{bmatrix}$$

As steady state all properties of the system, incluuding its mass, must be unchanging with time, hence cannot accumulate mass.

$$\text{mass in} = \text{mass out}$$

Law of Conservation of Energy

The law of conservation of energy can be written as

$$\begin{bmatrix}\text{energy in through}\\ \text{system boundaries}\end{bmatrix} - \begin{bmatrix}\text{energy out through}\\ \text{system boundaries}\end{bmatrix} = \begin{bmatrix}\text{energy accumulated}\\ \text{within the system}\end{bmatrix}$$

$$\boxed{\Delta E = Mi\left(u + e_k + e_p + pv\right)_i - M_o\left(u + e_k + e_p + pv\right)_o - Q + \omega_0}$$

where ΔE = Total change or Accumulation of energy in the system, Q = Energy have the system as heat, ω_0 = Shoft work, M_o = Mass leave the system, M_i = Mass enter the system

(a) For Steady-State Flow process

$$\Sigma_i M_h - \Sigma_o (M_h) - Q + w_s = 0$$

There is no accumulation or change in the system because all properties of the system invariant.

(b) For Adiabatic process

If the system is adeabatic ie no heat is transfered to or from the system under these condition, Q = O and equation (1) become

$$\Sigma_i (Mh) - \Sigma_o (Mh) + w_s = \Delta E$$

Enthalpy

The sum of internal energy and PV energy of any system, under a particular set of condition is known as enthalpy.

$$H = U + PV \text{ or } h = u + pv$$

where, H = Enthalpy, U = Internal energy, P = pressure, V = Volume

Irrespective of how enthalpy change occur certain convention are used in enthalpy calculation.

Reference State – Enthalpy change is independent of the reference state because H_{ref} cancel out in calculation

i.e., $$\Delta H = H_2 - H_1$$

State of Function

Enthalpy is a state function. Thus total enthalpy change for process is equal to the sum of change in the hypothetical path,

$$\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3$$

Factor Affect the Value of Enthalpy

Change in enthalpy can occur as a result of

(*i*) Temperature, (*ii*) Phase, (*iii*) Mixing and solution, (*iv*) Reation.

Temperature

Change in the enthalpy of a system due to variation in temperature called sensible heat change which is determined using a property of matter called heat capacity at constant pressure (C_P)

$$\Delta H = MC_P \Delta T$$

The value of C_P itself varies over the range of ΔT. Heat capacities arc often tabulated as polynomial function of temperature.

Such as $$C_P = a + bT + cT^2 + dT^3$$

where a, b, c, and d = coefficient of substance

Phase

Heat transfered to or from a system cause change of phase at constant temperature and pressure is known as latent heat. Type of latent heat are

(*i*) Latent heat of vaporisation Δh_v – heat requires to vaporise a liquid

(*ii*) Latent heat of fussion (Δh_f) – heat requires to melt a solid

(*iii*) Latent heat of sublimation (Δh_s) – heat required to directly vaporize a solid.

The change in enthalpy resulting from phase change is calculated directly from latent heat.

$$\Delta H = M \Delta h_v$$

where ΔH = change in enthalpy

Δh_v = latent heat of vapoarisation

M = mass

Mixing And Solution

The integral heat of solution Δh_m is defined as the change in enthalpy which occurs as one mole of solute is dissolved at constant temperature in given quantity of solvent. The enthalpy of a non ideal mixture of two component A and B is

$$H_{mix} = H_A + H_B + \Delta H_m$$

where ΔH_m = heat of mixing

Reaction

Heat of reaction ΔH_{rxn} is the energy released or absorved during reaction and is equal to the difference in enthalpy of reactant and product

$$\Delta H_{rxn} = \Sigma_{product} Mh - \Sigma_{reactant} Mh$$

where ΔH_{rxn} = heat of reaction,

M = Mass,

h = specific enthalpy,

Σ = Sum

Growth, Product Stoichiometry and Elemental Balance

Cell growth obeys the law of conservation of matter. All atoms of corbon, hydrogen, oxygen, nitrogen and other element consumed during growth are incorporated into new cells or excreted as product.

(*a*) C, H, O, and N Balance

We can write the following equation for aerobic cells growth

$$C_w H_x O_y N_z + aO_2 + bH_g O_h N_i \longrightarrow cCH_\alpha O_\beta N_\delta + d\,CO_2 + eH_2O$$

where $C_wH_xO_yN_z$ = chemical formula for substrate,

$H_gO_hN_i$ = chemical formula for nitrogen source,

$CH_\alpha O_\beta N_\delta$ = chemical formula for dry biomass,

O_2 = a mole O_2 consumed,

CO_2 = d mole CO_2 formed,

H_2O = e mole H_2O formed,

Once a formula for biomass is obtained, these coefficient can be evaluated using normal procedures for balancing equation i.e. element balance and solution of simultaneous equation.

C balance $w = c + d$

H balance $x + bg = c\alpha + 2e$

O balance $y + 2a + bh = c\beta + 2d + e$

N balance $z + bi = c\,\delta$

(*b*) Respiratory Quotient (RQ) A useful measurable parameter

$$RQ = \frac{\text{Mole of } CO_2 \text{ product}}{\text{Mole of } O_2 \text{ product}} = \frac{d}{a}$$

Electron Balance

Available elctrons refers to the number of electrons available for transfer to oxygen on combustion of a substance to CO_2, H_2O and nitrogen-containing compounds. The number of available electrons found in organic material is calculated from the valence of the various elements: 4 for C, 1 for H, –2 for O, 5 for P and 6 for S. The number of available electrons for N depends on the reference state:–3 if ammonia is the reference, 0 for molecular nitrogen N_2, and 5 for nitrate.

Degree of reduction γ, is defined as the number of equivalents of available electrons in that quantity of material containing 1 g atom carbon. Therefore, for substrate $C_wH_xO_yN_z$, the number of available electrons is $(4w + x - 2y - 3z)$. The degree of reduction for the substrate, γ_S is therefore $(4w + x - 2y - 3z)/w$.

Electrons available for transfer to oxygen are conserved during metabolism. In a balanced growth equation, number of available electrons is conserved by virtue of the fact that the amounts of each chemical element are conserved. the available-electron balance is :

$$w\gamma_S - 4a = c\,\gamma_B$$

where $\gamma_S - \gamma_B$ are the degrees of reduction of substrate and biomass, respectively.

Biomass Yield

$$Y_{XS} = \frac{g \text{ cells produced}}{g \text{ substrate consumed}}$$

$$Y_{XS} = \frac{c\ (\text{MW cells})}{(\text{MW substrate})}$$

Example Stoichiometric Coefficients for Cell Growth

Production of single-cell protein from hexadecane is described by the following reaction equation:

$$C_{16}H_{34} + aO_2 + bNH_3 \rightarrow cCH_{1.66}O_{0.27}N_{0.20} + dCO_2 + cH_2O$$

where $CH_{1.66}O_{0.27}N_{0.20}$ represent the biomass. IFRQ = 0.43, determine the stoichiometric coefficients.

Solution :

C balance	$16 = c + d$	(1)
H balance	$34 + 3b = 1.66c + 2e$	(2)
O balance	$2a = 0.27c + 2d + e$	(3)
N balance	$b = 0.20\,e$	(4)
RQ :	$0.43 = d/a.$	(5)
then	$d = 16 - c$	(6)
From (5);	$a = \frac{d}{0.43} = 2.326d.$	(7)

Combining (6) and (7) gives an expression for a in terms of c only;

$$a = 2.326\,(16 - c)$$

$$a = 37.22 - 2.326\,c. \quad (8)$$

Substituting (4) into (2) gives;

$$34 + 3(0.20\,c) = 1.66c + 2e$$
$$34 = 1.06c + 2e$$
$$e = 17 - 0.53\,c.$$

Substituting (8), (6) and (9) into (3) gives; (9)

$$2(37.22 - 2.326c) = 0.27\,c + 2(16 - c) + (17 - 0.53c)$$
$$25.44 = 2.39c; \quad c = 10.64$$

Using this result for c in (8), (4), (6) and (9) gives;

$a = 12.48$ $b = 2.13$

$d = 5.37$ $e = 11.36$

Check that these coefficient values satisfy Equation. (1) – (5)

The complete reaction equation is

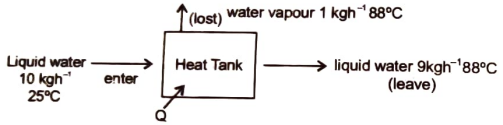
$$C_{16}H_{34} + 12.5\,O_2 + 2.13\,NH_3 \rightarrow 10.6\,CH_{1.66}O_{0.27}N_{0.20} + 5.37CO_2 + 11.4\,H_2O$$

Example – Water at 25°C enter on open heating tank at a rate of 10 kgh^{-1}, liquid water leaves the tank at 88°C at a rate of 9 kgh^{-1} water vapour is last from the system through evaporation. At steady state, what is the rate of heat input to the system?

Solution.

Step 1. Units in problem kg, h, kJ, °C

Step 2. Draw Block diagram



Step 3. Interpetation

(*i*) Process – Steady state

(*ii*) No leakage

(*iii*) Evaporation occur at 88°C

(*iv*) Shaft work – negligible

(*v*) No heat loss

(*vi*) No reaction occur

(*vii*) **Extra data need**

h for liquid water at 25°C = 104.8 kJ kg^{-1}

h for Saturated steam at 88°C = 2656.9 kJ kg^{-1}

h for liquid water at 88°C = 368.5.8 kJ kg

Step 4. Calculation

Write down energy balanced equation

$$\Sigma_i\,Mh - \Sigma_0\,Mh - \Sigma Mh_{vap\ out} - Q + W_s = 0 \text{ – for stedy state}$$

given, $W_s = 0$

∴ equation (1) become

$$\Sigma_i\,(Mh) - \Sigma_0\,(Mh) - \Sigma_{vap\ out}\,(Mh) - Q = 0$$

put the value, we get

$$(10\text{ kg})(104.8\text{ kJ kg}^{-1}) - (9\text{ kg})(368.5\text{ kJ kg}^{-1}) - (1\text{ kg})(2656.9\text{ kJ kg}^{-1}) - Q = 0$$
$$Q = -4925.4\text{ kJ}$$
$$Q = \text{Negative value}$$

Step 5. Result

The rate of heat input to the system = 4. 925 × 10^3 kJ h^{-1}

EXERCISE – I

MCQ TYPE QUESTIONS

1. Which of the following chemical is highly toxic and violently explosive if mixed with air, used for sterilization?

(*a*) Formaldehyde (*b*) Propylene oxide

(*c*) H_2O_2 (*d*) Ethylene oxide

2. Which microorganism is used for the production of collagenase?

(*a*) Pseudomonas sp (*b*) Flavobacterium sp

(*c*) Streptomyces sp (*d*) Clariceps purpurea

3. Tyndallization is useful where a medium is affected by

(*a*) Temperature of above 100°C

(*b*) Where spores are present

(*c*) Both (*a*) and (*b*)

(*d*) None

4. One of the major methods employed in depth filtration is _____

(*a*) Flocculation (*b*) Adsorption

(*c*) Interception (*d*) All of these

5. Which of the following is not the advantage of cartridge air filtration?

(*a*) High predictable performance

(*b*) High particle collection efficiency

(*c*) Easy to handle preformed media

(*d*) None of these

6. Hydrophobic compounds can be received by which filtration technique?

(*a*) Permeation (*b*) Micro-filtration

(*c*) Perstraction (*d*) Pervaporation

7. The doubling time (td) equation is _____

(*a*) $t_d = 0.693/$

(*b*) $t_d = 3.33\ (\log x_t - \log x_0)$

(*c*) $t_d = dx/dt$

(*d*) None of these

8. Mechanism of separation of contaminants present in air fibrous medicine is

(*a*) Inertial impaction (*b*) Interception

(*c*) Diffusion (*d*) All of these

9. Membranes used in filtration are made of

(*a*) Polyethersulfon (*b*) Polyvinyl chloride

(*c*) Polyarylamide (*d*) Cellulose

10. Which of the following methods is useful for sterilizing heat labile materials?

(*a*) Adsorption (*b*) Filtration

(*c*) Sedimentation (*d*) Diffusion

11. Match the types of depth filters with their mechanisms

Group I	Group II
P. Cellulose fibre	1. Interception
Q. Celiate	2. Diffusion
R. Porous	3. Sedimentation
S. Glass wool	4. Inertial impaction

(*a*) P-4, Q-1, R-2, S-3 (*b*) P-3, Q-1, R-4, S-2

(*c*) P-4, Q-3, R-1, S-2 (*d*) P-2, Q-3, R-1, S-4

12. The objective of mixing in fermentation is to

(*a*) Disperse the air bubbles

(*b*) To suspend the microorganism

(*c*) To enhance heat and mass transfer in the medium

(*d*) All of these

13. Which of the following statements is correct about the operation of fed-batch process?

(*a*) It eliminates the need for oxygen supply

(*b*) It helps controlling the repressive effects of the nutrient medium

(*c*) The volume of the culture increases with time

(*d*) Both (*b*) and (*c*)

14. If a chemical sterilant is to be used for glass wares etc., it should be

(*a*) Toxic and non-volatile

(*b*) Toxic and volatile

(*c*) Non-toxic and volatile

(*d*) Non-toxic and non-volatile

15. Best information on an industrial strain is obtained from

(*a*) Batch culture (*b*) Continuous culture

(*c*) Fed Batch (*d*) None of these

16. Importance of anti-foam in microbiological process is

(*a*) Antifoam reduces the surface tension in the foams

(*b*) Antifoam is used as carriers

(*c*) Antifoam destabilizes the protein films

(*d*) Both (*a*) and (*c*)

17. Which of the following nitrogen sources are used for industrial process?

(*a*) n-butane (*b*) Yeast extracts

(*c*) Cornsteep liquor (*d*) Dextrin

18. Thermal treatment is suitable for media sterilization because

(*a*) It is relatively low cost

(*b*) Effectiveness in achieving an acceptable level of sterility

(*c*) Both (*a*) and (*b*)

(*d*) None of these

19. The solubility of oxygen in aqueous solution as ambient temperature and pressure is only about
(a) 50 ppm (b) 10 ppm
(c) 25 ppm (d) 75 ppm

20. A fermentor sterilization in situ is less efficient than continuous heat sterilization because
(a) It uses higher temperature for long time
(b) It uses longer heating time during which heat is lost
(c) It uses steam as the heating source
(d) It uses larger volume and hence takes longer to cool the medium

21. Media containing spores and thermo labile constituents are sterilized by
(a) Pasteurization (b) UV radiation
(c) Dry heat (d) Tyndallization

22. Ultra-filtration process cannot be used for
(a) Desalting
(b) Fractional of protein
(c) Selective removal of solvents
(d) Harvesting of cells

23. Cell population is maintained in steady state in which type of culture?
(a) Batch (b) Fed Batch
(c) Continuous (d) None of these

24. Which of the following is used as antifoam?
(a) Oils (b) Silicon oil
(c) Both (a) and (b) (d) None of these

25. Usually the dilution rate is giving maximum
(a) Productivity (b) Specific growth rate
(c) Both (a) and (b) (d) None of these

26. The percent inhibition for this enzyme is calculated from the euquation
(a) {[cpm (control) - cpm (test)]/[cpm (control) - cpm (blank)]} × 100
(b) {[cpm (control) - cpm (test)]/[cpm (blank) - cpm (control)]} × 100
(c) {[cpm (test) - cpm (control)]/[cpm (control) - cpm (blank)]} × 100
(d) {[cpm (control) - cpm (blank)]/[cpm (test) - cpm (control)]} × 100

27. Which of the following is not true for fluidized bioreactors?
(a) Low shear rates
(b) Very good mixing between the three phases
(c) High energy requirements
(d) None of these

28. Most non-Newtonian fluids in bio-processing are
(a) Dilatant rheology
(b) Cassen body rheology
(c) Pseudo plastic rheology
(d) None of these

29. The relative magnitudes of the various mass-transfer resistances depend on the
(a) Composition and rheological properties of the liquid
(b) Mixing intensity
(c) Bubble size
(d) All of the above

30. Bubbles are generally beneficial because of the increase gas hold-up and large interfacial surface
(a) larger (b) small
(c) spherical (d) oval

31. Bubble size affect the value of
(a) K_L
b) Re(Reynold's number)
(c) Power number
(d) Mixing time

32. Antifoam compound affect the
(a) Surface chemistry of bubbles
(b) Tendency to coalesce
(c) Effect on K_La
(d) All of these

33. To maintain the non-cailesing character of the medium and high K_La value which method is preferred?
(a) Mechanical method of disrupting cell
(b) Chemical method of disrupting cell
(c) Both (a) and (b)
(d) None of these

34. Decrease in surface tension reduce the average diameter of bubbles thus prevalence
(a) higher value of a (b) lower value of a
(c) constant value (d) no affect

35. The temperature of aerobic fermentation affect
(a) Solubility of oxygen
(b) Mass transfer coefficient
(c) Both (a) and (b)
(d) None of these

36. Oxygen transfer is influenced by the
(a) presence of cells in broth
(b) temperature
(c) gas pressure and oxygen partial pressure
(d) All of the above

37. Polarographic electrode is used to measure the oxygen concentration is
(a) Dynamic method
(b) Direct measurement
(c) Dynamic gassing out technique
(d) Sodium Sulphite oxidation

38. The dissolved oxygen level of the fermenter is steady, then turn off the air supply, the oxygen concentration will decrease at rate of $dCi/dt = rO_2C_x$ measured by
(a) Dynamic method
(b) Direct measurement
(c) Dynamic gassing out technique
(d) Sodium Sulphite oxidation

39. For mixing, which type of fluid develop
(a) Turbulence fluid (b) Streamline fluid
(c) Both (a) and (b) (d) None of these

40. Match the following

Parameters	Measuring
P. Flow rate (gas)	I. Photometer
Q. Flow rate (liquid)	II. Rotameters
R. Viscosity	III. Magnetic inductive flow meters
S. Turbidity	IV. Rotational viscometers

(a) P-I, Q-III, R-II, S-IV (b) P-II, Q-III, R-I, S-IV
(c) P-II, Q-III, R-IV, S-I (d) P-IV, Q-III, R-II, S-I

41. Match the following

Parameter	Measuring devices
P. Temperature	I. Rubber sheated electrode
Q. Pressure	II. Strain gages
R. Agitator shaft power	III. Diaphragm gages
S. Foam	IV. Thermistor

(a) P-I, Q-II, R-III, S-IV (b) P-IV, Q-II, R-I, S-III
(c) P-IV, Q-III, R-II, S-I (d) P-IV, Q-III, R-II, S-I

42. Which equation has a saturation constant proportional to cell concentration?
(a) Blackmann equation
(b) Tessier equation
(c) Moser equation
(d) Contois equation

43. Contois equation is
(a) $\mu_g \alpha X^{-1}$ (b) $\mu_g = X^{-1}$
(c) $\mu_g \alpha \frac{1}{X^{-1}}$ (d) None of these

44. In non competitive substrate inhibition, if $K_1 > K_3$ then
(a) $\mu_g = \frac{\mu_m}{K_S + S + \frac{K_1}{S^2}}$ (b) $\mu_g = \frac{\mu_m S}{K_S + S + \frac{S^2}{K_1}}$
(c) $\mu_g = \frac{\mu_m S}{K_S\left[1 + \frac{P}{K_P}\right] + S}$ (d) All of these

45. Equation of non competitive product inhibition is
(a) $\mu_g = \frac{\mu_m S}{K_S + S + \frac{S^2}{K_1}}$ (b) $\mu_g = \frac{\mu_m}{\left[1 + \frac{K_S}{S}\right]\left[1 + \frac{P}{K_P}\right]}$
(c) $\mu_g = \frac{\mu_m S}{K_S\left[1 + \frac{P}{K_P}\right] + S}$ (d) None of these

46. What is an intrinsic concentration?
(a) The amount of a compound/unit cell
(b) The amount of a compound/cell volume
(c) Both (a) & (b)
(d) The amount of a compound/unit reactor volume

47. Which modeling approach has been developed primarily to predict growth, under conditions of several substrate availability?
(a) Cybernetic models
(b) Chemically structured models
(c) Both (a) & (b) (d) None of these

48. Complementary substrates in cybernetic model is/are
(a) Carbon (b) Nitrogen
(c) Both (a) & (b) (d) Glucose

49. Why glucose and lactose are substitutable substrates?
(a) Both compounds are complementary
(b) These compounds supply carbon and energy
(c) Both (a) & (b)
(d) Glucose supply carbon only and lactose supply energy only

50. One approach to model growth on multiple substrates is
(a) Cybernetic approach
(b) Models for transient behavior
(c) Both (a) & (b)
(d) Model with time delays

51. Particle counter method gives best results for
(a) Dilute cell (b) Particle suspensions
(c) Both (a) & (b) (d) Concentrated cell

52. Spectrophotometer is based on
(a) Absorption of light by suspended cells in sample culture media
(b) Absorption of light by Solid cells in sample culture media
(c) Absorption of light by scattered cells in sample culture media
(d) None of these

53. Which of following components can be measured as indirect measures of cell growth?
(a) DNA (b) RNA, proteins
(c) Proteins (d) All of these

54. Which of following components concentration remain constant during a batch growth cycle?
(a) DNA and RNA (b) RNA and protein
(c) DNA and protein (d) None of these

55. Which of following components concentration is used as a measure of microbial growth in a complex medium?
(a) RNA (b) Protein
(c) DNA (d) All of these

56. The ATP concentration in fermentation broth can be used as a measure of
(a) Biomass concentration
(b) Cell number density
(c) Substrate consumption
(d) None of these

57. Under anaerobic metabolism, ethanol and lactic acid are produced, among those which are growth associated product/s?
(a) Ethanol (b) Lactic acid
(c) Both (a) & (b) (d) None of these

58. When nitrate is used as the nitrogen source, hydrogen ions are removed from the medium, which results in
(a) Increase in pH (b) Decrease in pH
(c) No alteration in pH (d) Maintain constant pH

59. If the substrate is a biodegradable polymer, then the viscosity of the broth will
(a) Increase with time as bio-hydrolysis continues
(b) Decrease with time as bio-hydrolysis continues
(c) Constant with time as bio-hydrolysis continues
(d) None of these

60. Which of the following sentences are true regarding the microbial growth?
I. The increase in mass is a consequence of anabolism.
II. Cell mass thus increases at the same rate as cell number
III. A bacterial generation time is also known as its doubling time
(a) I, II (b) II, III
(c) I, III (d) I, II, III

61. How to minimize the duration of the lag phase?
(a) Cells should be adapted to the growth medium and conditions
(b) Cells should be young and active
(c) The inoculum size should be large
(d) All of these

62. During balanced growth, the net specific growth rate is determined by
(a) Cell number
(b) Cell mass would be the same
(c) Both (a) & (b)
(d) None of these

63. The stationary phase starts at the end of the deceleration phase, when the net growth rate is
(a) Increased (b) Zero
(c) Decreased (d) None of these

64. Generation time is formulated as
(a) G = t/n (b) G = 2t/n
(c) G =1/2t/n (d) G = 3t/n

65. Substrate consumption in lag phase of microbial growth is primarily used for
(a) Turn over of the cell material
(b) Increase in cell number
(c) Both (a) & (b)
(d) Motility

66. Maximum specific growth rate of a micro organism is calculated by taking the (In = log, X = biomass, t = time)
(a) Slope of In X v_S t during lag phase
(b) Slope of In X v_S t during the exponential growth phase
(c) Slope of X v_S t during stationary phase
(d) None of these

67. Match the following

Organism			$q0_2$ (mmol 0_2 / g dw – h)
P.	E. coli	(I)	$0.4\frac{\text{mmol } o_2/1-h}{10^6 \text{cells/m}^1}$
Q.	Saccharomyces cerevisiae	(II)	3 – 4
R.	Penicillium sp	(III)	8
S.	Animal cells Hela	(IV)	10 – 12

(a) P-I, Q-IV, R-II, S-III (b) P-IV, Q-III, R-II, S-I
(c) P-IV, Q-II, R-I, S-III (d) P-II, Q-III, R-IV, S-I

68. Match the following

Organism			$q0_2$ (mmol 0_2 / g dw – h)
P.	Azotobacter sp	(I)	Ca. 3
Q.	Aspergillus niger	(II)	30 – 90
R.	Acer pseudomonas (sycamore)	(III)	$0.15\frac{\text{mmol } o_2/1-h}{10^6 \text{cells/m}}$
S.	Diploid embryo WI-38	(IV)	0.2

(a) P-II, Q-III, R-IV, S-I (b) P-I, Q-III, R-II, S-IV
(c) P-II, Q-I, R-IV, S-III (d) P-III, Q-I, R-IV, S-II

69. Match the following

Transport process			Equation
P.	Flow	(I)	L^2/D
Q.	Mixing	(II)	$V_p C_p / UA$
R.	Heat transfer	(III)	$t_m = \frac{4V}{(1.5ND^3)}$ stirred vessel
S.	Diffusion	(IV)	L/V or V/Q

(a) P-II, Q-III, R-IV, S-I (b) P-II, Q-IV, R-I, S-III
(c) P-IV, Q-III R-II, S-I (d) P-III, Q-IV, R-I, S-II

70. Match the following

Conversion process			Equation
P.	Growth	(I)	1/
Q.	Chemical reaction	(II)	C/r
R.	Substrate consumption	(III)	$C_s / R_{max} (C_S >> K_S)$
S.	Heat production	(IV)	$\rho c_p \Delta kt / r\Delta H$

(a) P-I, Q-II, R-III, S-IV (b) P-IV, Q-III, R-II, S-I
(c) P-III, Q-IV, R-II, S-I (d) P-III, Q-I, R-IV, S-II

71. Match the following

Method	Type
P. Filtration	I. Centrifugation
Q. Adsorption	II. Charcoal
R. Sedimentation	III. Absolute depth
	IV. Flocculation

(a) P-IV; Q-II; R-I, III (b) P-III; Q-II; R-I, IV
(c) P-R; Q-IV; R-I, II (d) None of these

72. Match the following

Types of depth filters	Mechanisms
P. Ceramic	I. Sedimentation
Q. Celiate	II. Diffusion
R. Glass wool	III. Interception
S. Cellulose fiber	IV. Inertial impaction

(a) P-II; Q-III; R-IV; S-I
(b) P-IV; Q-II; R-III; S-I
(c) P-IV; Q-III; R-II; S-I
(d) P-I, Q-III, R-IV, S-II

73. Match the following

Type	Mechanisms
P. Ultrafiltration	I. Electrophoretic interaction
Q. Microporous	II. Adsorption
R. Macroporous	III. Bridging
S. Sintered metal	IV. Size exclusion

(a) P-I; Q-III; R-II; S-IV (b) P-I; Q IV; R-III; S-II
(c) P-IV; Q-III; R-II; S-I (d) P-IV; Q-I; R-III; S-II

74. Match the following

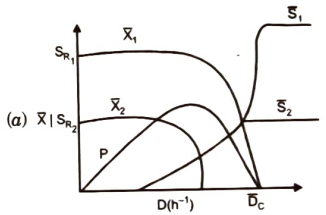
Micro organisms	Distribution of population
P. Molds	I. 10%
Q. Yeast	II. 40%
R. Gram positive bacteria	III. <1%
S. Gram negative bacteria	IV. 50%

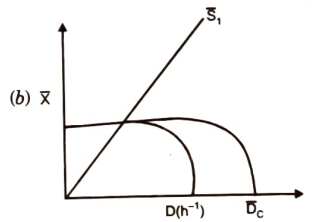
(a) P-IV; Q-III; R-I; S-II (b) P-II; Q-I; R-IV; S-III
(c) P-II; Q-IV; R-I; S-III (d) P-IV; Q-III; R-II; S-I

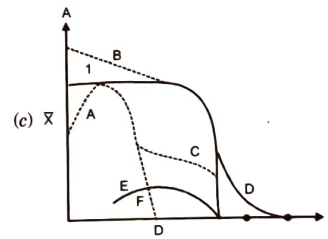
75. During steady state growth $\frac{dx}{dt} = 0$, consequently is
(a) = D (b) > D
(c) < D (d) = F

76. In chemostat under carbon limited conditions, a plot 1/y versus 1/D yields
(a) Straight line with slope m
(b) Straight line with slope 1/m
(c) Sigmoid line with slope m
(d) None of these

77. Deviations from ideal chemostat behaviour is

(a) 

(b) 

(c) 

(d) None of these

78. What is/are the advantages of continuous culture over batch culture?
(a) The growth rate can be controlled and maintained indefinitely
(b) Biomass conc. can be maintained constant by varying the dilution rate
(c) Both (a) and (b)
(d) Wall growth and cell aggregation can cause wash out

79. Which of the following statement/s is correct in case of continuous culture?
(a) Chemostats can be used to maintain a selection pressure for thermophiles/mutants with low growth rate and yields
(b) Chemostats can be used to maintain a selection pressure for thermophiles/mutants with high growth rate and yields
(c) Results obtained in continuous culture are frequently less reliable and reproducible
(d) All of these

80. What are the disadvantages of continuous culture compared to batch culture?
(a) Enables accurate determination of kinetic constant maintenance energy and true growth yields
(b) Growth over long periods may results in problems of contamination
(c) Both (a) and (b)
(d) None of these

81. Which of the following parameter, is used as scale up criteria
(a) Airflow case
(b) Agitator speed
(c) Diameter of impeller
(d) Volumetric mass transfer coefficient

82. Match the following bioreactor components in group-I with the most appropriate function given in group-II

Group-I	Group-II
P. Marine type impeller	I. Aeration of medium
Q. Draft tube	II. Animal cell cultivation
R. Diaphragm value	III. Animal cell cultivation
S. Sparger	IV. Sterile operation

(a) P-IV; Q-II; R-I; S-III (b) P-III; Q-I; R-IV; S-II
(c) P-III; Q-IV; R-II; S-I (d) P-II; Q-I; R-IV; S-III

83. Which of the following statements are true regarding bioreactors?
P. Continuous bioreactors provide less degree of control and uniform product quality than batch bioreactors
Q. Batch bioreactors are ideally suited for reaction with substrate inhibition
R. Choice of bioreactors is indicated by kinetic considerations
S. Fed batch bioreactors are also called semi batch bioreactors
(a) P, Q (b) Q, S
(c) R, S (d) P, R

84. Assume that experimental measurements for a certain organism have shown that cells can convert $\frac{2}{3}\left(\frac{wt}{wt}\right)$ of the substrate carbon (alkane/glucose) to biomass. Calculate the stoichiometric coefficients for the following biological reactions
(a) Hexadecane : $C_{16}H_{34} + aO_2 + bNH3 \rightarrow C(C_{4.4}H_{7.3}N_{0.86}O_{1.2}) + dH_2O + eCO_2$
(b) Glucose : $C_6H_{12}O_6 + aO_2 + bNH_3 \rightarrow C(C_{4.4}H_{7.3}N_{0.86}O_{1.2}) + dH_2O + eCO_2$
(a) a = 4.2, b = 1,473 (b) a = 12.427, b = 1.473
(c) a = 12.427, b = 0.557 (d) None of these

85. What are the equations are alternatives to the Monod equation
(a) Black man equation, Moser equation
(b) Tessier equation, ConG's equation
(c) Both (a) and (b)
(d) None of these

86. Match the following

Equation
1. Blackman equation 2. Tessier equation
3. Contois equation 4. Moser equation

Formula
A. $\mu_g = \frac{\mu_m S}{K_{SX}X + S}$
B. $\mu_g = \frac{\mu_m}{2K_S}S$, if $S < 2K_S$
C. $\mu_g = \mu_m\left(1 - e^{-ks}\right)$
D. $\mu_g = \frac{\mu_m S^n}{K_S + S^n} = \mu_m\left(1 + K_S S^{-n}\right)^{-1}$
(a) P-IV; Q-III; R-II; S-I (b) P-IV; Q-I; R-II; S-III
(c) P-II; Q-III; R-I; S-IV (d) P-I; Q-IV; R-II; S-III

87. For competitive substrate inhibition is
(a) $\mu_g = \frac{\mu_m}{\left[1 + \frac{K_S}{S}\right]\left[1 + \frac{S}{K_I}\right]}$
(b) $\mu_g = \frac{\mu_m S}{k_S\left[1 + \frac{S}{K_I}\right] + S}$
(c) $\mu_g = \frac{\mu_m S}{K_S + S + S^2/K_I}$
(d) All of these

88. In turbulent region, power is related to Reynold's number as
(a) Directly proportional
(b) Inversely proportional
(c) Independent
(d) None of these

89. Advantages of air lift bioreactor over stirred tank bioreactors is/are
(a) Low shear
(b) Low energy requirement
(c) Simple construction
(d) All of these

90. On scaling up air life system, the mixing time
(a) Increases (b) Decreases
(c) Increases slightly (d) Decreases slightly

91. Which of the following is not true regarding fluidized bioreactors?
(a) Very good mixing between 3 phases
(b) High energy requirements
(c) Lower shear rates
(d) None of these

92. The value of the $K_L a$ volumetric mass transfer coefficient can be controlled by
(a) Agitation conditions (b) Air flow rate
(c) Both (a) and (b) (d) None of these

93. Which of the following agitator is the most commonly used?
(a) Propeller (b) Turbine
(c) INTERMIG (d) M/G

94. Agitation is required in a bioreactor to
(a) Suspend the microbes and solid nutrients
(b) Disperse the air in a nutrient solution
(c) Disperse the immiscible liquids present
(d) All of these

95. Aeration in a bioreactor is provided by
(a) Impeller (b) Baffles
(c) Sparger (d) All of these

96. In a bioreactor, baffles are incorporated to
(a) Maintain uniform suspension of cells
(b) Minimize the size of air bubble for greater aeration
(c) Maintain uniform nutrient medium
(d) Prevent vortex and to improve duration efficiency

97. For a turbine aeration agitation unit, the power consumption
(a) Is same for gassed and ungassed systems
(b) Increases with decreasing turbine diameter
(c) Is greater for ungassed system than for gassed system
(d) Is greater for gassed system than for ungassed system

98. Which one of the following conditions is least important for anaerobic fermentation, to optimize the bioreactor system?
(a) Culture agitation to maintain O_2 supply
(b) Restriction of entry of contaminating organism
(c) Maintenance of constant culture volume
(d) Control of parameters like pH and temperature

99. The first and the primary stage of a typical SSF process is
(a) Aerobic (b) Anaerobic
(c) Both (a) & (b) (d) None of these

100. The 2nd stage of anaerobic submerged fermentation with a mixed bacterial culture is called
(a) Koji (b) Moromi
(c) Torulopsis (d) None of these

101. Fermentation in the 2nd stage may be realized by using
(a) The natural flora
(b) Externally added bacteria and yeasts
(c) Both (a) & (b)
(d) Koji

102. Which of the following strains are used as flavor producers in Soy sauce manufacture?
(a) Saccharomyces
(b) Torulopsis, pediococcus
(c) Saccharomyces, Torulopsis
(d) All of these

103. Moromi is usually fermented for
(a) 2-3 months (b) 2-6 months
(c) 8-12 months (d) 12-27 months

104. The major industrial use of Koji process is
(a) The production of enzymes by bacterial species
(b) The production of enzymes by fungal species
(c) The production of enzymes by viral species
(d) All of these

105. Which enzyme/s are produced by SSF of wheat bran by A. oryzae in a rotating drum fermenter?
(a) Bacterial Amyloses
(b) Fungal amylases
(c) Both (a) & (b)
(d) None of these

106. Which of the following enzyme/s is produced by Koji fermentations?
(a) Cellulases (b) Pectinases
(c) Both (a) & (b) (d) None of these

107. Which of the following fermenter is/are used by Trichoderma viride species for the production of cellulases from wheat bran?
(a) Rotating drum fermenter
(b) Rotary tray fermenter
(c) Both (a) & (b)
(d) Rotary durm Koji fermenter

108. Aflatoxins are produced by SSF of rice by
(a) Rhizopus (b) A. parasiticus
(c) Actinomucor (d) None of these

109. Which of the following fermenter is used for the production of ochratoxin's?
(a) Rotary drum fermenter
(b) Rotating alum fermenter
(c) Rotary drum Koji fermenter
(d) None of these

110. Microbial degradation of lignocellulosic can be accomplished by SSF's for
(a) Waste treatment purposes
(b) In biopulping of wood chips for use in paper manufacture
(c) Both (a) & (b)
(d) None of these

111. Match the following

Product	Common substrate
P. Soy sauce	I. Tofu
Q. Miso	II. Rice
R. Tempeh	III. Wheat
S. Sufu	IV. Soybean

(a) P-II; Q-III; R-IV; S-I (b) P-III; Q-II; R-IV;; S-I
(c) P-III; Q-II; R-IV; S-I (d) P-II; Q-IV; R-III; S-I

112. Which fermentation/s are usually realized in a controlled humidity air environment with air ventilation and agitation?
(a) Koji fermentations (b) Moromi fermentations
(c) Both (a) & (b) (d) None of these

113. Niemann – Pick disease is caused by a deficiency of the enzyme
(a) Glucosyl ceramide hydrolase
(b) Phosphatidase
(c) Sphingomyelinase
(d) None of these

114. Which of the following compound inhibits the polymerization of microtubules?
(a) Taxol (b) Colchicine
(c) Tubulin (d) Keratin

115. Which one of the following statements is not applicable to integral membrane proteins?
(a) They have compact three-dimensional structure
(b) They can be without α – helical region
(c) The main chain amide and carbonyl groups are engaged in hydrogen bonding.
(d) Polar side chains are exposed to the surroundings.

116. Inner mitochondrial membrane is rich in
(a) Cation transferases / Fatty acyl transferase
(b) Sulphite oxidase
(c) Choline Phosphotransferase
(d) Citrate synthetase

117. Cytochrome oxidase deficiency in mitochondria causes
(a) Menke's disease
(b) leber's Optic Neuropathy
(c) Kearns-Says syndrome
(d) None of these

118. Which one of the following condition is at leat important for an aerobic fermentation, to optimize the bioreactor?
(a) Restriction of the entry of contaminating organism
(b) Control of parameter like pH and temperature
(c) Cutture agitation to maintain O_2 supply
(d) Maintenance of constant culture volume

119. What are the advantages of using micro – organisms for SCP productions?
(a) The quality and quantity of protein content in micro – organisms are better compared to higher plants and animals.
(b) The culture conditions and fermentation processes are very simple
(c) Micro – organisms grow at a very rapid rate under optimal culture conditions
(d) All of these

120. Which technique is pre dominantly used for separating thief and non-thief containing proteins at the purification scheme?
(a) Reverse phase chromatography
(b) Hydrophobic interaction chromatography
(c) Covalent chromatography
(d) Ion exchange

NUMERICAL TYPE QUESTIONS

1. Annihilator is considered active if it causes more than 65% inhibition. The cpm valves respectively of control, test and blank samples for inhibitors W, X, Y and Z are given below. State which of the inhibitor is active?

2. The usual standard temperature/pressure employed in autoclaving is

3. is formulated as

4. Calculate the yield coefficients $Y_{x/s}$(g dw cells / g substrate). $Y_{x/O2}$(g dw cell / g O_2) for both reactions. Comment on the differences

5. Estimate the theoretical growth and product yield coefficients for ethanol fermentation by S. cerevisiae as described by the following overall reaction
$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$

6. $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$
The total number available electrons in 1 mole of glucose are 24. The cellular yield/available electron $Y_{x/s}$ is

7. Wheat bran is pretreated with formaldehyde, then the initial p^H of the bran is adjusted to

EXERCISE – II

(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2016

1. Prandtl number is the ratio of
(a) thermal diffusivity to momentum diffusivity
(b) mass diffusivity to momentum diffusivity
(c) momentum diffusivity to thermal diffusivity
(d) thermal diffusivity to mass diffusivity

2. Fed batch cultivation is suitable for which of the following?
P. Processes with substrate inhibition
Q. Processes with product inhibition
R. High cell density cultivation
(a) P and Q only (b) P and R only
(c) Q and R only (d) P, Q and R

3. Which one of the following is the most suitable type of impeller for mixing high viscosity (viscosity > 10^5 cP) fluids?
(a) Propeller (b) Helical ribbon
(c) Paddle (d) Flat b!ade turbine

4. Which of the following events occur during the stationary phase of bacterial growth?
P. Rise in cell number stops
Q. Spore formation in some Gram-positive bacteria such as *Bacillus subtilis*
R. Cell size increases in some Gram-negative bacteria such as *Escherichia coli*
S. Growth rate of bacterial cells nearly equals their death rate
T. Decrease in peptidoglycan crosslinking
(a) P, Q and S only (b) P, S and T only
(c) Q, R and S only (d) P, R and T only

5. A bioreactor is scaled up based on equal impeller tip speed. Consider the following parameters for small and large bioreactors:

Parameters	Small bioreactor	Large bioreactor
Impeller speed	N_1	N_2
Diameter of impeller	D_1	D_2
Power consumption	P_1	P_2

Assuming geometrical similarity and the bioreactors are operated in turbulent regime, what will be P_2/P_1?
(a) $(D_1/D_2)^2$ (b) $(D_2/D_1)^2$
(c) $(D_1/D_2)^5$ (d) $(D_2/D_1)^5$

2015

6. Which one of the following relations holds true for the specific growth rate (μ) of a microorganism in the death phase ?
(a) $\mu = 0$ (b) $\mu < 0$
(c) $\mu = \mu_{max}$ (d) $0 < \mu < \mu_{max}$

7. In a fed-batch culture, 200 g . L^{-1} glucose solution is added at a flow rate of 50 L . h^{-1}. The initial culture volume (at quasi steady state) and the initial cell concentration are 600 L and 20 g . L^{-1}, respectively. The yield coefficient ($Y_{x/s}$) is 0.5 g cell mass . g substrate^{-1}. The cell concentration (g . L^{-1}) at quasi steady state at t = 8 h is
(a) 40 (b) 52
(c) 60 (d) 68

8. The diameters of a large and a small vessel are 1.62 m and 16.2 cm, respectively. The vessels are geometrically similar and operated under similar volumetric agitated power input. The mixing time in the small vessel was found to be 15 s. Determine the mixing time (in seconds) in the large vessel.
(a) 15 (b) 30
(c) 61 (d) 122

9. Biomass is being produced in a continuous stirred tank bioreactor of 750 L capacity. The sterile feed containing 8 g.L^{-1} glucose as substrate was fed at a flow rate of 150 L.h^{-1}. The microbial system follows Monod's model with μ_m = 0.4 h^{-1}, K_s = 1.5 g.L^{-1} and $Y_{x/s}$ = 0.5 g cell mass-g substrate^{-1}. Determine the cell productivity (g.L^{-1}.h^{-1}) at steady state.
(a) 0.85
(b) 0.65
(c) 0.45
(d) 0.25

2014

10. The unit for specific substrate consumption rate in a growing culture is

(a) $\frac{g}{L.h}$ (b) $\frac{g}{h}$

(c) $\frac{g}{g.h}$ (d) $\frac{gmoles}{L.h}$

11. In a batch culture of *Penicillium chrysogenum*, the maximum penicillin synthesis occurs during the

(a) lag phase (b) exponential phase

(c) stationary phase (d) death phase

12. Since mammalian cells are sensitive to shear, scale-up of a mammalian cell process must consider, among other parameters, the following (given N = rotations/time, D = diameter of impeller)

(a) πND (b) πN^2D

(c) πND^2 (d) none of these

13. Lysine is being produced in a lab-scale reactor by a threonine auxotroph. After 2 weeks of operation it was observed that the concentration of lysine in the reactor was gradually decreasing. Microbiological assays of reactor samples showed absence of contamination and recorded data showed no change in the operating conditions. The most probable reason for decrease in lysine concentration may be attributed to

(a) accumulation of ethanol

(b) growth of revertants

(c) production of citric acid

(d) unutilized phosphoenol pyruvate

2013

Statement for Linked Answer (Q. 14 - 15):

During sterilization of a fermentation medium in a given bioreactor $\nabla_{heating} = 12.56$, $\nabla_{cooling} = 7.48$ and the total value of ∇ required for whole sterilization process is 52, where ∇ is the design criteria.

14. What is the value of $\nabla_{holding}$?

(a) 31.96 (b) 42.32

(c) 52.43 (d) 61.18

15. What is the holding period (min) at a k value of 3.36 min^{-1}?

(a) 10.6 (b) 9.5

(c) 8.4 (d) 7.2

2012

16. In an exponentially growing batch culture of Saccharomyces cerevisiae, the cell density is 20 gl^{-1} (DCW), the specific growth rate (μ) is 0.4h^{-1} and substrate uptake rate (v) is 16 $gl^{-1}h^{-1}$. The cell yield coefficient $Y_{x/s}$ will be

(a) 0.32 (b) 0.64

(c) 0.80 (d) 0.50

17. Match the downstream processes in Group I with the products in Group II.

Group I	Group II
P. Solvent extraction	1. Lactic acid
Q. Protein-A linked affinity chromatography	2. Penicillin
R. Extractive distillation	3. Monoclonal antibody
S. Salting out	4. Lipase

Codes :

	P	Q	R	S
(a)	2	3	1	4
(b)	4	1	2	3
(c)	4	1	3	2
(d)	2	4	1	3

18. A bacterial culture (200 μl containing 1.8×10^9 cells) was treated with an antibiotic Z (50 μg per ml) for 4 h at 37°C. After this treatment, the culture was divided into two equal aliquots.

Set A: 100 μl was plated on Luria agar.

Set B: 100 μl was centrifuged, the cell pellet washed and plated on Luria agar.

After incubating these two plates for 24 h at 37°C, Set A plate showed no colonies, whereas the Set B plate showed 0.9×10^9 cells. This experiment showed that the antibiotic Z is

(a) bacteriostatic (b) bacteriocidal

(c) bacteriolytic (d) apoptotic

Directions (Q. 19 – 20) : *β-Galactosidase bound to DEAE-cellulose is used to hydrolyze lactose to glucose and galac lose in a plug flow bioreactor with a packed bed of volume 100 liters and a voidage (ε) of 0.55. The K'_m and V'_{max} for the immobilized enzyme are 0.72 gl^{-1} and 18 $gl^{-1}h^{-1}$, respectively. The lactose concentration in the field stream is 20 gl^{-1}, and a fractional conversion of 0.90 is desired. Diffusional limitations may be ignored.*

19. The residence time required for the steady state reactor operation will be

(a) 0.1 h (b) 0.4 h

(c) 1.0 h (d) 1.1 h

20. The feed flow rate required for the above bioconversion will be

(a) 50 lh^{-1}

(b) 55 lh^{-1}

(c) 137 lh^{-1}

(d) 550 lh^{-1}

2011

21. In balanced growth phase of a cell

P. all components of a cell grow at the same rate

Q. specific growth determined by cell number or cell mass would be the same

R. the growth rate is independent of substrate concentration

S. the growth rate decreases with decreasing substrate concentration

(a) P, Q and S only (b) Q, R and S only

(c) P, Q and R only (d) P only

22. Substrate consumption in lag phase of microbial growth is primarily used for

P. turn over of the cell material

Q. maintenance of intracellular pH

R. motility

S. increase in cell number

(a) P, Q and S only (b) Q, R and S only

(c) P, Q and R only (d) S only

23. Wash out (as defined by $D = \mu_{max}$) of a continuous stirred tank fermenter is characterized by (X = biomass, S = substrate concentration in bioreactor, S_0 = substrate concentration in feed, P = product concentration in bioreactor)

(a) X = 0, S = 0, P = 0 (b) X = 0, S = S_0, P = 0

(c) X = 0, S < 0, P = 0 (d) X < 0, S < 0, P < 0

24. Match the microbial growth characteristics in **Group I** with the corresponding features in **Group II.**

Group I

P. Growth associated product formation

Q. Non growth associated product formation

R. Product inhibition

S. Substrate inhibition

Group II

1. Specific growth rate decreases with increasing product concentration
2. Specific product formation rate is constant
3. Specific product formation rate is proportional to specific growth rate
4. Specific growth rate decreases with increasing substrate concentration

	P	Q	R	S
(a)	1	2	4	3
(b)	3	2	1	4
(c)	2	1	3	4
(d)	2	3	4	1

25. In a well aerated and agitated microbial culture, the 'supply' of oxygen is equal to 'demand' (uptake) of the growing culture. The K_La for such a system will be (K_La = volumetric mass transfer coefficient, C^* = dissolved oxygen concentration in liquid in equilibrium with gaseous oxygen, C = instantaneous value of dissolved oxygen concentration, 'r' = specific oxygen uptake rate per unit weight of cells, X = dry weight of the cells per unit volume)

(a) $(r\,X)/(C^* - C)$

(b) $(r)/X(C^* - C)$

(c) $(C^* - C)/(r\,X)$

(d) $(X)/r(C^* - C)$

26. Structured William's model

P. can describe the changes in intracellular components of the cell during growth

Q. cannot describe the death phase of the cells

R. can describe the variation of size of cells in the different phases of growth

S. cannot describe the lag period of growth

Which one of the following is correct?

(a) P, Q and S only

(b) P, Q and R only

(c) Q, R and S only

(d) P, R and S only

27. Determine the correctness or otherwise of the following **Assertion (A)** and the **Reason (R).**

Assertion : In synchronous culture, majority of the cells move to next phase of the cell cycle simultaneously.

Reason : Synchronous culture could be obtained by starving cells for essential nutrient components.

(a) both (A) and (R) are true and (R) is the correct reason for (A)

(b) both (A) and (R) are true but (R) is not the correct reason for (A)

(c) (A) is true but (R) is false

(d) (A) is false but (R) is true

28. Maximum specific growth rate (μ_{max}) of a microorganism is calculated by taking the (ln = $\log_e$, X = biomass, t = time)

(a) slope of ln X *vs* t of the growth cycle

(b) slope of ln X *vs* t during the exponential growth phase

(c) slope of X *vs* t

(d) slope of X *vs* t during the exponential phase of growth

Common Data for Questions 29 and 30:

A microorganism grows in a continuous 'chemostat' culture of 60 m^3 working volume with sucrose as the growth limiting nutrient at dilution rate, D = 0.55 h^{-1}. The steady state biomass concentration is 4.5 Kg dry biomass m^{-3} and the residual sucrose concentration is 2.0 Kg m^{-3}. The sucrose concentration in the incoming feed medium is 10.0 Kg m^{-3}.

29. What would be the yield $Y_{X/S}$ (Kg biomass/Kg substrate)?

(a) 0.562 (b) 0.462

(c) 0.362 (d) 0.162

30. What would be the sucrose concentration in the input feed for the output to be 45 Kg biomass h^{-1}?

(a) 3.225 Kg m^{-3} (b) 4.425 Kg m^{-3}

(c) 5.115 Kg m^{-3} (d) 6.525 Kg m^{-3}

2010

31. A culture of bacteria is infected with bacteriophage at a multiplicity of 0.3. The probability of a single cell infected with 3 phages is

(a) 0.9 (b) 0.27

(c) 0.009 (d) 0.027

32. In a chemostat operating under steady state, a bacterial culture can be grown at dilution rate higher than maximum growth rate by

(a) partial cell recycling

(b) using sub-optimal temperature

(c) pH cycling

(d) substrate feed rate cyling

33. A roller bottle culture vessel perfectly cylindrical in shape having inner radius (r) = 10 cm and length (l) = 20 cm was fitted with a spiral film of length (L) = 30 cm and width (W) = 20 cm. If the film can support 10^5 anchorage dependent cells per cm^2, the increase in the surface area after fitting the spiral film and the additional number of cells that can be grown respectively are

(a) 1200 cm^2 and 12×10^7 cellls

(b) 600 cm^2 and 6×10^7 cells

(c) 600 cm^2 and 8300 cells

(d) 1200 cm^2 and 8300 cells

34. Thermal death of microorganisms in the liquid medium follows first order kinetics. If the initial cell concentration in the fermentation medium is 10^8 cells / ml and the final acceptable contamination level is 10^{-3} cells, for how long should 1m^3 medium be treated at temperature of 120° (thermal deactivation rate constant = 0.23 / min) to achieve acceptable load?

(a) 48 min (b) 11 min

(c) 110 min (d) 20 min

Common Data for Question 35 and 36:

A culture of *Rhizobium* is grown in a chemostat (100 m^3 bioreactor). The feed contains 12 g/L sucrose, K_s for the organism is 0.2 g/L and $\mu_m = 0.3\ h^{-1}$.

35. The flow rate required to result in steady state concentration of sucrose as 1.5 g/L in the bioreactor will be

(a) 15 $m^3\ h^{-1}$ (b) 26 $m^3\ h^{-1}$

(c) 2.6 $m^3\ h^{-1}$ (d) 150 $m^3\ h^{-1}$

36. If $Y_{X/S}$ = 0.4 g / g for the above culture and steady state cell concentration in the bioreactor is 4 g/L the resulting substrate concentration will be

(a) 2 g/L (b) 8 g/L

(c) 4 g/L (d) 6 g/L

2009

37. A culture vessels in which physical, physicochemical and physiological conditions, as well as cell concentration, are kept constant is known as

(a) Cell concentrator (b) Biostat

(c) Batch bioreactor (d) Incubator

38. Which of the following statements are true about bioreactors?

P. Continuous bioreactors provide less degree of control and uniform product quality than batch bioreactors.

Q. Batch bioreactors are ideally suited for reaction with substrate inhibition.

R. Choice of a bioreactor is dictated by kinetic considerations.

S. Fed batch bioreactors are also called semibatch bioreactors.

(a) P, Q

(b) Q, S

(c) R, S

(d) P, R

39. Determine the correctness or otherwise of the following Assertion (a) and Reason (r):

Assertion (a): Bacterial growth is called synchronous when majority of the cells are in same stage of the bacterial cell cycle.

Reason (r) : Synchronous culture can be obtained by growing bacteria in an enriched medium.

(a) Both (a) and (r) are true and (r) is the correct reason for (a)

(b) Both (a) and (r) are true but (r) is not the correct reason for (a)

(c) (a) is true but (r) is false

(d) (a) is false but (r) is true

2008

40. Diauxic pattern of biomass growth is associated with

(P) multiple lag phases

(Q) sequential utilization of multiple substrates

(R) simultaneous utilization of multiple substrates

(S) absence of lag phase

(a) P, R (b) P, Q

(c) R, S (d) Q, S

41. Match the bioreactor components in group 1 with the most appropriate function given in group 2

Group 1	Group 2
P. Marine type impeller	1. Recirculation of medium
Q. Draft tube	2. Aeration of medium
R. Diaphragm valve	3. Animal cell cultivation
S. Sparger	4. Sterile operation

	P	Q	R	S
(a)	4	2	1	3
(b)	3	1	4	2
(c)	3	4	2	1
(d)	2	1	4	3

42. A bacterial culture with an approximate biomass composition of $CH_{1.8}O_{0.5}N_{0.2}$ is grown aerobically on a defined medium containing glucose as the sole carbon source and ammonia being the nitrogen source. In this fermentation, biomass is formed with a yield coefficient of 0.35 gram dry cell weight per gram of glucose and acetate is produced with a yield coefficient of 0.1 gram acetate per gram of glucose. The respiratory coefficient for the above culture will be

(a) 0.90 (b) 0.95

(c) 1.00 (d) 1.05

43. A bacterial culture having a specific oxygen uptake rate of 5 mmol O_2 $(g\text{-}DCW)^{-1}\ hr^{-1}$ is being grown aerobically in a fed-batch bioreactor. The maximum value of the volumetric oxygen transfer coefficient is $0.18s^{-1}$ for the stirred tank bioreactor and the critical dissolved oxygen concentration is 20% of the saturation concentration (8mg/ml). The maximum density to which the cells can be grown in the fed-batch process without the growth being limited by oxygen transfer, is approximately.

(a) 14 g/l (b) 26 g/l

(c) 32 g/l (d) 65 g/l

2007

44. The specific growth rate (μ) of a microorganism in death phase is

(a) 0 (zero) (b) μ_{max}

(c) less than zero (d) greater than zero

45. Main functions of baffles in a bioreactor are

P. to prevent a vortex

Q. to increase aeration

R. to reduce interfacial area of oxygen transfer

S. to reduce aeration rate

(a) P, Q (b) Q, R

(c) R, S (d) P, S

46. How many kilograms of ethanol is produced from 1 kilogram of glucose in ethanol fermentation?

(a) 2.00 (b) 0.20

(c) 0.51 (d) 0.05

47. Downstream processing of an industrial process yielded a highly purified bioactive protein. This protein was subjected to cleavage by trypsin. Chromatographic separation of products resulted in 4 peptides (P, Q, R, S) with the following amino acid sequences

P. phe-val-met-val-arg

Q. ala-ala-try-gly-lys

R. val-phe-met-ala-gly-lys

S. phe-gly-try-ser-thr

Chemical cleavage of the same protein with cyanogenbromide and chromatographic separation resulted in three peptides (*i*, *ii*, *iii*) with the following sequences

(*i*) ala-gly-lys-phe-gly-try-ser-thr

(*ii*) ala-ala-pry-gly-lys-phe-val-met

(*iii*) val-arg-val-phe-met

The order of the peptides that gives the primary structure of the original protein is

(a) P, Q, R, S (b) Q, P, R, S

(c) Q, R, P, S (d) R, Q, P, S

Statement for Linked Answer Questions 48 and 49 :

In a Fed-batch culture glucose solution is added with a flow rate of 2 m^3/day. The initial volume of the culture is 6m^3.

48. The volume of culture at the end of second day (neglect loss due to vaporization) is

(a) 6 m^3

(b) 8 m^3

(c) 10 m^3

(d) 12 m^3

49. What would be the dilution rate of the system at the end of second day?

(a) 2.00

(b) 0.20

(c) 0.02

(d) 0.01

Statement for Linked Answer Questions 50 & 51 :

Absence of cellulosic cell wall, high β-carotene content and GRAS status make *Dunaliella salina* a good model system for producing edible vaccines. 10^9 Cells of D. *salina* were electroporated with a high expression DNA vector containing an antigenic gene.

50. If 10^3 cells survived after electroporation, how many cells were killed during this process (round of to the nearest number) ?

(a) 10^9 (b) 10^8

(c) 10^6 (d) 10^5

51. The antigen is expressed as transmemb-rane protein with a single epitope on its extracellular domain. The cells that survived (assume 100% transfection and expression of protein) were incubated with a radio labeled Fab fragment (specific activity: 100 cpm/picomole) against this epitope. After washing, the cell pellet has 1000 cpm. The average number of epitopes present on a single recombinant alga are

(a) 6×10^9 (b) 1×10^9

(c) 6×10^3 (d) 1×10^6

2006

52. During cultivation of microorganisms in a fermenter, various parameters are controlled by appropriate sensor (probe). Match each probe in group 1 with the appropriate response mechanism in group 2.

Group 1 (Probe)	Group 2 (Response)
P. Thermistor	1. Activation of acid alkali pump
Q. Oxygen electrode	2. Activation of vegetable oil pump
R. Metal rod	3. Activation of hot cold water pump
S. pH electrode	4. Increase/decrease in stirrer motor speed

(a) P-2, Q-3, R-1, S-4 (b) P-1, Q-2, R-4, S-3

(c) P-3, Q-2, R-4, S-1 (d) P-3, Q-4, R-2, S-1

53. Which one of the following monolayer culture systems have the highest surface area: medium ratio?

(a) Roux bottle (b) Spiracell roller bottle

(c) Hollow fibres (d) Plastic bag/film

Statement for Linked Answer Questions 54 & 55 :

A bioreactor of working volume 50 m^3 produces a metabolite (X) in batch culture under given operating conditions from a substrate (S). The final concentration of metabolite (X) at the end of each run was 1.1 kg m^{-3}. The bioreactor was operated to complete 70 runs in each year.

54. What will be the annual output of the system (production of metabolite (X)) in kg per year ?

(a) 55 (b) 3850

(c) 45.5 (d) 77

55. What will be the overall productivity of the system in kg $year^{-1}$ m^{-3} ?

(a) 19250 (b) 38.50

(c) 3850 (d) 77

NUMERICAL TYPE QUESTIONS

2016

1. The power required for agitation of non-aerated medium in fermentation is _____ kW.

Operating conditions are as follows:

Fermentor diameter = 3 m

Number of impellers = 1

Mixing speed = 300 rpm

Diameter of the Rushton turbine = 1 m

Viscosity of the broth = 0.001 Pa.s

Density of the broth = 1000 $kg.m^{-3}$

Power number = 5

2. The empirical formula for biomass of an unknown organism is $CH_{1.8}O_{0.5}N_{0.2}$. To grow this organism, ethanol (C_2H_5OH) and ammonia are used as carbon and nitrogen sources, respectively. Assume no product formation other than biomass. To produce 1 mole of biomass from 1 mole of ethanol, the number of moles of oxygen required will be _____.

3. *Saccharomyces cerevisiae* is cultured in a chemostat (continuous fermentation) at a dilution rate of 0.5 h^{-1}. The feed substrate concentration is 10 $g.L^{-1}$. The biomass concentration in the chemostat at steady state will be _____ $g.L^{-1}$.

Assumptions: Feed is sterile, maintenance is negligible and maximum biomass yield with respect to substrate is 0.4 (g biomass per g ethanol).

Microbial growth kinetics is given by

$$\mu = \frac{\mu_m S}{K_S + S}$$

where μ is specific growth rate (h^{-1}), $\mu_m = 0.7\ h^{-1}$, $K_s = 0.3\ g.L^{-1}$ and s is substrate concentration ($g.L^{-1}$).

4. Decimal reduction time of bacterial spores is 23 min at 121°C and the death kinetics follow first order. One liter medium containing 10^9 spores per mL was sterilized for 10 min at 121°C in a batch sterilizer. The number of spores in the medium after sterilization (assuming destruction of spores in heating and cooling period is negligible) will be _____ $\times 10^7$.

5. An enzyme converts substrate A to product B. At a given liquid feed stream of flow rate 25 $L.min^{-1}$ and feed substrate concentration of 2 $mol.L^{-1}$, the volume of continuous stirred tank reactor needed for 95% conversion will be _____ L.

Given the rate equation: $-r_A = \frac{0.1C_A}{1+0.5C_A}$

where $-r_A$ is the rate of reaction in $mol.L^{-1}.min^{-1}$ and C_A is the substrate concentration in $mol.L^{-1}$

Assumptions: Enzyme concentration is contant and does not undergo any deactivation during the reaction.

2015

6. A synchronous culture containing 1.8×10^5 monkey kidney cells was seeded into three identical flasks. The doubling time of these cells in 24 h. After 24 h, the cells from all the three flasks were pooled and dispensed equally into each well of three 6-well plates. The number of cells in each well will be _____ $\times 10^4$.

7. Oxygen transfer was measured in a stirred tank bioreactor using dynamic method. The dissolved oxygen tension was found to be 80% air saturation under steady state conditions. The measured oxygen tensions at 7s and 17s were 55% and 68% air saturation, respectively. The volumetric mass transfer coefficient K_{La} is _____ s^{-1}

8. Samples of bacterial culture taken at 5 PM and then the next day at 5 AM were found to have 10^4 and 10^7 cells.mL^{-1}, respectively. Assuming that both the samples were taken during the log phase of cell growth, the generation time of this bacterium will be _____ h.

9. *Saccharomyces cerevisiae* produces ethanol by fermentation. The theoretical yield of ethanol from 2.5 g of glucose is _____ g.

2014

10. A T-flask is seeded with 10^5 anchorage-dependent cells. The available area of the T-flask is 25 cm^2 and the volume of the medium is 25 ml. Assume that the cells are rectangles of size 5 μm × 2 μm. If the cells grow to monolayer confluence after 50 h, the growth rate in number of cells/(cm^2.h) is _____ $\times 10^5$.

11. Consider a continuous culture provided with a sterile feed containing 10 mM glucose. The steady state cell density and substrate concentration at three different dilution rates are given in the table below.

Dilution rate (h^{-1})	Cell density ($g\ L^{-1}$)	Substrate concentration (mM)
0.05	0.248	0.067
0.5	0.208	1.667
5	0	10

The maximum specific growth rate μ_m (in h^{-1}), will be _____.

2013

12. A callus of 5 g dry weight was inoculated on semi-solid medium for growth. The dry weight of the callus was found to increase by 1.5 fold after 10 days of inoculation. The growth index of the culture is _____

13. A chemostat is operated at a dilution rate of 0.6 h^{-1}. At steady state, the biomass concentration in the exit stream was found to be 30 $g\ l^{-1}$. The biomass productivity ($g\ l^{-1}h^{-1}$) after 3h of steady state operation will be _____

14. A batch bioreactor is to be scaled up from 10 to 10,000 liters. The diameter of the large bioreactor is 10 times that of the small bioreactor. The agitator speed in the small bioreactor is 450 rpm. Determine the agitator speed (rpm) of the large bioreactor with same impeller tip speed as that of the small bioreactor. _____

15. Calculate the percentage sequence identity for the pairwise alignment given below._____

HELLO-

YELLOW

16. In a batch culture, the specific rate of substrate utilization is 0.25 g (g cell mass)$^{-1}$ h^{-1} and specific rate of product formation is 0.215 g (g cell mass)$^{-1}$ h^{-1}. Calculate the yield of product from the substrate (Yp/s). _____

17. The maximum cell concentration ($g\ l^{-1}$) expected in a bioreactor with initial cell concentration of 1.75 $g\ l^{-1}$ and an initial glucose concentration of 125 $g\ l^{-1}$ is ($Y_{x/s}$ = 0.6 g cell/g substrate)_____

18. A fed batch culture was operated with intermittent addition of glucose solution at a flow rate of 200 ml h^{-1}. The values of K_s, μ_m and D are 0.3 $g\ l^{-1}$, 0.4 h^{-1} and 0.1 h^{-1}, respectively. Determine the concentration of growth limiting substrate (gl^{-1}) in the reactor at quasi-steady state. _____

ANSWERS

EXERCISE – I

MCQ Type Questions

1. (d)	2. (b)	3. (c)	4. (c)	5. (d)	6. (c)	7. (a)	8. (d)	9. (a)	10. (b)
11. (b)	12. (d)	13. (c)	14. (b)	15. (b)	16. (d)	17. (c)	18. (c)	19. (b)	20. (c)
21. (d)	22. (c)	23. (c)	24. (b)	25. (a)	26. (a)	27. (c)	28. (c)	29. (d)	30. (b)
31. (a)	32. (d)	33. (a)	34. (a)	35. (b)	36. (d)	37. (c)	38. (a)	39. (a)	40. (c)
41. (d)	42. (a)	43. (a)	44. (b)	45. (b)	46. (c)	47. (a)	48. (c)	49. (b)	50. (a)
51. (c)	52. (a)	53. (d)	54. (c)	55. (c)	56. (a)	57. (a)	58. (a)	59. (b)	60. (d)
61. (d)	62. (c)	63. (b)	64. (a)	65. (c)	66. (b)	67. (b)	68. (c)	69. (c)	70. (a)
71. (b)	72. (c)	73. (c)	74. (b)	75. (a)	76. (a)	77. (c)	78. (c)	79. (b)	80. (b)
81. (d)	82. (b)	83. (c)	84. (b)	85. (c)	86. (c)	87. (b)	88. (c)	89. (d)	90. (c)
91. (b)	92. (c)	93. (b)	94. (d)	95. (d)	96. (d)	97. (b)	98. (a)	99. (a)	100. (b)
101. (c)	102. (d)	103. (c)	104. (b)	105. (b)	106. (c)	107. (c)	108. (b)	109. (c)	110. (c)
111. (b)	112. (a)	113. (c)	114. (b)	115. (b)	116. (a)	117. (c)	118. (c)	119. (d)	120. (c)

Numerical Type Questions

1. Z-7200, 2800, 200 2. 121°C/15 psi 3. $0.693/t_d$

4. For hexadecane; $Y_{x/O_2} = 0.557$ gdw cells / g O_2; For glucose; $Y_{x/O_2} = 1.76$ gdw cells / g O_2

5. $Y_{x/s} = 0.117$ gdw / g glucose, $Y_{p/s} = 0.51$ ethanol / g glucose, $Y_{CO_2/s} = 0.49$ g ethanol / g glucose

6. $Y_{x/s} = 76$ gdw cell / mol 7. $p^H = 3.5$ to 4.0

EXERCISE – II

MCQ Type questions

1. (c)	2. (b)	3. (b)	4. (a)	5. (b)	6. (b)	7. (b)	8. (c)	9. (b)	10. (c)
11. (c)	12. (a)	13. (b)	14. (a)	15. (b)	16. (d)	17. (a)	18. (a)	19. (*)	20. (*)
21. (a)	22. (c)	23. (b)	24. (b)	25. (a)	26. (b)	27. (b)	28. (b)	29. (a)	30. (*)
31. (d)	32. (a)	33. (b)	34. (c)	35. (b)	36. (a)	37. (b)	38. (c)	39. (c)	40. (b)
41. (b)	42. (b)	43. (d)	44. (a)	45. (a)	46. (c)	47. (c)	48. (c)	49. (b)	50. (a)
51. (a)	52. (d)	53. (c)	54. (b)	55. (d)					

Numerical Type Questions

1. (625.0 : 625.0) 2. (1.9 : 2.0) 3. (3.65 : 3.75) 4. (3.6 : 3.8) 5. (4986.0 : 4989.0) 6. (2)
7. (0.07339) 8. (1.2121) 9. (1.20) 10. (2 to 2) 11. (0.795 to 0.805) 12. (0.5)
13. (18) 14. (45) 15. (66.5 to 66.7) 16. (0.81 to 0.91) 17. (76.7 to 76.9) 18. (0.1)

EXPLANATIONS

EXERCISE – I

MCQ TYPE QUESTIONS

1. The major use for ethylene oxide is as a chemical intermediate in industry. The acute (short-term) effects of ethylene oxide in humans consist mainly of central nervous system (CNS) depression and irritation of the eyes and mucous membranes. Chronic (long-term) exposure to ethylene oxide in humans can cause irritation of the eyes, skin, and mucous membranes, and problems in the functioning of the brain and nerves. Relatively small amounts of ethylene oxide are used as a fumigant, a sterilant for food (spices) and cosmetics, and in hospital sterilization of surgical equipment and plastic devices that cannot be sterilized by steam.

2. Collagenases are endopeptidases that digest native collagen in the triple helix region. They are mainly produced by the bacterium Flavobacterium sp. Collagens are the major fibrous component of animal extracellular connective tissue. Bacterial collagenases differ from vertebrate collagenases in that they exhibit broader substrate specificity.

3. Tyndallization is an old process for sterilizing food. A simple, effective, modern sterilizing method is to heat the substance being sterilized to 121°C for 15 minutes in a pressured system. Tyndallization essentially consists of heating the substance for 15 minutes for three days in a row (usually by boiling it). On the second day most of the spores that survived the first day will have germinated into bacterial cells. These cells will be killed by the second day's heating.

4. Depth filters comprise fibres or fibrous materials that trap particles as they pass through the tortuous path between fibres. Interception of particles is common for large particles. If a large enough particle follows the streamline that lies very close to the media surface it will hit the media grain and be captured. The theory is based on the general phenomenological expression for the rate of particle deposition on the membrane surface, in which the deposition rate is proportional to the product of the particle deposition coefficient and the concentration of suspended particles, with the deposition coefficient being an arbitrary function of the mass of deposited particles and permeate velocity.

5. More dust manufacturers and end users are choosing air filter cartridge for high filtration efficiency, reasonable size and cost of equipment in the field of industrial dust. When we design the air filter cartridge, we firstly need to determine the de-dusting, material recovery or air purification of all. Filter selection needs to consider the following factors: temperature, dust concentration, water, pH, conductivity and so on.

6. Perstraction is a separation process in which membrane permeation and extraction phenomena occur by contacting the downstream with an extracting solvent. It is a method for separating a noble metal catalyst from a crude reaction product of a noble metal-catalyzed hydroformylation reaction which comprises: (a) contacting the crude reaction product under perstraction conditions with a membrane capable of allowing a substantial portion of unreacted olefin feed and hydroformylation reaction product to pass there through as permeate while retaining a substantial portion of the catalyst as retentate; (b) removing the permeate by sweeping it away from the membrane by means of a sweep stream which is the same as the olefin feed used in the hydroformylation reaction; and (c) retaining the catalyst as retentate.

7. The mean doubling time td of growth of biomass is given by: $t_d = 0.693/$

The mean generation time or "doubling time" (g) is the average time required for all the components of the culture to double. During the exponential (or logarathmic) growth phase, a bacterial culture mimics a first-order chemical reaction, i.e. the rate of increase of cells is proportional to the number of bacteria present at that time. The constant of proportionality, , is an index of the growth rate and is called the growth rate constant:

Rate of increase of cells = × number of cells

The value of can be determined from the following equation:

$\ln Nt - \ln N_0 = (t - t_0)$

8. Inertial impaction is the deposition of large aerosol particles on the walls of an airway conduit. The impaction tends to occur where the airway direction changes. Small particles have less inertia and are more likely to be carried around corners and continue in the path of the airflow. Interception of particles is common for large particles. If a large enough particle follows the streamline that lies very close to the media surface it will hit the media grain and be captured. Diffusion is the flow of solute particles from a solution of higher concentration to a solution of lower concentration. Thus, all of these mechanisms are involved in the separation of contaminants.

9. Polyethersulfone (PES) membrane is a hydrophilic membrane constructed from pure polyethersulfone polymer membrane. PES membrane filters are designed to remove particulates during general filtration. Their low protein and drug binding characteristics also make polyethersulfone (PES) membrane disc filters ideally suited for use in life science applications.

10. Filter sterilization is useful for sterilizing heat labile liquids. The filters have pores of specific sizes down to 0.01 micron in diameter. Three types of filters that are used:

Depth filters - routine filters made of paper, asbestos or glass fibres.

Membrane filters - cellulose acetate or cellulose nitrate materials. It contains lots of tiny holes that act live sieves to retain bacteria.

Nucleopore filters - polycarbonate filters. Very uniform pore sizes. Great for microscopy work since the filters are flat in nature.

11. Cellulose fibres serve as mediums for holding the large particulate matter and allow the fluid form to pass through it by sieving which is the process of sedimentation. Celite has high porosity, because it is composed of microscopically small, coffin-like, hollow particles and is used as a filtration aid, to filter very fine particles that would otherwise pass through or clog filter paper. Porous materials are used for filtration of fine particles through inertial impaction. Glass wool serves the purpose of filtration by diffusing the materials from high to low concentration.

12. Mixing is an essential process in fermentation. There are two sources of energy in a fermentation process: Mixing impellers provide circulation of fluid in the tank, and the exoansion and velocity of air passing through a fermenter impart fluid motion.

13. Two basic approaches to the fed-batch fermentation can be used: the constant volume fed-batch culture - Fixed Volume Fed-Batch - and the Variable Volume Fed-Batch. In fixed volume type of fed-batch, the limiting substrate is fed

without diluting the culture. A variable volume fed-batch is one in which the volume changes with the fermentation time due to the substrate feed. The way this volume changes it is dependent on the requirements, limitations and objectives of the operator. Fed-batch fermentation is a production technique in between batch and continuous fermentation.

14. High-level disinfectants are chemicals that can act as sterilants if an increased exposure time is utilised. High-level disinfectants for endoscope reprocessing need to have wide anti-bacterial properties together with the ability to kill relevant viruses including HIV, HBV and HCV. But for glasswares, a sterilant should be highly toxic and extremely volatile.

15. Continuous cultures are ongoing cultures which continue to grow from batch to batch without any replacement of the inoculums and constant supply of the nutrient media. Continuous cultures are more efficient and beneficial for industrial purposes. Any industrial strain for a particular process can be best known from its continuous culture growth kinetics.

16. A force is generated at the water/air interface that is based on forces of attraction (dipole-dipole interactions and hydrogen bonds) between the water molecules. Since each water molecule at the surface interacts more strongly with the other water molecules below it, i.e. inside the liquid, and beside it in the boundary layer than it does with the molecules above it in the air, this leads to what is known as surface tension. It shows up as the attempt by the liquid to minimize its surface area. Antifoams are substances that lower the surface tension of a liquid. At the water's surface, the antifoam particles align themselves such that the hydrophilic head projects into the water and the hydrophobic tail projects out of the surface of the water. Since the forces of attraction between the non-polar hydrocarbon groups of the antifoam particles are much lower than between the water molecules, the surface tension of the water is lowered.

17. Certain industrial nitrogen sources include soybean meal, corn steep liquor, $(NH_4)_2SO_4$ and NH_4NO_3. The corn steep liquor being rich in nutrients finds a use as a feedstock for the organisms used in the industrial production of antibiotics such as penicillin.

18. Thermal treatment is mostly done for sterilization of the media. This is so because, in addition to being economical, heat treatment allows a high degree of sterility to the media destroying a majority of the microbes that may be present in the media and contaminate it. The sterilized media is majorly free from microbes on the application of high levels of temperature.

19. The solubility of oxygen in aqueous solution is approximately about 10ppm for ambient temperature and pressure.

20. In situ sterilization of a fermentor involves the passing of steam water through the fermentor thereby de-contaminating the fermentor from a lot of microbes and hence preparing the fermentor for further processes. And in continuous heat sterilization, heating coils are surrounding the fermentor which raises the temperature to higher levels and thus reduces the level of contamination. But in case of in situ sterilization, the steam passed may have varying temperature and the heating effect is not uniform. Also, steam can corrode the material of the fermentor.

21. The process used for the purpose of sterilization of the media containing thermo labile substances and spores is known as Tyndallization. A simple, effective, modern sterilizing method is to heat the substance being sterilized to 121°C for 15 minutes in a pressured system. Tyndallization essentially consists of heating the substance for 15 minutes for three days in a row (usually by boiling it).

22. Ultrafiltration is a low-pressure membrane process used to separate high molecular weight compounds from a feed stream. Ultrafiltration has larger pores than nanofiltration and reverses osmosis and is therefore the least costly of the three to operate. Ultrafiltration is useful for the separation of delicate materials since it is a non-denaturing method of separation. Cross flow membrane filtration is commonly used for the concentration and purification of proteins and other biological particles from suspensions. Other uses include production of ultra pure water, egg and animal blood processing, clarification of juice, downstream processing, membrane bioreactors, treatment of bleach plant effluents, and recovery of lignin compounds in the pulp and paper industry.

23. In steady state, the population of the cell or biomass in a fermentor during a fermentation reaction is maintained mainly in case of a continuous type of culture. This is because with the formation of primary and secondary metabolites and the cell reaching the stationary phase of growth, more exponential phase cells are inoculated with fresh and sterilized media which revive and restore the cell growth and number, thereby maintaining the steady population of the cells.

24. Silicone-based defoamers are polymers with silicon backbones. The silicone compound consists of an hydrophobic silica dispersed in a silicone oil. Emulsifiers are added to ensure that the silicone spreads fast and well in the foaming medium. The silicone compound might also contain silicone glycols and other modified silicone fluids. Silicone based defoamers are also suitable in non-aqueous foaming systems like crude oil and oil refining. For very demanding applications fluorosilicones may be suitable.

29. Mass transfer in a bioreactor is mostly affected by the intensity of mixing of the impeller and the sparging of air through the sparger. Also, bubble size of the fluids in the media, particularly in a bubble column reactor also affects the efficiency of mass transfer. Apart from that, composition of the media and the feed stream along with density and viscosity also matter in resistance towards the free flowing mass transfer in the reactor vessel.

30. Bubbles in an air reactor column are essential for effective mass transfer throughout the vessel through the dispersion or diffusion of oxygen and other solutes. Bubbles have a high retention power of gas and can store gas molecules for a long interval of time. The interfacial surface is small which leads to easy diffusion of the gases.

31. The determination of K_La of a fermenter is essential in order to establish its aeration efficiency and to quantify the effects of operating variables on the provision of oxygen. The volumetric mass transfer coefficient, K_La, indicates the rate of oxygen used for fermentation, taking into account all oxygen-consuming molecules in the medium. The size of the bubble mostly affects the value of K_L of the reactor.

32. An additive which reduces the surface tension of a solution or emulsion, thus inhibiting or modifying the formation of foam is called and antifoam. Commonly used agents are insoluble oils, dimethyl polysiloxanes and other silicones, certain alcohols, stearates and glycols. The additive is used to prevent formation of foam or is added to break foam already formed.

33. To maintain the non-coalescing character of the medium and to ensure a high value of the volumetric feed rate, majorly mechanical method of disrupting the cells are used so that the bubbles do not coalesce with each other and the volumetric mass transfer coefficient, K_La remains unaffected. Various mechanical methods like use of stirrer, impeller foam breaker can be employed to maintain the high fidelity of the reactor vessel.

34. Due to reduction of the surface tension in bubbles, there is a considerable reduction in the diameter of the bubbles as well. Due to reduction in the diameter of the bubbles, the bubbles regain a small spherical shape having very low interfacial surface area. This results in increase in the overall surface area i.e., 'a' of the smaller spherical bubble which increases the extent of gas holding capacity of the bubble.

35. High temperature of the fermentation broth and the fermentation vessel affect the mass transfer and the oxygen transfer rate of the medium. The rise in temperature results in vaporizing and uprising of the bubbles which hampers the mass transfer coefficient as a whole.

36. In aerobic bioprocesses, oxygen is a key substrate; due to its low solubility in broths (aqueous solutions), a continuous supply is needed. The oxygen transfer rate (OTR) must be known, and if possible predicted to achieve an optimum design operation and scale-up of bioreactors. The gas-liquid mass transfer in a bioprocess is strongly influenced by the hydrodynamic conditions in the bioreactors. These conditions are known to be a function of energy dissipation that depends on the operational conditions, the physicochemical properties of the culture, the geometrical parameters of the bioreactor and also on the presence of oxygen consuming cells. The dissolved oxygen concentration in a suspension of aerobic microorganisms depends on the rate of oxygen transfer from the gas phase to the liquid, on the rate at which oxygen is transported into the cells (where it is consumed), and on the oxygen uptake rate (OUR) by the microorganism for growth, maintenance and production.

37. In a polarographic system the measurement device (transducer) is a Clark oxygen electrode. The processing/coupling device is an oxygen monitor. The recording device is a flatbed strip chart recorder, or perhaps a computer-assisted data acquisition system.The solution or suspension to be assayed is placed in a sealed chamber that is exposed to the surface of a Clark oxygen electrode. This dissolved oxygen (D.O.) chamber contains one or more access ports for adding/removing materials. The medium is stirred to ensure homogeneity and to ensure that oxygen can freely diffuse into the electrode. The presence of oxygen causes the electrode to deliver a current to the oxygen monitor, which amplifies the current and converts it to a voltage output that is directly proportional to the concentration of oxygen in the chamber. The overall method is a gassing out technique to find out the oxygen concentration.

38. There are many dynamic methods for measuring the volumetric mass transfer coefficient. The "gas out-gas in" method can directly determine the volumetric mass transfer coefficient in a bioreactor system and provide estimates of the volumetric microbial oxygen uptake rate and the average oxygen saturation concentration at the gas-liquid interface. The oxygen transfer rate is dependent upon the volumetric mass transfer coefficient, K_La, and the driving force, $C_L^* - C_L$, (where C_L, is the dissolved oxygen concentration and C_L^* is the oxygen saturation concentration in the liquid phase at the gas-liquid interface).

39. In turbulent flow vortices, eddies and wakes make the flow unpredictable. Turbulent flow happens in general at high flow rates and with larger pipes. Shear stress for turbulent flow is a function of the density-ρ. The Reynolds number (Re) is proportional to inertial force divided by viscous force. The flow is laminar when Re < 2300; transient when 2300 < Re < 4000; turbulent when 4000 < Re. Thus, for mixing to human, turbulent fluid flow is required.

Q. 42 & 43

According to Contois equation, the specific growth rate decreases with decreasing substrate concentrations & eventually becomes inversely proportional to the cell concentration in the medium ($_g \alpha x^{-1}$)

44. Substrate inhibition – At high substrate concentrations, microbial growth rate is inhibited by the substrate.

The major substrate inhibition pattern & expression.

Non-competitive substrate inhibition :

$$g = \frac{\mu_m}{\left(1+\frac{K_S}{S}\right)\left(1+\frac{S}{k_1}\right)}$$

or if $K_1 >> K_S$, then : $g = \frac{\mu_m S}{k_s + s + \frac{s^2}{k_1}}$

45. Product inhibition – High concentration of product can be inhibitory for microbial growth. Product inhibition may be competitive / noncompetitive,

Competitive product inhibition :

$$g = \frac{\mu_m s}{k_S\left(1+\frac{P}{k_P}\right)+S}$$

Non – competive product inhibition :

$$g = \frac{\mu_m}{\left(1+\frac{k_S}{S}\right)\left(1+\frac{P}{k_P}\right)}$$

46. All reactions should be expressed in terms of intrinsic concentration. An intrinsic concentration is the amount of a compound/unit cell/cell volume. Extrinsic concentrations cannot be used in kinetic expression.

47. Cybernetic models approach has been developed primarily to predict growth under conditions when substrates are available. One approach to modeling growth on multiple substrates is a cybernetic approach. Cybernetic means that a process is goal seeking (e.g. maximization of growth rate).

Q. 48 & 49

In cybernetic model the substrates may be complementary (e.g: carbon/nitrogen) or substitutable. For example, glucose & lactose would be substitutable, as these compounds both supply carbon & energy.

52. Spectrophotometry rapid method (Direct method) for determining cell mass concentration is based on the absorption of light by suspended cells in sample culture media. The intensity of the transmitted light is measured using a spectrophotometer.

53. Intracellular components of cells such as RNA, DNA, & protein can be measured as indirect measures of cell growth.

54. During a batch growth cycle, the concentrations of RNA, DNA & protein intracellular components change with time. Concentration of RNA (RNA/ cell weight) varies significantly during a batch growth cycle; however, DNA & protein concentrations remain fairly constant. Therefore, in a complex medium, DNA concentration can be used as a measure of microbial growth.

56. The intracellular ATP concentration (mg ATP/mg cells) is approximately constant for a given organism. Thus, the ATP concentration in a fermentation broth can be used a measure of biomass concentration. The method is based in luciferase activity, which catalyzes oxidation of luciferin at the expense of O_2 & ATP with the emission of light.

57. Certain products produced under anaerobic conditions, such as ethanol & lactic acid, can be related nearly stoichiometrically to microbial growth. Products must be eight growth associated (ethanol) or mixed growth associated (lactic acid) to be correlated with microbial growth.

58. In some cases, changes in the pH/acid – base addition to control pH can be used to monitor nutrients uptake & microbial growth when nitrate is used as the nitrogen source, hydrogen ions are removed from the medium, resulting in an increase in pH. In this case, the amount of acid added is proportional to nitrate uptake & therefore to microbial growth.

59. In some fermentation process, as a result of mycelial growth/extra cellular polysaccharide formation, the viscosity of the fermentation broth increases during the course of fermentation. If the substrate is biodegradable polymer, such as starch/cellulose, then the viscosity of the broth decreases with time as biohydrolysis continues changes in the viscosity of the fermentation broth can be correlated with the next to microbial growth.

61. To minimize the duration of the lag phase, cells should be adapted to the growth medium & conditions before inoculation, & cell should be young (or exponential phase cells) & active, & the inoculation size should be large (5% to 10% by volume). The nutrient medium may need to be optimized & certain growth factors induced to minimize the lag phase.

62. During balanced growth, the net specific growth rate determined from either cell number/cell mass would be the same. Since the nutrient concentrations are large in this phase, the growth rate is independent of nutrient concentration.

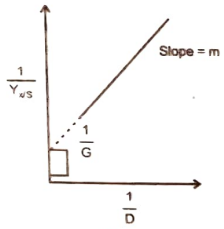
63. The stationary phase starts at the end of the deceleration phase, when the net growth rate is zero (no cell division)/ when the growth rate is equal to the death rate. Even through the net growth rate is zero during the stationary phase.

64. The rate of exponential growth of a bacterial culture is expressed as **generation time**, also the **doubling time** of the bacterial population. Generation time (G) is defined as the time (t) per generation (n = number of generations). Hence, G = t/n is the equation from which calculations of generation time (below) derive.

65. Increase in cell number is because grow by binary fusion doubling in cell number usually occurs at the same rate individuals cells grow & divide.

75. During steady – state growth $\frac{dx}{dt} = 0$, consequent $= D$

Equation shows that simply by varying the medium supply rate the growth rate can be varied. This holds until D > $_m$ (the maximum specific growth rate)



Determination of $\frac{1}{D}$ the growth yield, Y_G, & maintenance energy requirement.

This slope does not pass through the origin due to the requirement for substrate consumption for the maintenance of cell integrity & viability, the slope also shown that the yield coefficient indeed changes with growth rate. Especially the maintenance energy is high

77. Frequently, deviations from this ideal behavior of chamostats are found in particle which are due to the Monod equation being on approximation, only, in, addition to variations in substrate & cell conc. which are not accounted for in the simple theory.

78. Advantages of continuous culture over batch culture are

(*a*) The growth rate can be controlled & maintained indefinitely

(*b*) Biomass conc. can be maintained constant by varying the dilution rate

(*c*) Secondary metabolite production can be maintained simultaneously with growth

(*d*) Substrate – limited growth can be maintained & changes in cell composition & metabolic activity studied by changing the growth – limiting nutrient

80. Disadvantages of continuous culture compared to Batch culture

(*a*) Production of some non-growth related products can not always be achieved & often requires fed – batch culture techniques

(*b*) Wall growth & cell aggregation can cause was – out / prevent true steady – state growth

(*c*) Growth over long periods may result in problems of contamination

(*d*) Continous growth over long periods can result in the loss of the original strain due to take – over by a faster growing stain

81. For the scale up of gas liquid contractor, the volumetric mass transfer coefficient K_L a can be used as scale up criterion

84. For hexadecame

Amount of carbon in 1 mole of substrate

$= 16(2) = 192g$

Amount of carbon converted to biomass

$= 192\ (2/3) = 128g$

Then, $128 = C(4.4)\ (12)$; $C = 2.42$

Amount of carbon converted to

$CO_2 = 191 - 128 = 64g$

$64 = e(12).\ e = 5.33$

The nitrogen balance yields

$14b = C(0.86)\ (14)$

$b = (2.42)\ (0.86) = 2.085$

The hydrogen balance is

$34(1) + 3b = 7.3C + 2d$

$d = 12.43$

For oxygen balance yields

$2a(16) = 1.2C\ (16) + 2e\ (16) + d(16)$

$a = 12.427$

For glucose

Amount of carbon in 1 mole of subtrase = 72g

Amount of carbon converted to biomass

$= 72\ (2/3) = 48g$

Then, 48 = 4.4C (12); C = 0.909

Amount of carbon converted to

CO_2 = 72 – 48 = 24kg

24 = 12es e = 2

The nitrogen balance yields

L4b = 0.86C(14)

b = 0.782

The hydrogen balance is

12 + 3b = 7.3C + 2d

d = 3.854

The C_2 balance yields

6(16) + 2(16)a = 1.2(16) C + 2(16) C +16d

a = 1.473

Q. 85-87

The following equations are alternatives to the Monod equations:

Blackman equation : g = m if S ≥ 2KS

$$g = \frac{\mu m}{2ks} S, \text{ iff } S < 2\ kS$$

Tessier equation : $g = m(1 - e^{kS})$

Moser equation : $\mu g = \frac{\mu m s^n}{k_s + s^n} = (1 + k_g S^{-n})^{-1}$

Contois equation : $g = \frac{\mu m s}{k_s \times X + S}$

89. The main advantages of air – lift bioreactors compared with stirred tank bioreactors are

(*i*) Low shear

(*ii*) Relatively low energy requirements

(*iii*) Simpler construction, as seals are not required around the agitator shaft

90. On scaling – up – air – lift systems; the mixing the only increases slightly with size because the liquid circulation rate can be kept almost constant. This is turn is because the driving force pressure difference caused by the gas hold up increases with height, whereas the wall friction decreases with increasing diameter. Consequently the metabolic performance of air – lift reactor should not be reduced by scale – up

91. The main advantages of a fluidized bioreactor system is relatively low energy requirements.

93. There are 3 main types of agitators that are used in bioreactors they are

(*i*) Turbine

(*ii*) Propeller

(*iii*) MIG/INTERMIG

The Rushton turbine is the most common of the 3 types of agitator.

99. A typical SSF process involves 2 stages. The first & primary stage is an aerobic, fungal, solid - state fermentation of grains called the koji.

100. The 2nd stage of anaerobic submerged fermentation with a mixed bacterial culture called the moromi.

101. Fermentation in the second stage (moromi) may be realized by using the natural flora, or usually, with externally added bacteria & yeasts.

102. Some strains of Saccharomyces, Torulopsis & Pediococcus are used as flavour producers in soy sauce manufacture.

103. The moromi is usually fermented for 8 to 12 months. Forever, the processing time can be reduced to 6 months by temperature profiling.

Q. 104 & 105

The major industrial use of the Koji process is for the production of enzymes by fungal species. Fungal amylases are produced by SSF wheat from by A. oryzae in a rotating – drum fermenter.

106. Enzymes other than amylases, such as cellulose, pectinase, protease & lipases, can also be produced by Koji fermentations.

Q. 108 & 109

Some secondary metabolites, such as antibacterial agents, produced by Rhizopus & Actinomucor species in some koji processes. Certain mycotoxins, such as aflatoxins were produced by SSF of rice (40% moisture) by A. parasiticus ochratoxins were also produced by Aspergillus species on wheat in a rotary– drum koji fermenter.

110. Microbial degradation of lignocellulosic can also be accomplished by solid – state fermentations for waste – treatment purpose / in biopulping of wood chips for use in paper manufacture.

112. Koji fermentations are usually realized in a controlled – humidity air environment with air ventilation & agitation.

113. Niemann – pick disease is one of a group of lysosome storage disease that affect person's metabolism. It is caused by the deficiency of the enzyme sphingomyelinase.

114. Colchicines inhibits the polymerization of microtubules, thus blocking cell processes such as cell division that depend on functioning microtubules.

115. In integral membrane proteins, the hydrophobic membrane-spanning region consists mainly of amino acid residues with hydrophobic side – chains that folded in a α – helical conformation.

116. Inner mitochondrial membrane is rich in action translocases fatty acyl tranferase.

117. Cytochrome oxidase deficiency in mitochondria causes Kearns says syndrome.

120. Covalent chromatography is mainly used for separating thief containing proteins from non-thief containing proteins.

NUMERICAL TYPE QUESTIONS

2. Autoclaving is the most effective and most efficient means of sterilization. All autoclaves operate on a time/temperature relationship. The usual standard temperature/pressure employed is 121°C/15 psi for 15 minutes.

4. For hexadecane

$$Y_{X/S} = \frac{2.42\ (MW)_{biomass}}{(MW)_{substrate}}$$

$$Y_{X/S} = \frac{2.42\ (91.34)}{226}$$

$$= 0.98 \text{ gdw cells / g substrate}$$

$$Y_{X/02} = \frac{2.42\ (MW)_{biomass}}{12.43(MW)_{02}}$$

$$Y_{X/02} = \frac{2.42\ (91.34)}{(12.43)\ (32)} = 0.557 \text{ gdw cells / go}_2$$

For glucose

$$Y_{X/S} = \frac{(0.409)\ (91.34)}{180}$$

$$= 0.461 \text{ gdw cells / g substrate}$$

$$Y_{X/02} = \frac{(0.909)\ (91.34)}{(1.473)\ (32)} = 1.76 \text{ gdw cells / go}_2$$

5. Since Y_X/ATP = 10.5 gdw/mol ATP & since glycolysis yield 2ATP/mole of glucose in yeast

$$Y_{X/S} = 10.5 \text{ golw/mol ATP } 2\ \frac{\text{moles ATP}}{180\text{g glucose}}$$

or

$Y_{X/S}$ = 0.117 gdw / glucose

For complete conversion of glucose to ethanol by the yeast pathway, the monomial yield would be $Y_{P/S} = \frac{2(46)}{180}$ = 0.51g ethanol / g glucose.

While for CO_2 the maximum yield is

$$Y_{CO2/S} = \frac{2(44)}{180} = 0.49 \text{ g ethanol / g glucose}$$

6. Consider the aerobic catabolism of glucose

$C_6H_{12}O_6 + 60_2 \rightarrow 6CO_2 + 6H_2O$

The total number of available electrons in 1 mole of glucose is 24. The cellular yield/available electron is 1 × 1s = 24(3.14) = 76 gdw cells/mol.

7. Wheat bran is pretreated with formaldehyde, & the initial pH of the bran is adjusted to pH = 3.5 to 4.0 to reduce chance of contamination. Usually, perforated forms, rotating dreams, or packed beds with air ventilation are used.

EXERCISE – II

MCQ TYPE QUESTIONS

1. Prandlt No is the ratio of Momentum Diffusivity and Thermal Diffusivity.

2. A fed-batch culture is a semi-batch operation in which the nutrients necessary for cell growth and product formation are fed either intermittently or continuously via one or more feed streams during the course of an otherwise batch operation. The types of bioprocesses for which fed-batch culture is effective can be summarized as follows:

1. Substrate inhibition,
2. High cell density (High cell concentration)
3. Glucose effect (Crabtree effect)
4. Catabolite repression
5. Auxotrophic mutants
6. Expression control of a gene who has repressible promoter in recombinant cells
7. Extension of operation time, supplement of water lost by evaporation, and decreasing viscosity of culture broth.

3. For mixing high viscosity fluids 10000cP helical ribbon impeller is generally used.

4. Stationary phase of Bacterial growth characterises of cell size increase of certain gram +ve bacteria, cell divion is halted, spore formation occurs in gram +ve bacteria and the growth rate becomes equal to the death rate owing to nutrient limitation, spore formation, starvation util and unless the culture is stirred. Peptidogylcan cross linking is also remarkably reduced.

5. Scaled up criteria is equal to impeller tip speed

So $N_1D_1 = N_2D_2$

$$\therefore \quad \frac{N_2}{N_1} = \frac{D_1}{D_2} \qquad \ldots(i)$$

Now, power equation is given by $P_2 \propto N_2^3 D_2^5$

$$\frac{P_2}{P_1} = \frac{N_2^3 D_2^5}{N_1^3 D_1^5} = \left(\frac{N_2}{N_1}\right)^3 \left(\frac{D_2}{D_1}\right)^5$$

$$= \left(\frac{D_1}{D_2}\right)^3 \left(\frac{D_2}{D_1}\right)^5 \quad \left(\because \frac{N_2}{N_1} = \frac{D_1}{D_2}\right)$$

$$\frac{P_2}{P_1} = \left(\frac{D_2}{D_1}\right)^2$$

6. $\ln \frac{x_t}{x_0} = \mu_t$ if $x_t = 5 \times 10^5$, $x_0 = 6.5 \times 10^5$

then μ is –ve $\quad \therefore \ \mu < 0$

7. For Quasi steady state

$$x = x_0 + Fy_{X/S} S_0 t$$

$$x_0 = 20 \times 600 = 12000$$

$F = 50$

$Y_{x/s} = 0.5$

$S_0 = 200$

$t = 8$

$\therefore \quad x = 52000$ g/1000 lit = 52 g/lit.

8. $$\frac{t_2}{t_1} = \left(\frac{D_2}{D_1}\right)^{1/6}\left(\frac{N_1}{N_2}\right)^{2/3}$$

Now, $\frac{P_1}{V_1} = \frac{P_2}{V_2}$

$\Rightarrow \quad N_1^3 D_1^2 = N_2^3 D_2^2$

$\therefore \quad \frac{N_1}{N_2} = \left(\frac{D_2}{D_1}\right)^{2/3} = 4.6415$

$\therefore \quad t_2 = 15(10)^{1/6}\,(4.6415)^{2/3} \approx 61$

9. Cell productivity $= DY_{x/s}\left(S_R - \frac{K_s D}{\mu_m - D}\right)$

$D = \frac{150}{750} = 0.2$

$S_R = 8$

$K_S = 1.5$

$\mu_m = 0.4$

$Y_{x/s} = 0.5$

Then after calculation cell productivity = 0.65

10. Specific substrate concentration is defined as the amount of substrate consumed in grams per unit gram of substrate provided to the growing culture per unit time or per hour. So, the unit of specific substrate concentration is $gg^{-1}h^{-1}$.

11. In batch cultures of microorganisms, especially bacteria, the exponential phase is concerned with the multiplication and high metabolic activities of the microbes along with growth. But as the microbe advances to the stationary phase, metabolic activities reach the maximum limit with stagnancy in multiplication and growth of the bacteria. During this phase, there occurs maximum production of secondary metabolites such as toxins. So, Penicillium chrysogenum synthesizes penicillin at maximum concentrations during the stationary phase of the batch culture.

12. Shear speed of an impeller along with the diameter of the impeller blade have profound impact on the shear stress generated in the bioreactor system. Mammalian cells are very shear sensitive. So, for scale up of a small scale bioreactor where mammalian cells are used as feed, the major parameters to be considered would be the shear speed and agitator diameter. This is because, while scaling up, impeller speed has to be considered for controlling the shear speed which is given by: $T_i = \pi ND$

13. In a lab-scale bioreactor, lysine is produced from threonine auxotroph but it was found that concentration of lysine was gradually decreasing. The most probable reason was found to be revertants. Reversion is the process by which mutant DNA is changed so that the protein inactivated by the first mutation is reactivated by the second. There are three classes of reversion: Class I, class II and class III. Revertant is a mutant gene, individual, or strain that regains a former capability (as the production of a particular protein) by undergoing further mutation.

14. Given, $\nabla_{heating} = 12.56$, $\nabla_{cooling} = 7.48$

Now we have, $\nabla_{total} = \nabla_{heating} + \nabla_{cooling} + \nabla_{holding}$

$\Rightarrow \quad 52 = 12.56 + 7.48 + \nabla_{holding}$

$\Rightarrow \quad \nabla_{holding} = 31.96$

15. Given k value = 3.36 min^{-1}

Holding value of design criteria, $\nabla_{holding} = 31.96$

Therefore, time of holding, $t = \nabla_{holding}/k$

$= 31.96/3.36$ min $= 9.512$ min.

16. Yield coefficient = mass of new cells formed/ mass of substrate consumed

New cells mass formed = cell density X specific growth rate $= 20 \times 0.4 = 8$

Substrate consumed = 16 $g/l/h$

Yield coefficient $= \frac{8}{16} = 0.5$.

17. Solvent extraction is a method for processing materials by using a solvent to separate the parts. It can be used for penicillin extraction.

Affinity chromatography is the preferred method of bioselective adsorption and subsequent recovery of a compound from an immobilized ligand. Each is designed for highly specific and efficient purification of proteins and related compounds like monoclonal antibody.

Extractive distillation is defined as distillation in the presence of a miscible, high boiling, relatively non-volatile component, and the solvent that forms no azeotrope with the other components in the mixture. It can be used for lactic acid recovery.

Salting out is a purification method that utilizes the reduced solubility of certain molecules in a solution of very high ionic strength. Crude lipase can be purified by using salting out method.

18. Bacteriostatic antibiotics limit the growth of bacteria by interfering with bacterial protein production, DNA replication, or other aspects of bacterial cellular metabolism. Since after incubation, plate A showed no colonies and plate B showed $0.9 * 10^9$ colonies (half of the bacterial culture before inoculating), thus it shows that antibiotic Z is bacteriostatic which has stopped the bacteria from reproducing.

21. Balanced growth of the cell depends upon the concentration of the medium component concentration. The components are been uptaken in uniform rate which yields the specific growth rate constant. When it goes to decline phase, concentration of the component decreases.

22. In lag phase, environmental condition is adapted by bacteria themselves. In this phase each bacterium is matured enough but cannot divide. Synthesis of enzymes and other molecules occur in this phase. So, the substrate consumed by microbes in lag phase is involved in the motility of the micro-organisms, maintenance of intracellular pH and cell materials required for growth and physiology of the microbes in the upcoming phases.

23. Dilution rate is defined as rate of flow of medium into a continuous culture system. It is equal to the rate of cell division when the number of cell remains constant in vessels because the cells that are replaced by medium outflow by cell division equally. In continuous culture, rate of dilution regulates the rate of microbial growth through concentration of growth limiting nutrient in the medium. This is the advantage of this culture system.

24. At growth associated product formation specific product formation rate is proportional to specific growth rate when Non growth associated product formation shows the constant sp. Growth rate. Product induced inhibition can be seen at increase in product concentration with decrease in sp growth rate, this phenomena is vice versa with substrate inhibition..

25. The "gas out-gas in" method can directly determine the volumetric mass transfer coefficient in a bioreactor system and provide estimates of the volumetric microbial oxygen uptake rate and the average oxygen saturation concentration at the gas-liquid interface. The rate of oxygen transfer from the bubble air to the liquid phase is described by the equation:

$$dC_L/dt = K_L a\,(C^* - C_L)$$

$$\Rightarrow \quad K_L a = dC_L/dt * 1/(C^* - C_L)$$

$$\Rightarrow \quad K_L a = r^*X/(C^* - C_L)$$

Where $r^*X = dC_L/dt$, r is the specific oxygen uptake rate per unit weight of cells and X is the dry weight of cells per unit volume.

For reliable measurements, deconvolution of the oxygen probe measurements must be made.

26. Williams proposed a simple structured model that divides the biomass into two parts: Part 1 is composed of intermediates, enzymes, and other entities involved in the formation of materials used for synthesis of structural and genetic material; Part 2 is the structural and genetic part of the biomass. It can be used to describe the change in intracellular components during cell growth. In the different phases of cell growth, this model enables analytical depiction and explanation of the variation in the size of the cells. But the model is not structured for reasoning with the death phase of the cells.

27. Synchronous growth of a bacterial population is that during which all bacterial cells of the population are physiologically identical and in the same stage of cell division cycle at a given time. Synchronous growth helps studying particular stages or the cell division cycle and their interrelations. A synchronous culture can be obtained either by manipulating environmental conditions such as by repeatedly changing the temperature or by adding fresh nutrients to cultures as soon as they enter the stationary phase, or by physical separation of cells by centrifugation or filtration. Synchronous cultures rapidly lose synchrony because not all cells in the population divide at exactly the same size, age or time.

28. By Monod model, ln X Vs t graph is linear and slope = μ, where μ is specific growth rate, μ_{max} is maximum specific growth rate. It represents the log phase of the growth.

29. We know,

Yield of biomass,

$Y_{X/S}$ = Biomass produced/ Substrate consumed

Now,

Substrate consumed,

S = Initial substrate concentration (S_i)
– Final substrate concentration (S_f)
$= S_i - S_f$

$S_i = 10$ kg/ m^3 and $S_f = 2$ kg/ m^3

Biomass produced, X = 4.5 kg dry biomass/ m^3

Therefore,

$Y_{X/S} = X/(S_i - S_f) = 4.5\,(10 - 2)$

$= 0.562$ Kg of biomass/ kg of the substrate.

30. Data is insufficient to solve the question.

31. Probability of bacteria getting infected by a single bacteriophage= 0.3 i.e. P(A)= 0.3

Given, 3 phages, So, probability of bacteria getting infected by 3 phages = P(A) * P(B) * P(C)

$= 0.3 * 0.3 * 0.3 = 0.027$

32. Chemostat is operated by supplying an essential growth limiting nutrient at a constant rate with the result that the culture density and growth rate adjust themselves to the supply.

At steady state,

dx/dt= 0 Therefore, μ= D

This equation shows that by simply varying the medium supply rate the growth rate can be varied.This equation holds true if μ eD or μ–D = positive.

Here, Dc = μ max, i.e. the critical dilution rate is defined as the lowest dilution rate at which washout occurs.

Thus, for a bacterial culture to grow at dilution rate higher than maximum growth rate, the partial cell recycling is needed under steady state condition.

33. Given,

Inner radius (r) = 10 cm

Length (l) = 20 cm

Length of spiral film (L)= 30 cm

Width (W) = 20 cm

Thus, the increase in the surface area after fitting of surface film = 30 * 20 = 600 cm^2

Also, the film can support 10^5 anchorage dependent cells per cm^2

Thus, the film will add = 10^5 cells/cm^2 * 600 cm^2

= 6 * 10^7 cells.

34. Initial number of cells= 10^8 cells/ml * $1m^3$

But, we know,

$1m^3 = 10^6 cm^3$

Also, $10^3 cm^3$ = 1000ml

Therefore, total initial number of cells

= 10^8 cells * 10^6

= 10^{14} cells

Final acceptance of contamination

= 10^{-3} cells/ml or 10^3 cells/m^3

Thus, using the first order rate constant equation:

$t = 2.303/K \log_{10} [10^{14}]/10^3$

$= 2.303/0.23$ per min $\log_{10} (10^{14}/10^3)$

$= 10 * 11 \log_{10} (10)$

t = 110 min.

35. By using the Monod Equation under steady state,

We have,

$= {}_{max} [S]/[S] + K_S$

$= 0.3\ hr^{-1} * 1.5g/l / [1.5g/l] + [0.2g/l]$

$= 0.2648\ hr^{-1}$

D = (by using the Monod model)

Also, D = F/V

Therefore, equating the two, we get,

F = V = 0.2648 * 10

$F = 26.5\ m^3 hr^{-1}$

36. We know,

Yield, $Y_{x/s} = -\Delta X/\Delta S$

$Y_{x/s} = 0.4$ g/L

Therefore,

$\Delta X = X_{final} - X_{initial} = 4$ g/L

From the equation,

$Y_{x/s} = -\Delta X/\Delta S$

$0.4 = -[4/(S_{final} - 12)]\ 0.4\ S_{final} - 4.8$

$= -4,\ 0.4\ S_{final} = 0.8,\ S_{final} = 2$

Final substrate concentration,

$S_{final} = 2$ g/L

37. Biostat keeps all the given parameters constant.

38. Bioreactor choice should be dependent upon the kinetic properties if the reactor and its stability condition. Fed batch reactors are semi batch reactor as it holds the nutrient media for a specific retention time.

39. Assertion: A synchronous or synchronized culture is a microbiological culture or a cell culture that contains cells that are all in the same growth stage.

Reason: Synchronous cultures can be obtained in several ways:

- External conditions can be changed, so as to arrest growth of all cells in the culture, and then changed again to resume growth.
- Cell growth can also be arrested using chemical growth inhibitors.
- Cells in different growth stages have different physical properties. Cells in a culture can thus be physically separated based on their density or size, for instance. This can be achieved using centrifugation (for density) or filtration (for size).
- In the Helmstetter-Cummings technique, a bacterial culture is filtered through a membrane. Most bacteria pass through, but some remain bound to the membrane. Fresh medium is then applied to the membrane and the bound bacteria start to grow. Newborn bacteria that detach from the membrane are now all at the same stage of growth; they are collected in a flask that now harbors a synchronous culture.

40. When both the substrates are present in a batch bioreactor, microbial cells do not consume both substrates simultaneously or additively but instead grow on the sugars sequentially. The sequential consumption results in an interesting pattern of the cells first consuming one preferred substrate and then after an intermediate lag phase, consuming the remaining less preferred substrate. That is, the less preferred substrate is not utilized until the preferred substrate is completely consumed and is no longer available in the growth medium. This sequential utilization of two substrates in batch cultures has been observed in numerous experiments by Monod, who has termed this phenomenon as the "diauxic pattern of growth". It has multiple lag phases.

41. Sparger is being used for the aeration of the medium in bioreactor. Diaphram valve is being used for the air sterilization in large scale animal or microbial cell culture condition. Draft tube has been used for the feed circulation in semi fedbatch culture. Marine type turbine is being used to reduce shear stress on fluid medium in animal cell culture.

42. $Y_{x/s} = 0.35$ g/g, $Y_{P/S} = 0.1 \frac{\text{g acetate}}{\text{g glucose}} \frac{\text{g dry cell}}{\text{g glucose}}$

The respiratory coefficient

$$\text{Respiratory quotient} = \frac{\text{Moles of } CO_2 \text{ formed}}{\text{Moles of } CO_2 \text{ consumed}}$$

$$= \frac{5}{4.95} = 1.01$$

$$C_6H_{12}O_6 + b\,NH_3 + O_2 \rightarrow CH_{1.8}O_{0.5}N_{0.2} + eH_2O + fCO_2$$

C : $6a = 1 + f$ ...(1)

H : $12a + 3b = 1.8 + 2e$ ...(2)

O : $6a + 2d = 0.5 + e + 2F$ ...(3)

$b = 0.2$

N : $b = 0.2$

$\therefore \quad 12a = 1.2 + 2e$

$6a = 1 + F$

Let if $a = 1$, i.e. F = 5

$e = 5.4$

$d = 4.95$

$b = 0.2$

$a = 1$

43. $K_L a = 0.185\ S^{-1} = 5.139 \times 10^{-5}\ hr^{-1}$

Critical dO_2 is 20% of the saturate conc.(8 mg/ml)

Cell density. $C_L = 1.6 \times 10^3$.m/l C = (8×10^3 mg/l)

$dO_2 = K_L a\,(C - C_L)$

$$OUR = \frac{1/\mu}{1/O_2} = q_{O2} X.$$

q_{o2} = mmo2/gm sw. hr.

$= 5 \times 10^{-3}$ gm dm hr

$\therefore K_{La} (C - C_L) = q_{02} X.$

$$\Rightarrow \quad X = \frac{K_{La}(C - C_L)}{q_{02}}$$

$$\therefore \quad X = \frac{5.139 \times 10^{-5}(8 - 1.6) \times 10^3}{5 \times 10^{-3}}$$

= 65.779 g/lit.

44. Specific growth rate is defined as the increase in cell mass per unit time. In death phase, no cellular growth is observed. During this, the cell population continuously decreases with time and the population is said to have entered into death phase of the growth cycle. Therefore the value of specific growth constant for death phase is zero.

45. The main roles of a baffle in a bioreactor are

- Increase aeration rate
- to prevent a vortex

46. $C_6H_{12}O_6$ + Zymase → $2\,C_2H_5OH + 2\,CO_2$ use mass balance process to calculate the ethanol production.

180g of Glucose yields = 92 g of ethanol

Therefore,

1000g of glucose yields = 92/180 * 1000 = 0.51 kg

47. Cyanogen bromide hydrolyzes peptide bonds at the C-terminus of methionine residues. This reaction is used to reduce the size of polypeptide segments for identification and sequencing. The aspartate residue (Asp 189) located in the catalytic pocket (S1) of trypsins is responsible for attracting and stabilizing positively charged lysine and/or arginine, and is, thus, responsible for the specificity of the enzyme. This means that trypsin predominantly cleaves proteins at the carboxyl side (or "C-terminal side") of the amino acids lysine and arginine except when either is bound to a N-terminal proline. Although large-scale mass spectrometry data suggest cleavage occurs even with proline. Trypsins are considered endopeptidases, i.e., the cleavage occurs within the polypeptide chain rather than at the terminal amino acids located at the ends of polypeptides.

48. Final volume = initial volume

+ Flow rate × Number of day

$= 6m^3 + 2m^3\ day^{-1}$ (2day) = $10m^3$

49. Dilution rate at end of 2nd day

$$= \frac{\text{Flow rate}}{\text{Total volume at end of } 2^{nd} \text{ day}}$$

$$= \frac{2m^3}{10m^3} = 0.2$$

50. Number of cell died

= Total cell – Number of cell survived

$= 10^9 - 10^3$

$= 9.9 \times 10^8 = 10^9$

51. 10^3 cpm in pellet is due to F_{ab} binding to epitope present on recombinant living cells/algae

Total F_{ab} or epitope

$$= \frac{\text{Radioactivity in pellet}}{\text{Specific activity}}$$

$$= \frac{10^3 cpm}{10^2 cpm(picomole)^{-1}}$$

$$= \frac{10^3 \times 10^{-12}}{10^2}$$

$= 10 \times 10^{-12}$ mole = 10 picomole

These 10 picomole epitopes are present on 10^3 cells (assuming) 100% transfection and expression of protein).

Average number of epitopes on single recombinant alga

$$= \frac{10 \times 10^{-12}}{10^3} \text{ mole}$$

$= 10^{-14}$ mole

$= 6.023 \times 10^{23} \times 10^{-14} = 6 \times 10^9$

52. Thermistors monitor temperature and as per requirements very precisely activate hot/cold water pump to heat or cool down the system. The oxygen electrode directly measures the concentration of dissolved oxygen in the liquid phase and so as to regulate the oxygen level it increases the stirrer motor speed for more aeration whereas decreases speed if oxygen concentration has to be decreased. Metal rod is used to switch on/off the vegetable oil pump as vegetable oil has defoming action but its higher concentration can as well has negative effects. Thus metal rod maintains appropriate oil concentration in the fermentor. pH electrode measures the pH of the system liquid and thus varies acid/alkali pump to increase and decrease pH respectively.

53. Hollow fibers also provide a tremendous amount of surface area in a small volume. Cells grow on and around the fibers at densities of greater than 1 X 108 per mL. Hollow fiber cell culture is the only means to culture cells at in vivo like cell densities.

54. In single run metabolite (X) concentration is

$= 1.1$ kg m^{-3}.

Since, total number of runs = 70

Total concentration of (X)

$= 70 \times 1.1$ kg m^{-3} run^{-1}

$= 70 \times 1.1$ kg m^{-3}

$= 77$ kg m^{-3}

Now, total volume of batch culture = 50 m^3.

Total amount will be

$= 50\ m^3 \times 77$ kg m^{-3}

= 3850 kg per year

55. Overall productivity

= Total number of runs × final concentration at each end

$= 70$ runs $\times 1.1$ kg m^{-3}

$= 77$ kg m^{-3} $year^{-1}$

NUMERICAL TYPE QUESTIONS

1. $N_P = \dfrac{P}{Sn^3D^5}$

$\therefore P = N_P Sn^3 D^5$

$$= 5 \times 1000 \times \left(\frac{300}{60}\right)^3 \times 1^5 = 625000\text{W}$$

P = 625 kW

Where

N_P = Power No. = 5

P = Power required

S = Density 1000 kg/m^3

n = Speed in Hz $= \dfrac{300}{60} = 5$ Hz

D = Diameter of turbine = 1 m

2. The equation for biomass production is

$$C_2H_5OH + a\,NH_3 + bO_2 \rightarrow CH_{1.8}O_{0.5}N_{0.2} + dCO_2 + eH_2O$$

Given 1 : ? 1

Now making elemental balance on both sides.

LHS C = 2

N = a i.e. a = 0.2

RHS C = 1 + d = 2 i.e. d = 1

N = 0.2

Now, putting value of a & d in equation (i)

$$C_2H_5OH + 0.2NH_3 + bO_2 \rightarrow CH_{1.8}O_{0.5}N_{0.2} + Co_2 + eH_2O$$

Now, H balance LHS

H = 6 + 0.6 = 6.6

RHS

H = 1.8 + 2e = 6.6

$\therefore$ e = 2.4

and finally O_2 balance

LHS

O = 1 + 2b

RHS

0 = 0.5 + 2 + 2.4 (i.e. e = 2.4) = 4.9

hence 1 + 2b = 4.9

$\therefore$ 2b = 3.9

$\therefore$ b = 1.95

$\therefore$ Final equation

$$C_2H_5OH + 0.2\,NH_3 + 1.95O_2 \rightarrow CH_{1.8}O_{0.5}N_{0.2} + CO_2 + 2.4H_2O$$

3. (1) $\bar{X} = Y_{X/S}\left(S_R - \dfrac{k_s D}{\mu_m - D}\right)$

(2) Here $Y_{X/S} = 0.4 \dfrac{\text{g biomass}}{\text{g ethanol}}$

$S_R = 10\ gL^{-1}$

$k_S = 0.3\ g\ L^{-1}$

$D = 0.5\ h^{-1}$

$\mu_m = 0.7\ h^{-1}$

(3) $\therefore \bar{X} = 0.4\left(10 - \dfrac{(0.3)(0.5)}{(0.7-0.5)}\right)$

$\bar{X} = 3.70$

4. $D_r = \dfrac{2.303}{K_d}$

$\therefore K_d = \dfrac{2.303}{D_r}$, $D_r = 23$ min

$\Rightarrow K_d = 0.100\ min^{-1}$

Now $N_0 = 10^5 \times 1000 = 10^8$ spores in 1 litre

$\therefore \ln \dfrac{N_0}{N_1} = k_d t$, t = 10 min

$\therefore \ln \dfrac{10^8}{N_1} = 0.1 \times 10$

$\Rightarrow \dfrac{10^8}{N_1} = e^1 = 2.718$

$\therefore N_1 = 3.678 \times 10^7$ spores

5. For CSTR

(i) $\dfrac{V}{\nu_0} = \dfrac{C_{AO}X_A}{-r_A}$

here $\nu_0 = 25$ L/min

$C_{AO} = 2$ ml/L

$X_A = 0.95$

$-r_A = \dfrac{0.1C_A}{1+0.5C_A}$

(ii) $V = \dfrac{\nu_0 C_{AO} X_A}{\left(\dfrac{0.1C_A}{1+0.5C_A}\right)}$

further $C_A = C_{AO}(1 - X_A)$

$\therefore V = \dfrac{\nu_0 C_{AO} X_A}{\left(\dfrac{0.1C_{AO}(1-X_A)}{1+0.5C_{AO}(1-X_A)}\right)}$

Putting values & solve

V = 4987.5 L

6. Total cells after 24 hours

$= 1.8 \times 10^5 \times 2 = 3.6 \times 10^5$

Total number of wells = 6 × 3 = 18

$\therefore$ Number of cells in each well

$$= \frac{3.6 \times 10^5}{18} = 2 \times 10^4$$

7. $K_{La} = \dfrac{\ln\left(\dfrac{80-55}{80-68}\right)}{17-7} = 0.07339$

8. $G = \dfrac{t}{3.3 \log \dfrac{N_2}{N_1}}$

$N_2 = 10^7$

$N_1 = 10^4$, $t = 12$hr

G = generation time

Then G = 1.2121 h

9. $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$

180 2×46 = 92

So, Yield = $\dfrac{92}{180}$ g ethanol/g glucose

So, for 2.5 gm = $2.5 \times \dfrac{92}{180} = 1.27$ gm

10. Given, Seeding of T-flask with = 10^5 no. of anchorage dependent cells

Area of flask available = 25cm^2 and volume of flask = 25mL

Size of cells = 5μm × 2μm

Area of the T-flask covered by a single cell

$= 10\mu m^2 = 10 \times 10^{-8}\ cm^2$

Now, anchorage dependent cells multiply every hour per unit area

So, the growth rate of the cells = 2×10^5 cells/cm^2h

11. At dilution rate = $5h^{-1}$, cell density = 0. This means cell wash out occurs at this dilution rate

The mass balance on the microorganisms in a CSTR of constant volume is:

[Rate of accumulation of cells, g/s] = [Rate of cells entering, g/s] – [Rate of cells leaving, g/s] + [Net rate of generation of live cells, g/s]

So, maximum specific growth rate $\mu_{max} = 0.8\ h^{-1}$

12. Given, Initial dry weight = 5 g

After inoculation for 10 days, dry weight

$= 1.5 \times 5 = 7.5$ g

Therefore Growth index of the callus

=(Difference or change in weight)/ (Originalweight)

$= (7.5 - 5)/5 = 0.5$

13. Given, Biomass concentration in the exit stream
$= 30$ g/L

Dilution rate of the chemostat = 0.6 h^{-1}

Therefore, Biomass productivity of the chemostat
$= 30 \times 0.6\ gL^{-1}h^{-1} = 18\ gL^{-1}h^{-1}$

14. Let the agitator speed in the large bioreactor be V_L

Now, we have for scale up:

$$V_L \times D_L = V_S \times D_S$$

Where,

D_L=diameter of the large bioreactor,

V_S = agitator speed of the small bioreactor = 450 rpm,

D_S = diameter of the small bioreactor,

n = scale up ratio = 10,000/10 = 1000

Now, $D_L = 10\ D_S$

Therefore, $V_L \times 10\ D_S = 450 \times D_S$

$\Rightarrow \quad V_L = 45$

Hence the agitator speed of the large bioreactor is 45 rpm.

15. Given, two sequences HELLO- and YELLOW

The % identity between the two sequences would be: No. of identical positions/ Total no. of positions

No. of identical positions = 4 (ELLO); Total no. of positions = 6

Therefore, percentage identity
$= 4/6 \times 100 = 66.667\ \%$

16. Given, for a batch culture

Specific rate of substrate utilization = 0.25 g/g/h

Specific rate of product formation = 0.215 g/g/h

Yield of product ($Y_{P/S}$) = 0.215/.25 = 0.86

17. Given, Initial cell concentration, X = 1.75 g/L

Initial concentration of substrate glucose,
S = 125 g/L

Given, Yield of biomass, $Y_{X/S}$ = 0.6 g/g

Now, Concentration of biomass produced
$= Y_{X/S} \times S = 125 \times 0.6$ g/L = 75 g/L

Therefore, maximum concentration of cell expected in the bioreactor
$= 75 + 1.75$ g/L = 76.75 g/L

18. Given, flow rate for the glucose solution, F = 200 mL/hr

K_S = 0.3 g/L; $_m$ = 0.4 h^{-1}; D = 0.1 h^{-1}

Therefore, Volume of the reactor,

V = F/D = 200/0.1 mL = 2000 mL = 2L

We have, from Monod's equation of substrate consumption

$= (\ _{max} * S)/(K_S + S)$

where,

= rate of substrate consumed per unit time.

$_{max}$ = maximum rate of substrate consumed.

K_S = Monod's constant.

S = steady state substrate concentration.

In steady state, mass balance equation is given by:-

Input – Output – Consumption = 0

$\Rightarrow (F/V)*S_R - \quad = 0$

$\Rightarrow D(S) - \{(\ _{max}*S)/(K_S + S)\} = 0$

Therefore,

$D(S) = \{(\ _{max}*S)/(K_S + S)\} = \{(\ _{max}*S)/(K_S + S)\}$
$= \{(0.4S)/(0.3 + S)\}$

$\Rightarrow DS = 0.4S/(0.3 + S)$

$\Rightarrow 0.1 = 0.4S/(0.3 + S)$

$\Rightarrow 0.4S = 0.03 + 0.1S$

$\Rightarrow 0.3S = 0.03$

$\Rightarrow S = 0.1$

Therefore, the concentration of the limiting substrate = 0.1 g/L